

ASUS ZS670KS

RF Hardware Trouble Shooting Guide

V1

2020/07/16

ZS670KS Supported Bands

2G:	GSM	GSM850/EGSM/DCS/PCS
3G:	WCDMA	B1/B2/B3/B4/B5(B6/B19)/B8
4G:	LTE-FDD	B1/B2/B3/B4/B5/B7/B8/B12/B17/ B18/B19/B20/B26/B28/B29
	LTE-TDD	B38/B39/B40/B41
	4*4MIMO	B3/B7
5G:	NR-FDD	N1/N2/N3/N5/N7/N8/N12/N20/ N28
	NR-TDD	N38/N77/N78
	4*4MIMO	N3/N7/N77/N78

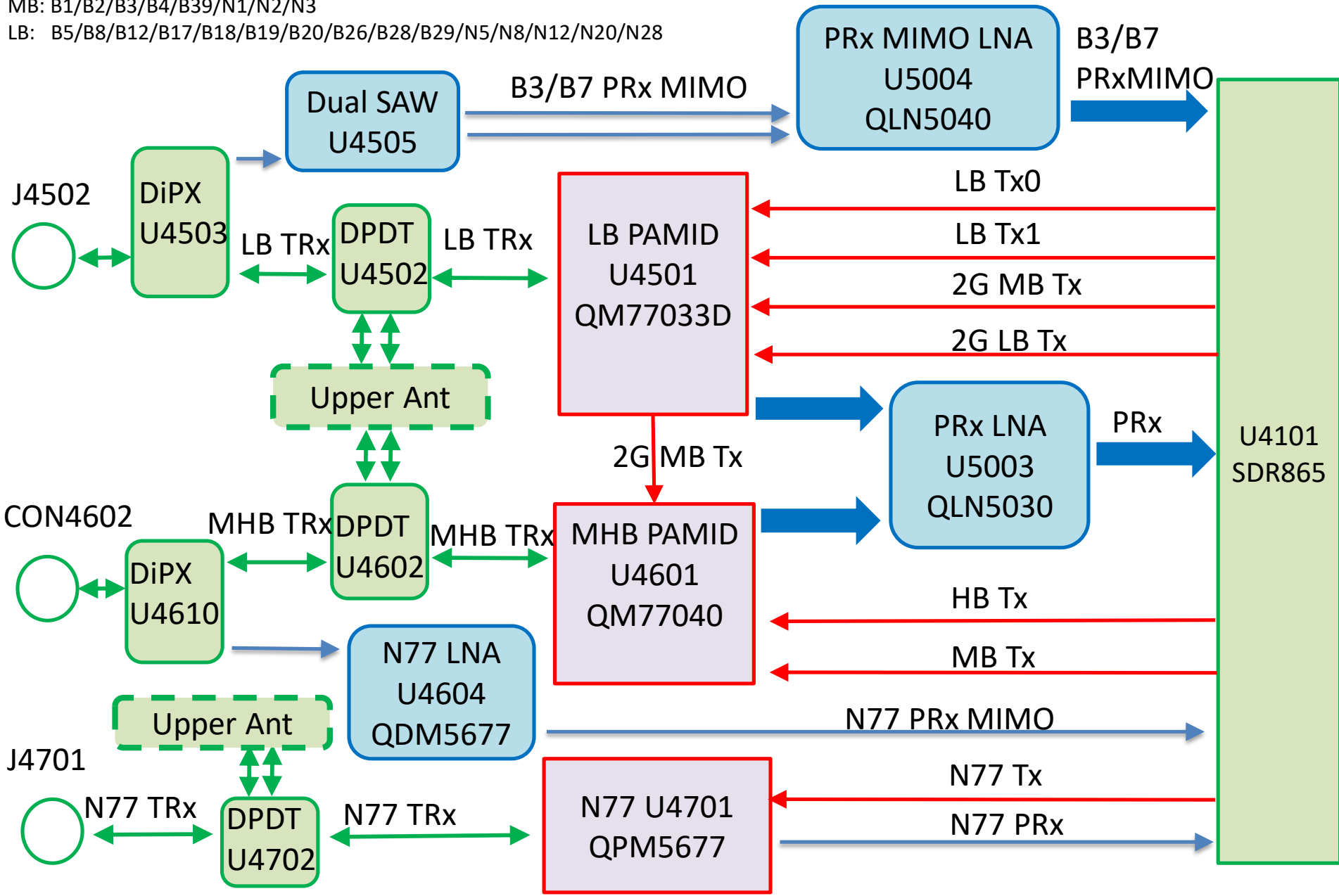
Tx/PRx path block diagram

N77: N77/N78

HB : B7/B38/B40/B41/N7/N38

MB: B1/B2/B3/B4/B39/N1/N2/N3

LB: B5/B8/B12/B17/B18/B19/B20/B26/B28/B29/N5/N8/N12/N20/N28



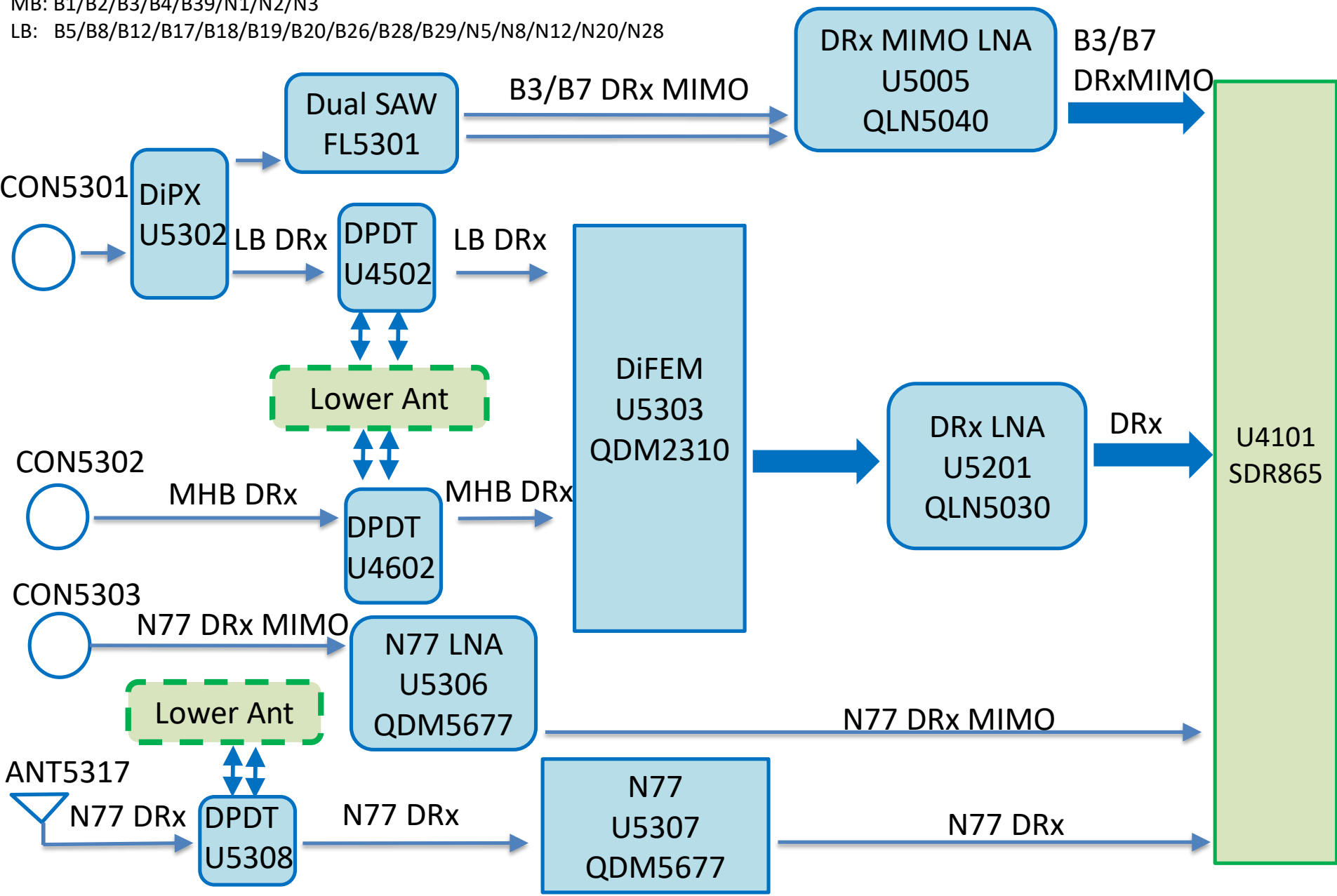
DRx path block diagram

N77: N77/N78

HB : B7/B38/B40/B41/N7/N38

MB: B1/B2/B3/B4/B39/N1/N2/N3

LB: B5/B8/B12/B17/B18/B19/B20/B26/B28/B29/N5/N8/N12/N20/N28

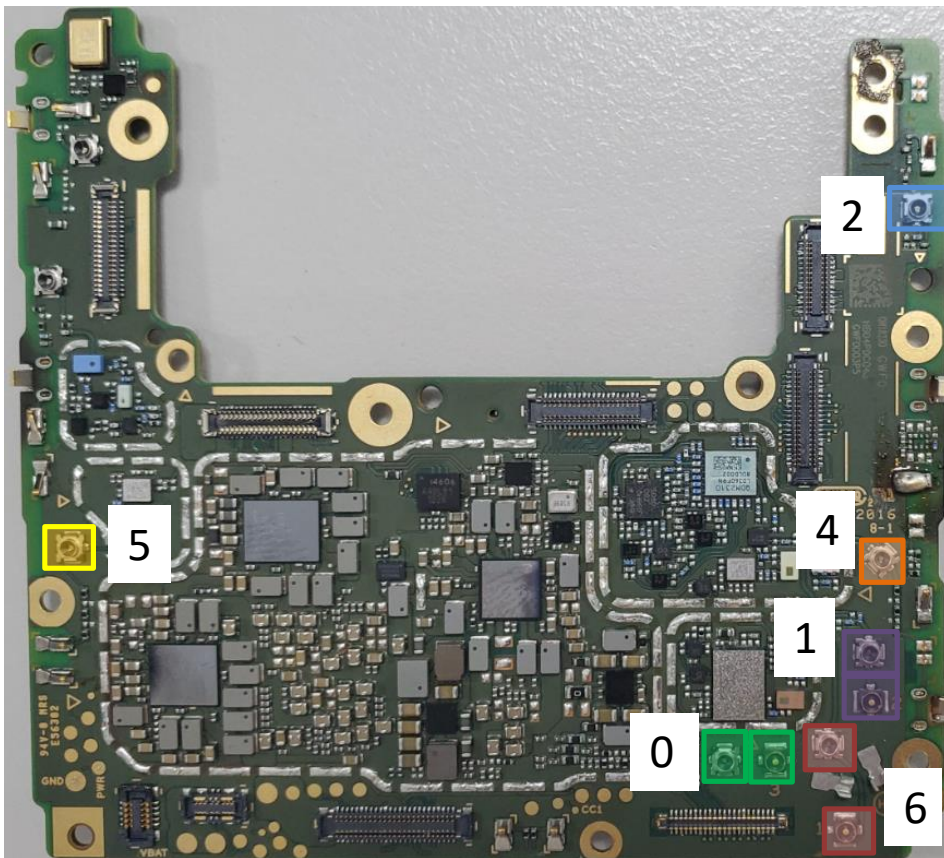


Antenna Port

PARAMETERS: Rx Gain Calibration Parameters

Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)
LTE	5	0	65	4	2	1	0	

bottom



- 0. LB TRx & B3/B7 PRx MIMO + NR
- 1. MHB TRx & N77 PRx MIMO + NR
- 2. MHB DRx + NR
- 4. LB DRx & B3/B7 DRx MIMO + NR
- 5. N77 DRx MIMO
- 6. N77 TRx

B3 PRx/DRx MIMO

B3 PRx	PRx	<table><tr><th>Technology</th><th>Band</th><th>Sub Band</th><th>Signal Path</th><th>RFM Device</th><th>Antenna Path</th><th>PLL ID</th><th>Antenna Number</th><th>Time (s)</th></tr><tr><td>LTE</td><td>3</td><td>0</td><td>35</td><td>16</td><td>0</td><td>4</td><td>1</td><td></td></tr><tr><th>Technology</th><th>Band</th><th>Sub Band</th><th>Signal Path</th><th>RFM Device</th><th>Antenna Path</th><th>PLL ID</th><th>Antenna Number</th><th>Time (s)</th></tr><tr><td>LTE</td><td>3</td><td>0</td><td>36</td><td>2</td><td>0</td><td>0</td><td>1</td><td></td></tr><tr><th>Technology</th><th>Band</th><th>Sub Band</th><th>Signal Path</th><th>RFM Device</th><th>Antenna Path</th><th>PLL ID</th><th>Antenna Number</th><th>Time (s)</th></tr><tr><td>LTE</td><td>3</td><td>0</td><td>37</td><td>6</td><td>0</td><td>2</td><td>1</td><td></td></tr><tr><th>Technology</th><th>Band</th><th>Sub Band</th><th>Signal Path</th><th>RFM Device</th><th>Antenna Path</th><th>PLL ID</th><th>Antenna Number</th><th>Time (s)</th></tr><tr><td>LTE</td><td>3</td><td>0</td><td>38</td><td>12</td><td>0</td><td>3</td><td>1</td><td></td></tr></table>	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	LTE	3	0	35	16	0	4	1		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	LTE	3	0	36	2	0	0	1		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	LTE	3	0	37	6	0	2	1		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	LTE	3	0	38	12	0	3	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)																																																																	
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	LTE	3	0	38	12	0	3	1																																																																			
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Technology		Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)																																																																		
LTE		3	0	125	14	14	4	0																																																																			
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LTE		3	0	126	10	14	3	0																																																																			
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LTE	3	0	127	8	14	2	0																																																																				
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LTE	3	0	128	0	14	0	0																																																																				
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		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)																																																																	
		LTE	3	0	85	17	4	4	2																																																																		
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		LTE	3	0	86	3	4	0	2																																																																		
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	LTE	3	0	88	13	4	3	2																																																																			
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Technology		Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)																																																																		
LTE		3	0	129	15	15	4	4																																																																			
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LTE		3	0	130	11	15	3	4																																																																			
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LTE	3	0	131	9	15	2	4																																																																				
Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)																																																																			
LTE	3	0	132	1	15	0	4																																																																				

B7 PRx MIMO

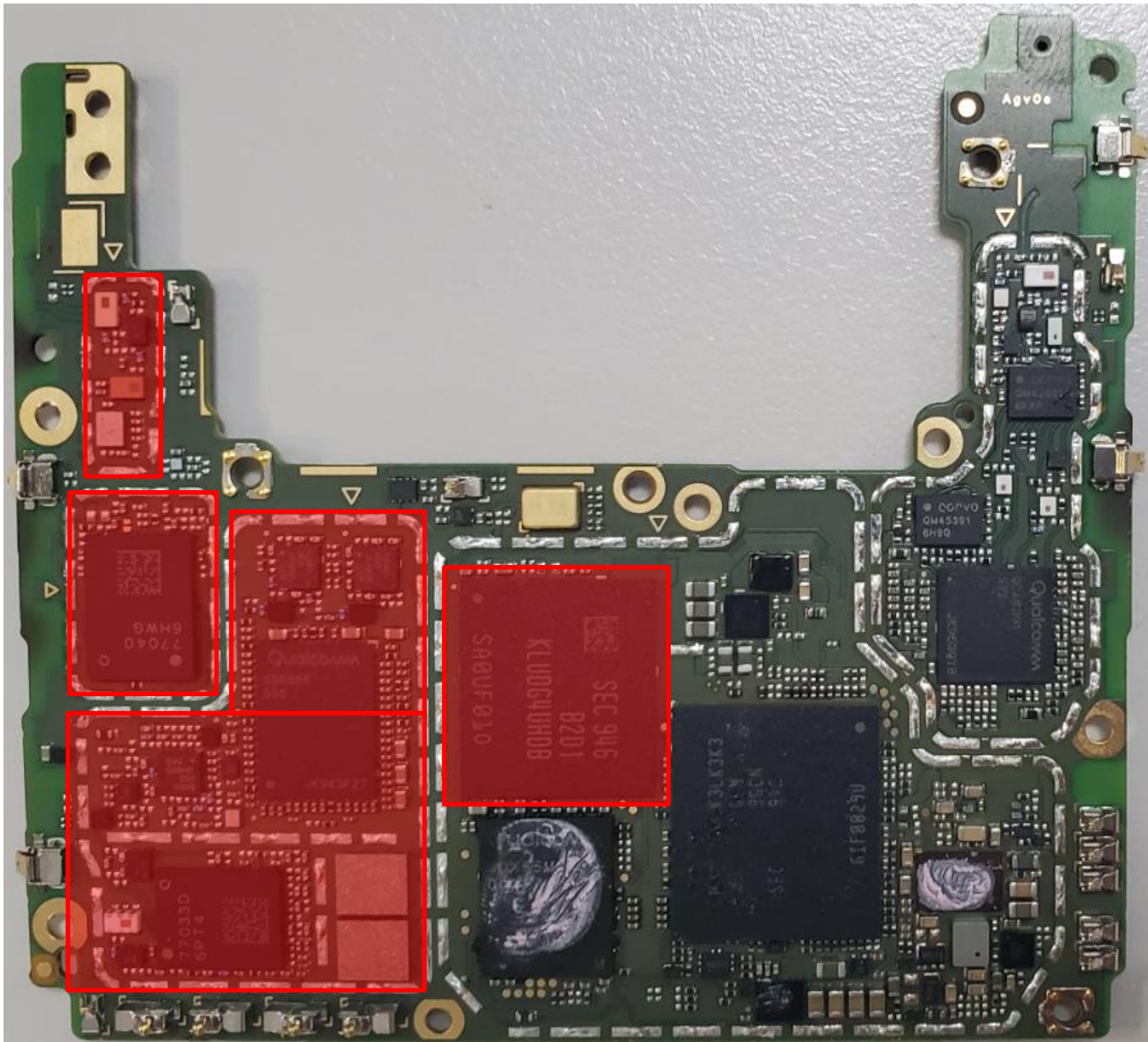
B7 PRx	PRx	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	39	16	0	4	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	40	2	0	0	1		
	CA B7+B38 (走B38 path)	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	41	6	0	2	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	42	12	0	3	1		
	PRx MIMO	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	43	16	0	4	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	44	2	0	0	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	45	6	0	2	1		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	46	12	0	3	1		
	PRx MIMO	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	133	14	14	4	0		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	134	10	14	3	0		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	135	8	14	2	0		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	136	0	14	0	0		

B7 DRx MIMO

B7 DRx	DRx	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	69	17	4	4	2		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	70	3	4	0	2		
	CA B7+B38 (走B38 path)	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	71	7	4	2	2		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	72	13	4	3	2		
	DRx MIMO	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	73	17	4	4	2		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	74	3	4	0	2		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	75	7	4	2	2		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	76	13	4	3	2		
	DRx MIMO	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	137	15	15	4	4		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	138	11	15	3	4		
	DRx MIMO	Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	139	9	15	2	4		
		Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)	
		LTE	7	0	140	1	15	0	4		

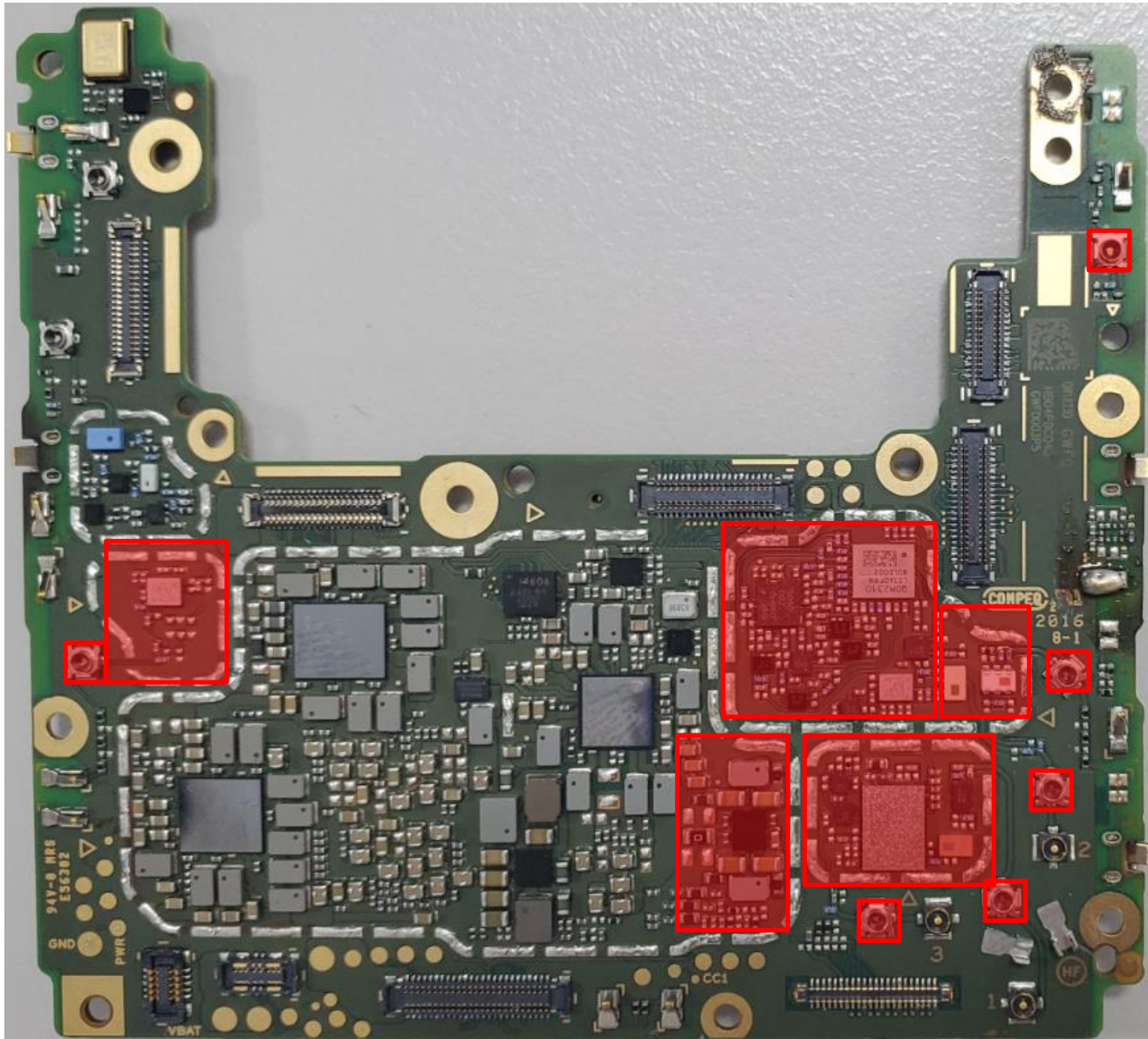
RF TOP side

After replacing/reheating these parts in the red block, you have to do RF calibration.



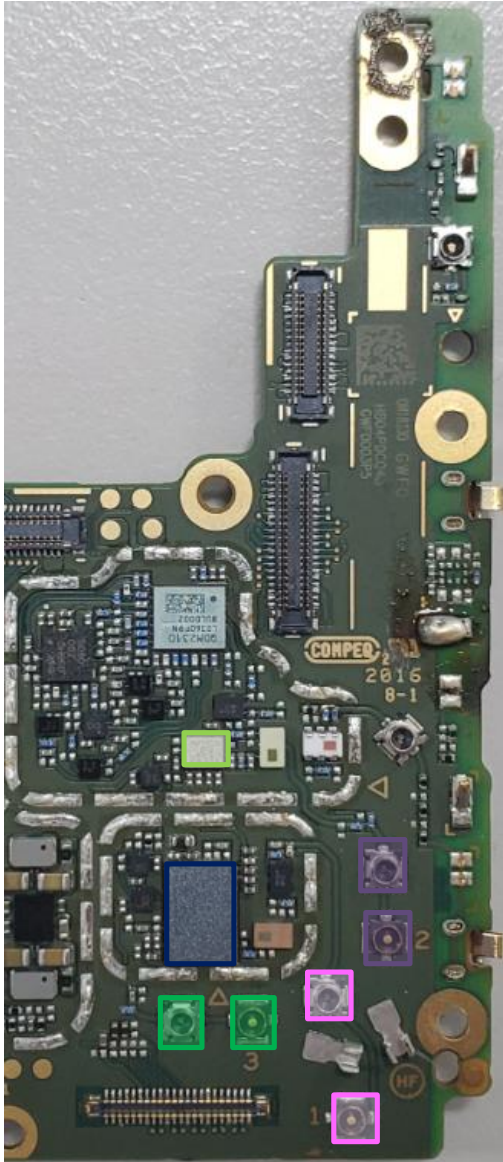
RF BOTTOM side

After replacing/reheating these parts in the red block, you have to do RF calibration.

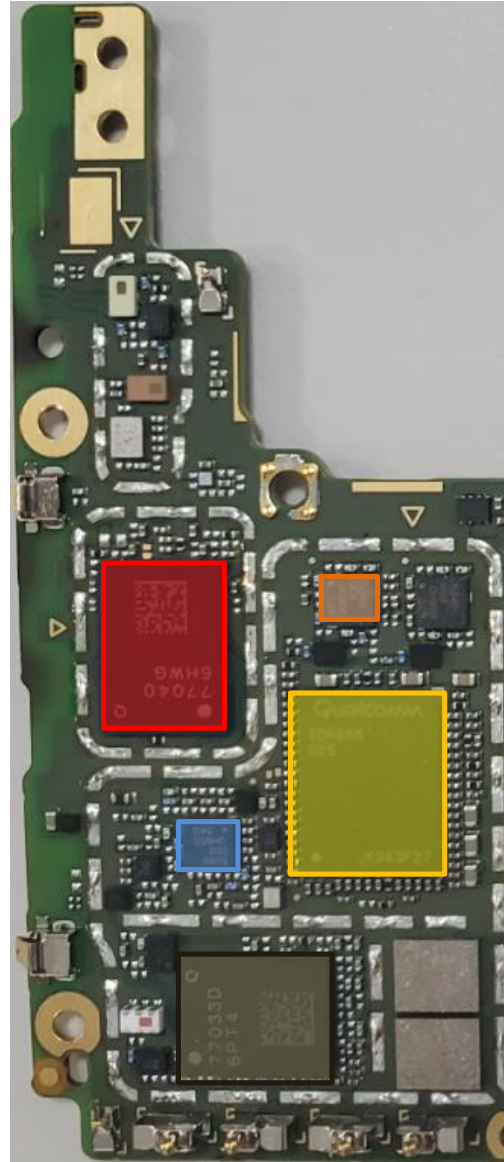


Key Components : Tx/PRx

bottom



top



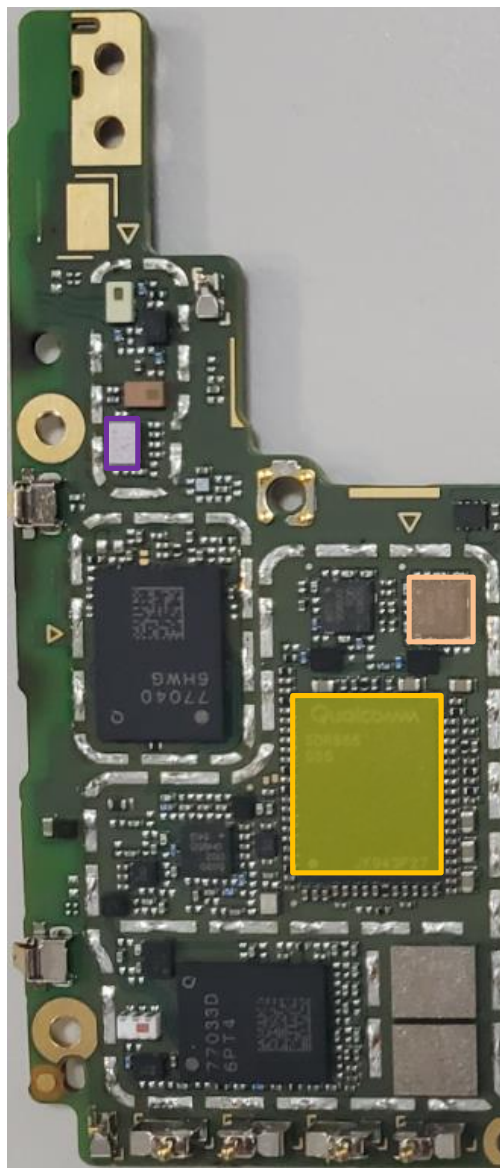
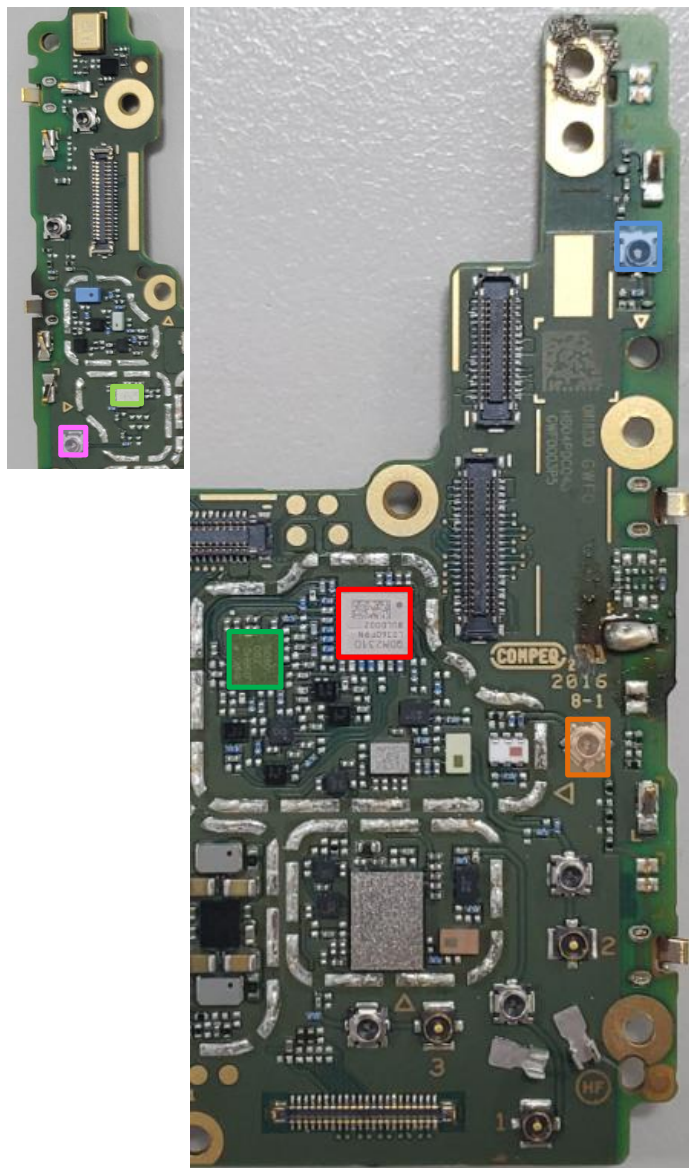
-  RF Connector N77
J4701 CON4701
-  RF Connector MHB
CON4601 CON4602
-  RF Connector LB
J4501 J4502
-  MHB PAMID QM77040
U4601
-  LB PAMID QM77033D
U4501
-  PRx MIMO QLN5040
U5004
-  PRx QLN5030
U5003
-  N77 TRx QPM5677
U4701
-  N77 PRx MIMO QDM5677
U4604
-  SDR865
U4101

Key Components : DRx

bottom

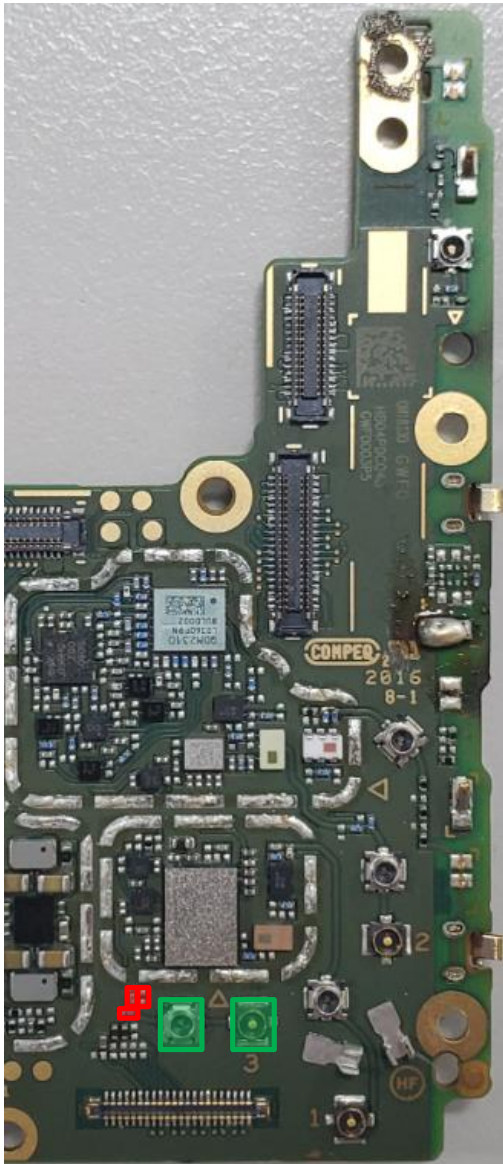
top

-  Div. RF Connector N77
CON5303
-  Div. RF connector LB
CON5301
-  Div. RF connector MHB
CON5302
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  DRx MIMO QLN5040
U5005
-  N77 DRx QDM5677
U5307
-  N77 DRx MIMO QDM5677
U5306
-  SDR865
U4101

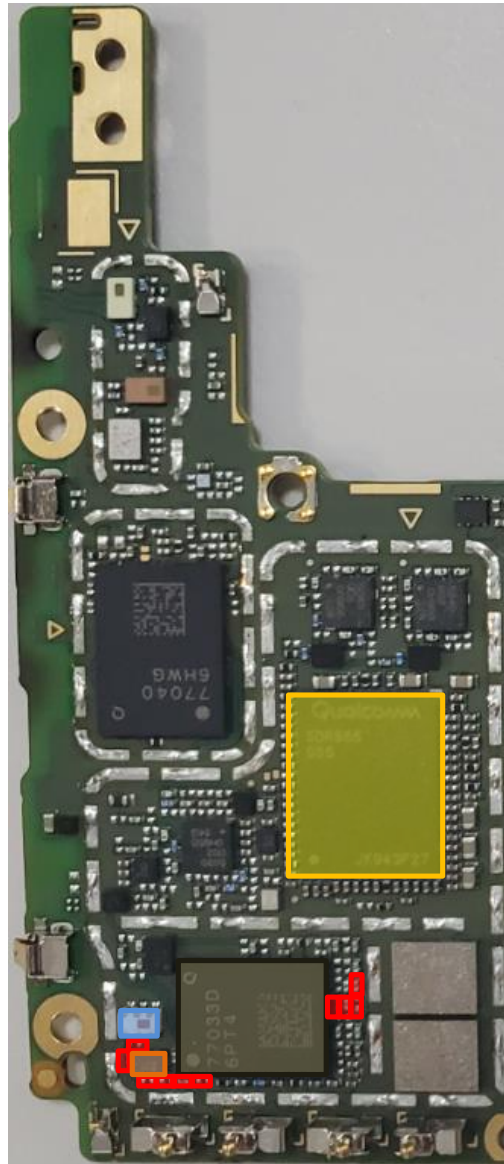


TX Path : GSM 850/900

bottom



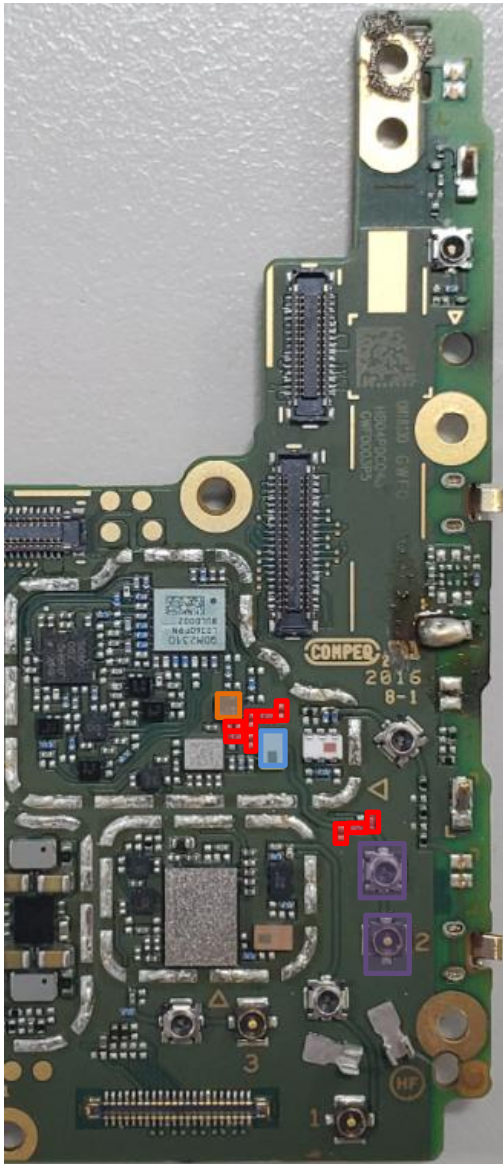
top



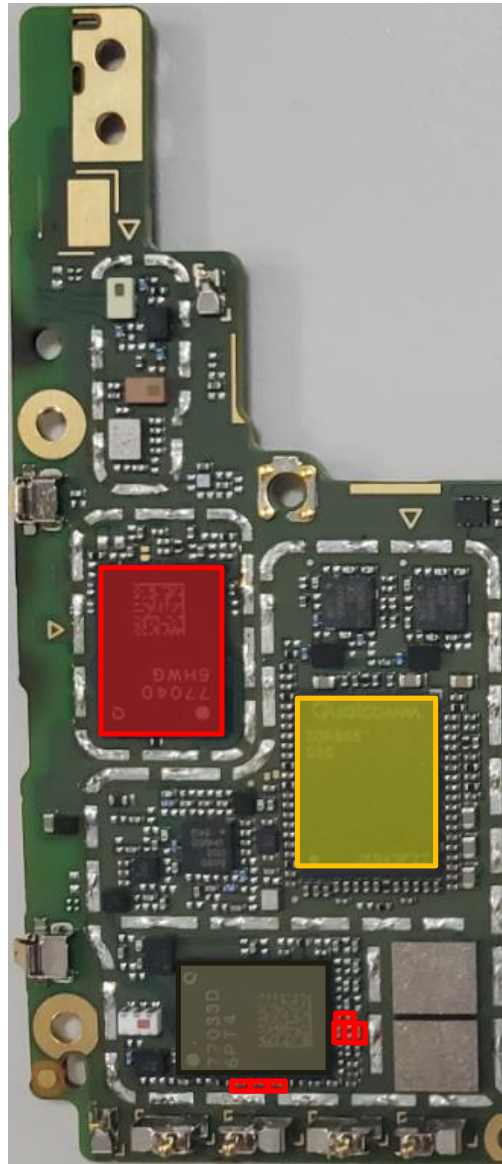
-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  SDR865
U4101

TX Path : GSM DCS/PCS

bottom



top



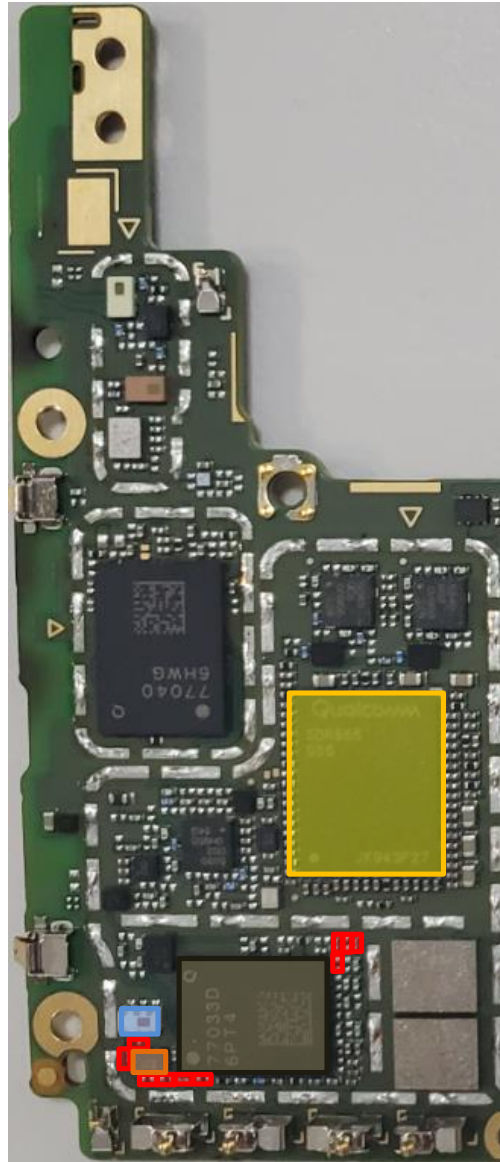
-  RF Connector MHB
CON4601 CON4602
-  LB PAMID QM77033D
U4501
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

LB Tx0 Common Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

 RF Connector LB
J4501 J4502

 LB PAMID QM77033D
U4501

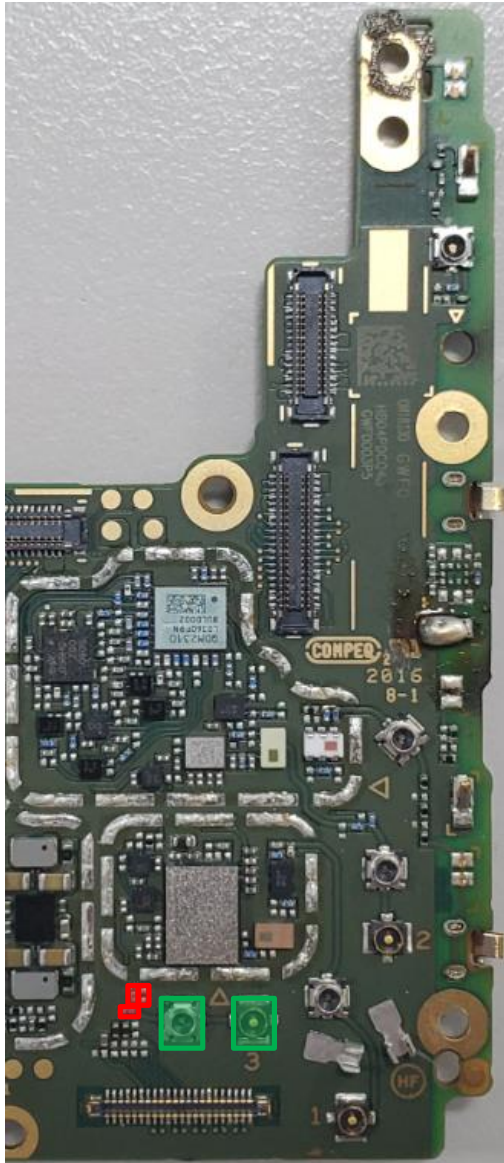
 LB DPDT
U4502

 LB + B3&B7 DiPX
U4503

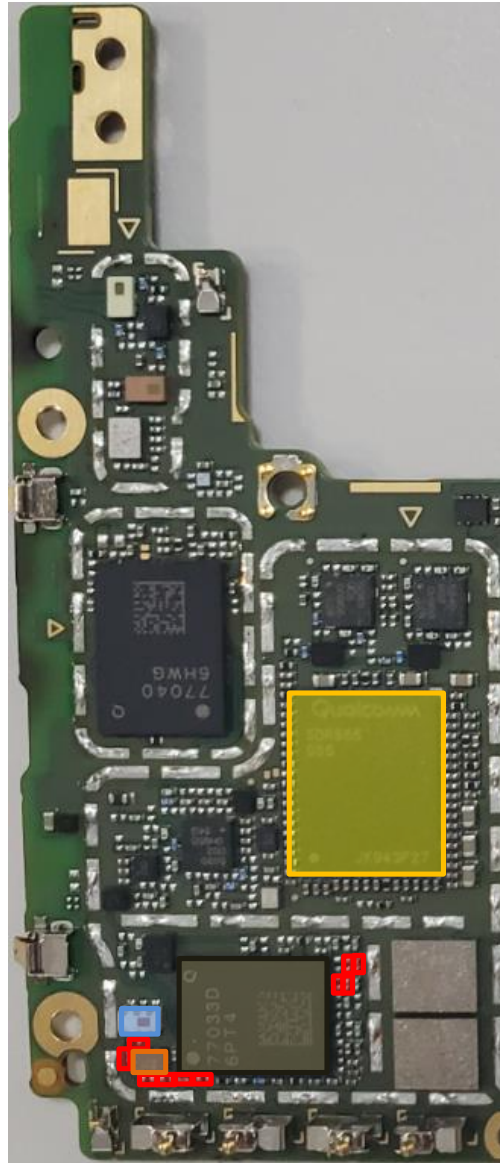
 SDR865
U4101

LB Tx1 Common Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

 RF Connector LB
J4501 J4502

 LB PAMID QM77033D
U4501

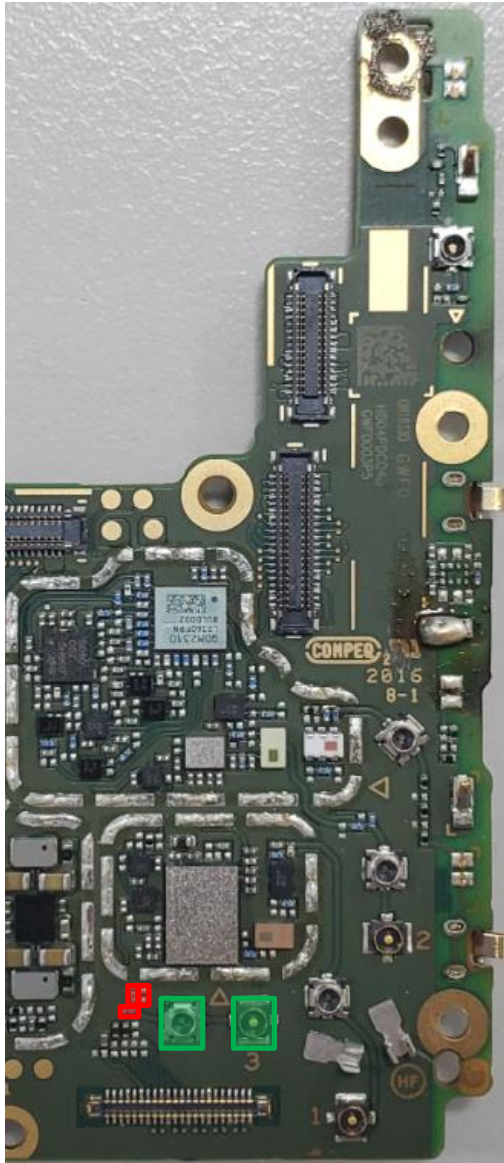
 LB DPDT
U4502

 LB + B3&B7 DiPX
U4503

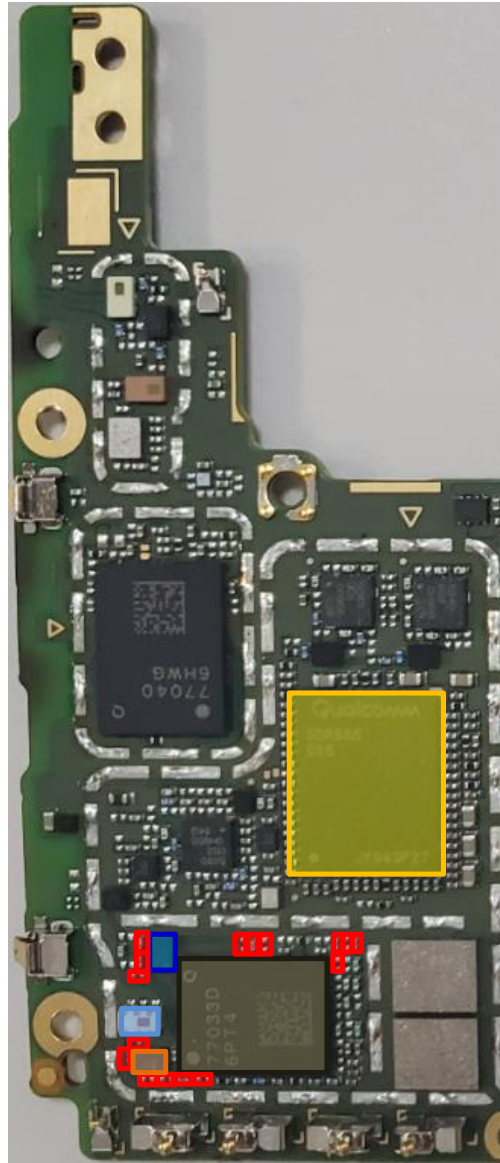
 SDR865
U4101

B28/N28 Tx0 Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

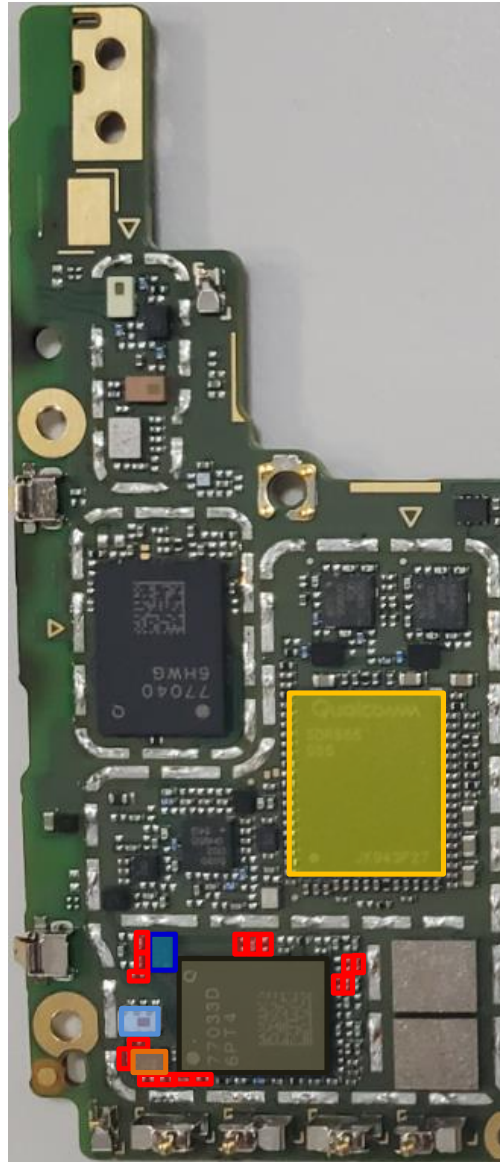
-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  B28 DUX FILTER
U4504
-  SDR865
U4101

B28/N28 Tx1 Path

bottom



top

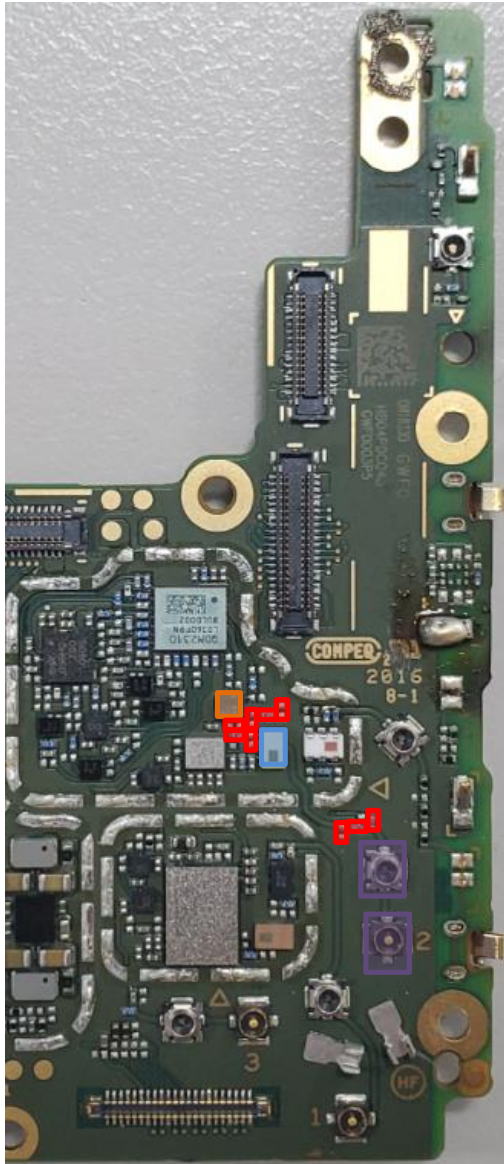


LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

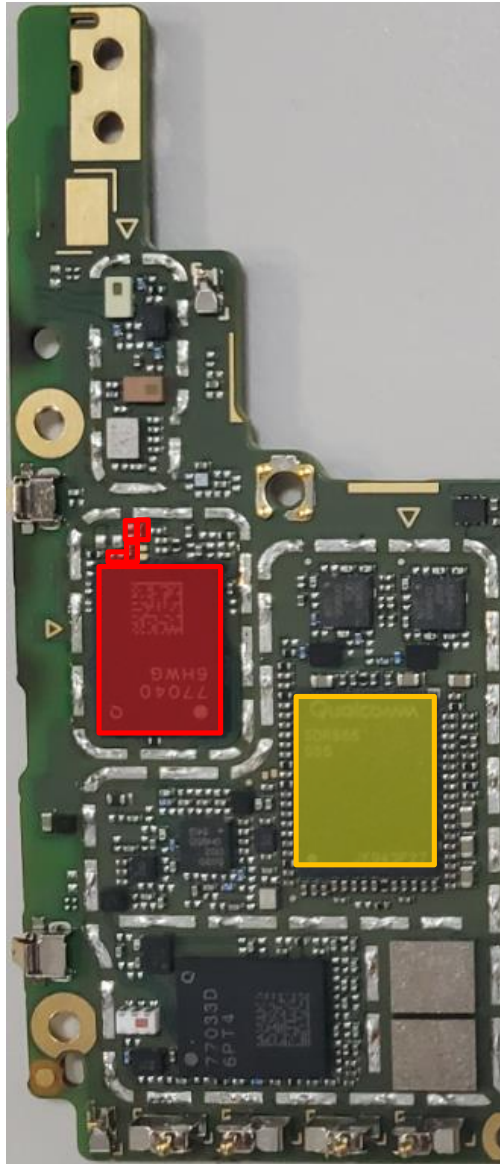
-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  B28 DUX FILTER
U4504
-  SDR865
U4101

MB Tx Common Path

bottom



top

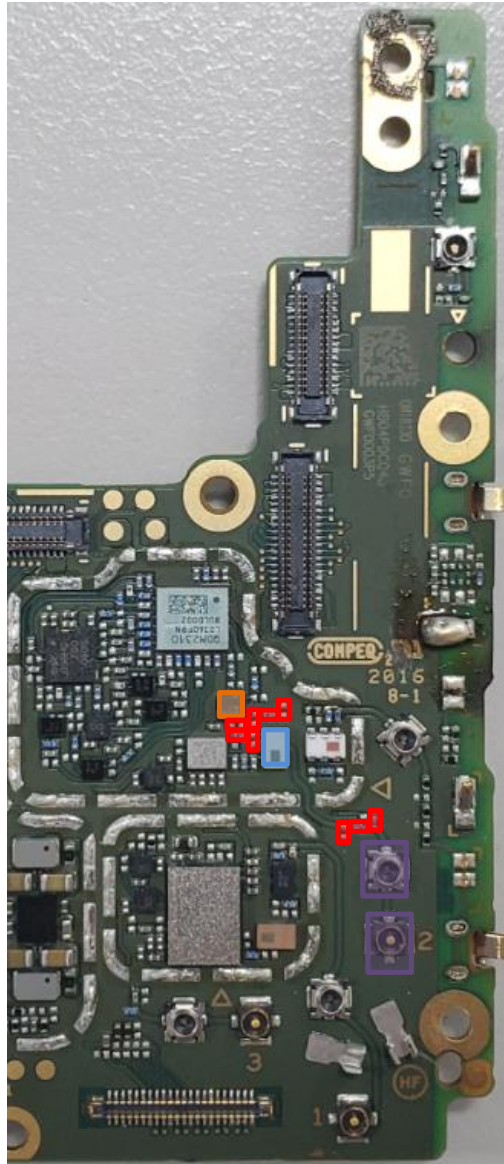


MB: B1/B2/B3/B4/B39/
N1/N2/N3

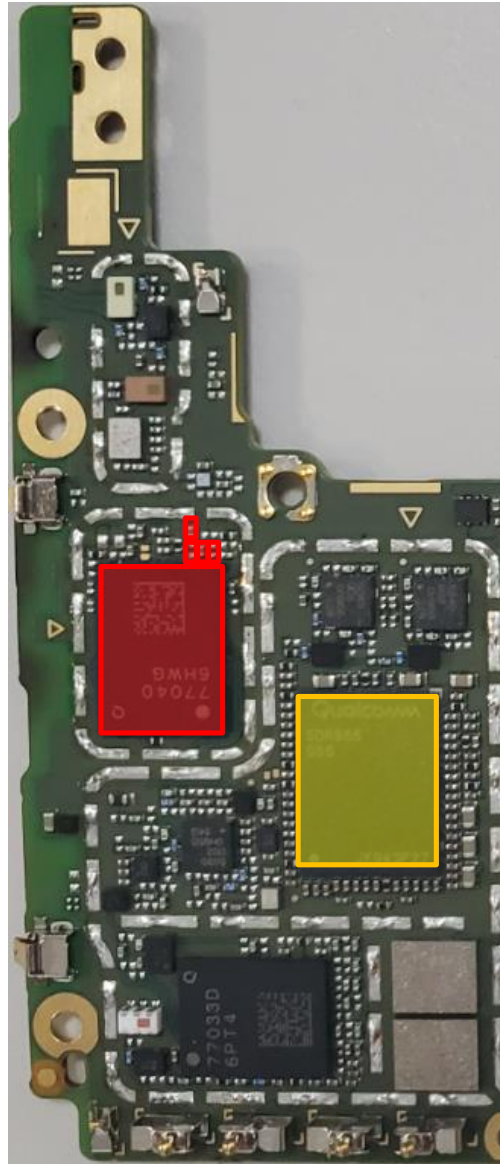
-  RF Connector MHB
CON4601 CON4602
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

HB Tx Common Path

bottom



top

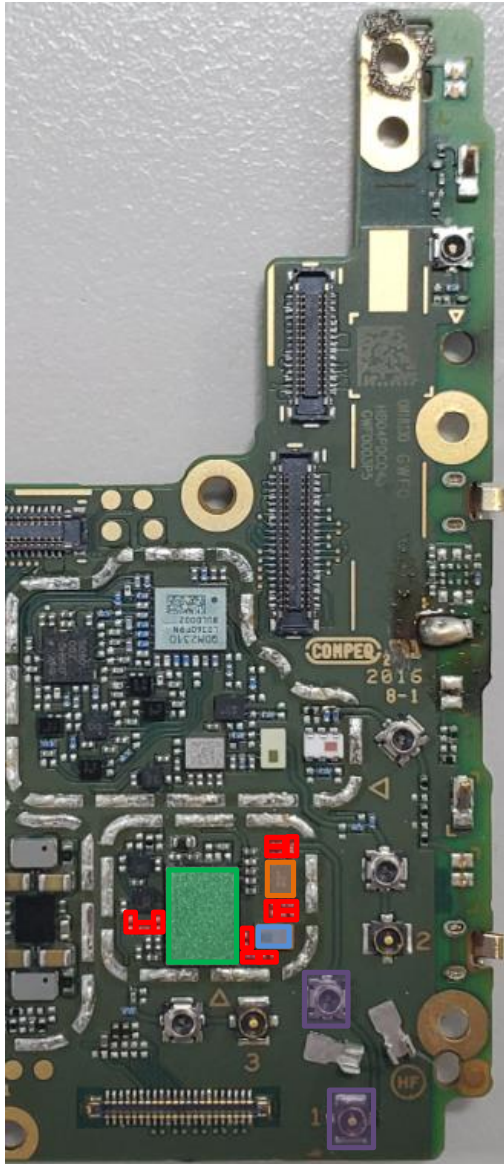


HB : B7/B38/B40/B41/
N7/N38

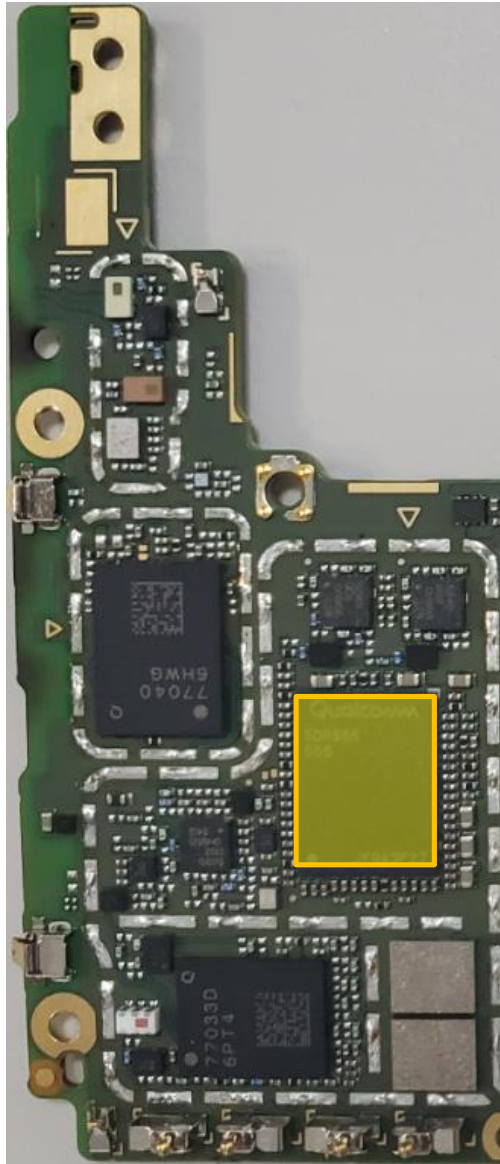
-  RF Connector MHB
CON4601 CON4602
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

N77 Tx Common Path

bottom



top



N77 : N77/N78

RF Connector N77
J4701 CON4701

N77 PA QPM5677
U4701

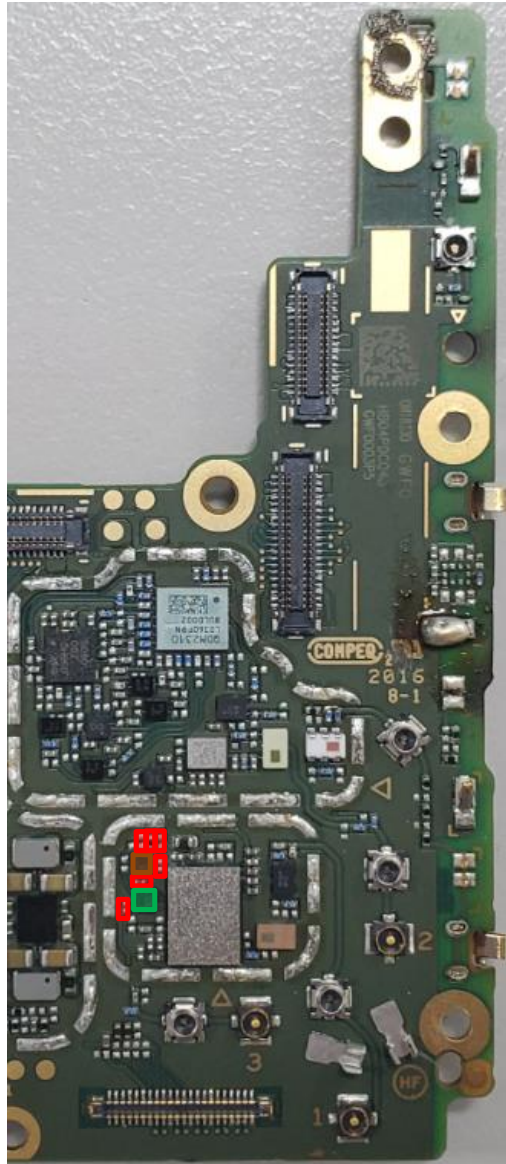
N77 DPDT
U4702

N77 BPF
U4705

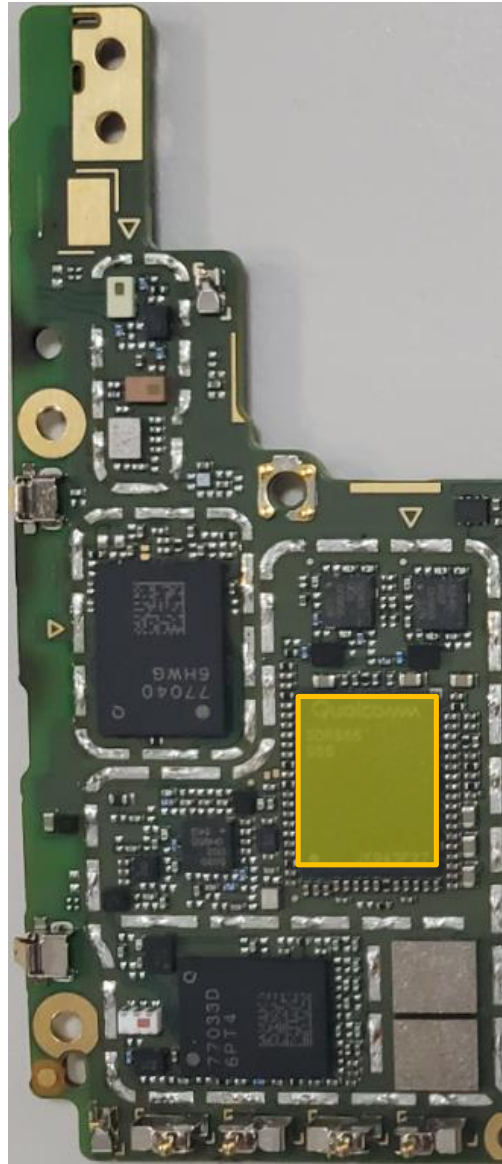
SDR865
U4101

FBRx Path

bottom



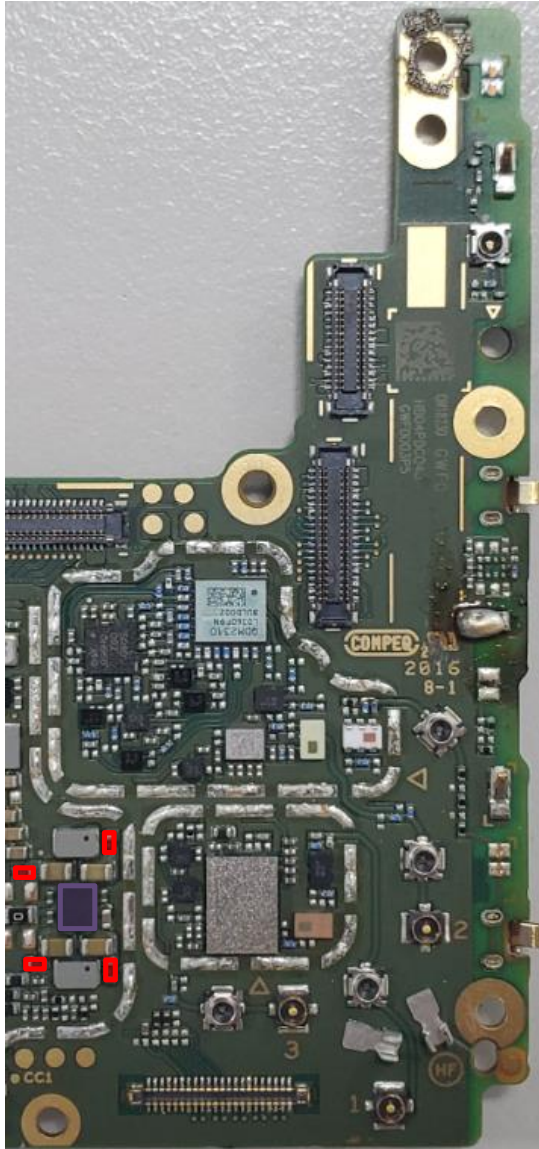
top



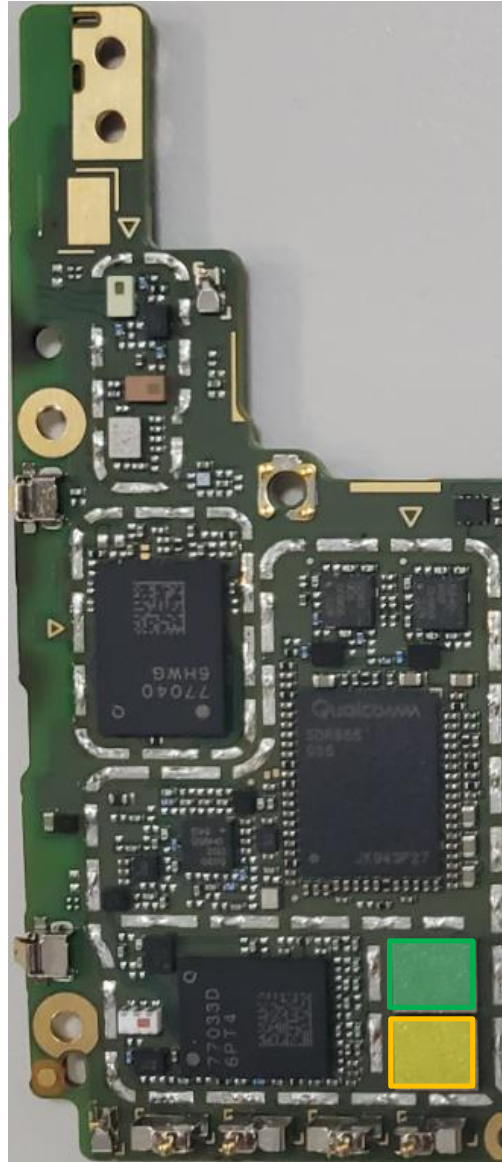
-  SPDT
U4606
-  SPDT
U4611
-  SDR865
U4101

DCDC

bottom



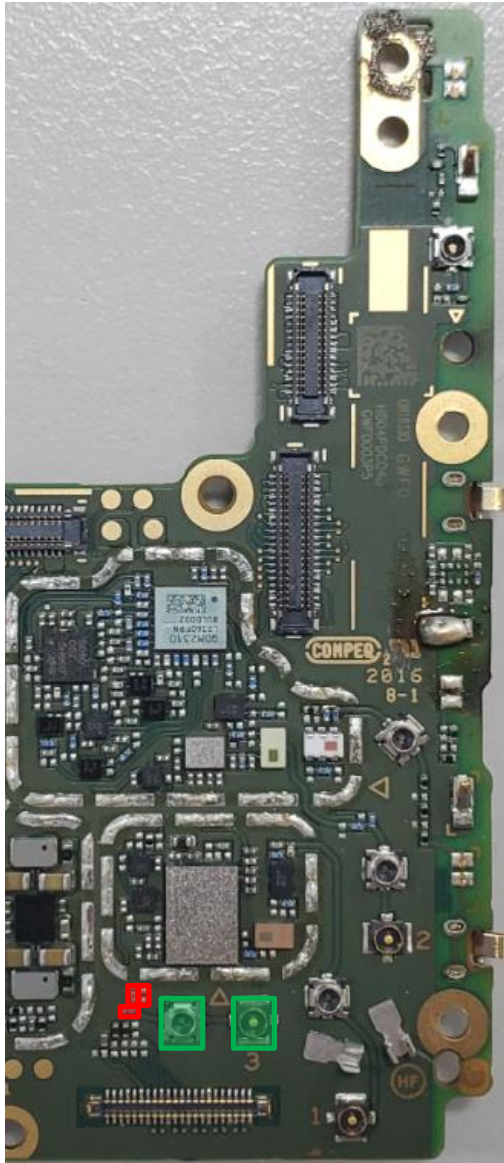
top



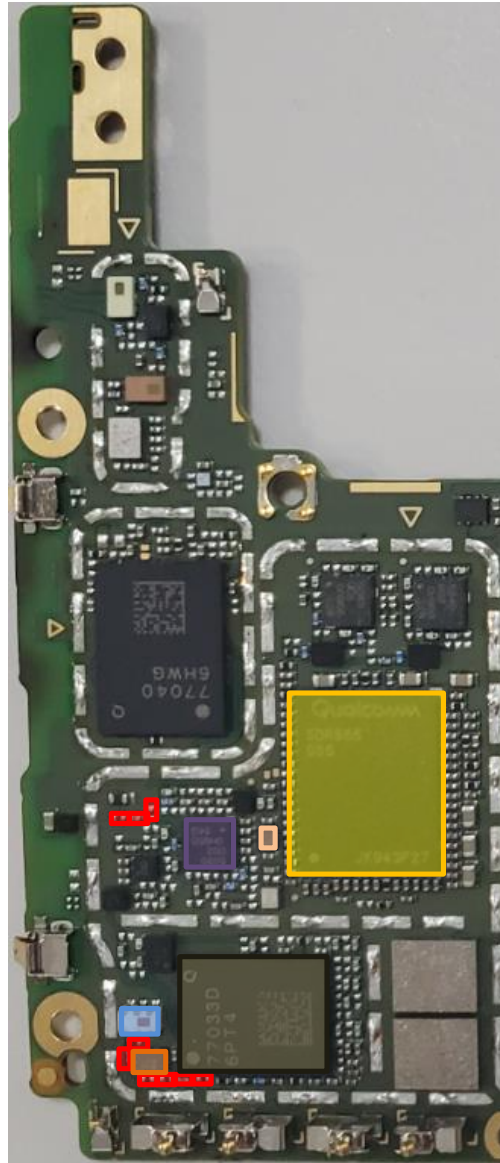
-  QET5100M for MHB
U4801
-  QET5100M for LB
U4802
-  QET6100 for N77
U4901

LB B8 PRx Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

 RF Connector LB
J4501 J4502

 LB PAMID QM77033D
U4501

 PRx LNA QLN5030
U5003

 LB DPDT
U4502

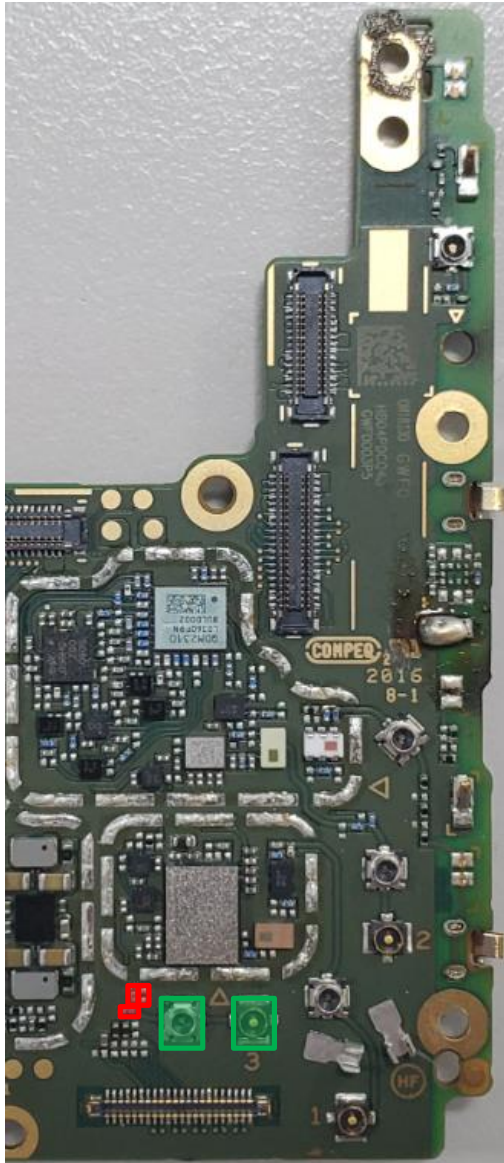
 LB + B3&B7 DiPX
U4503

 LB LNA OUTPUT SPDT
U5007

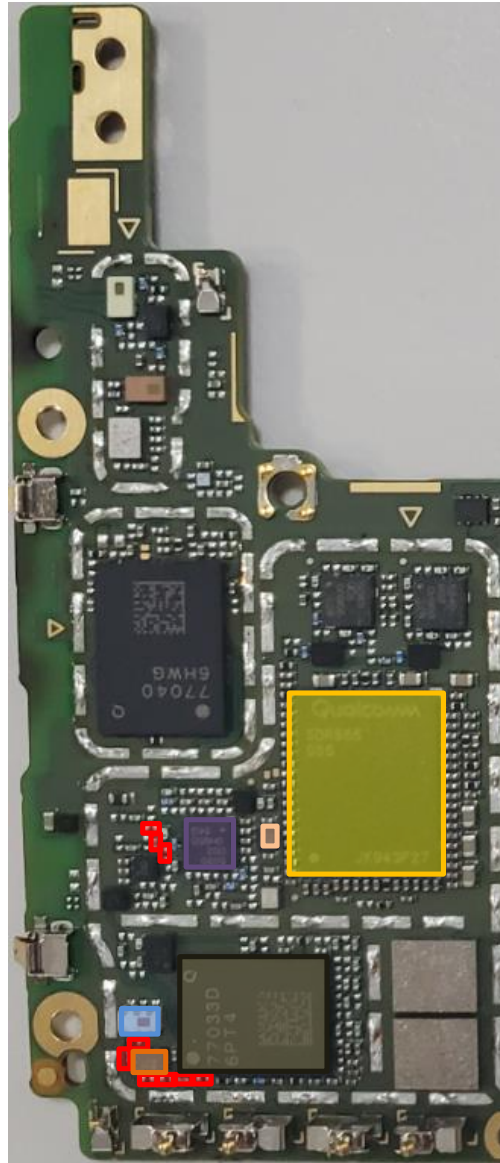
 SDR865
U4101

LB B20 PRx Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

 RF Connector LB
J4501 J4502

 LB PAMID QM77033D
U4501

 PRx LNA QLN5030
U5003

 LB DPDT
U4502

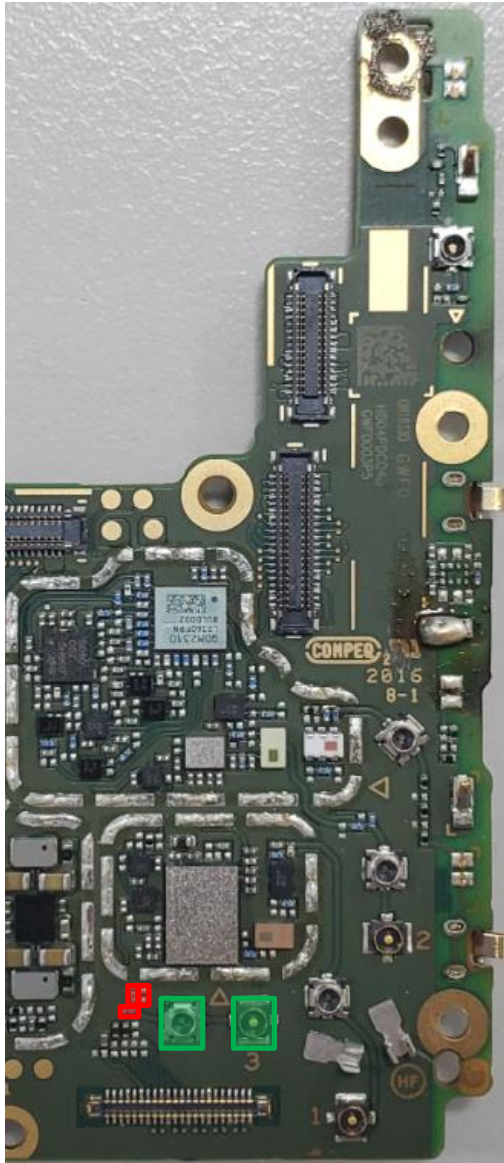
 LB + B3&B7 DiPX
U4503

 LB LNA OUTPUT SPDT
U5007

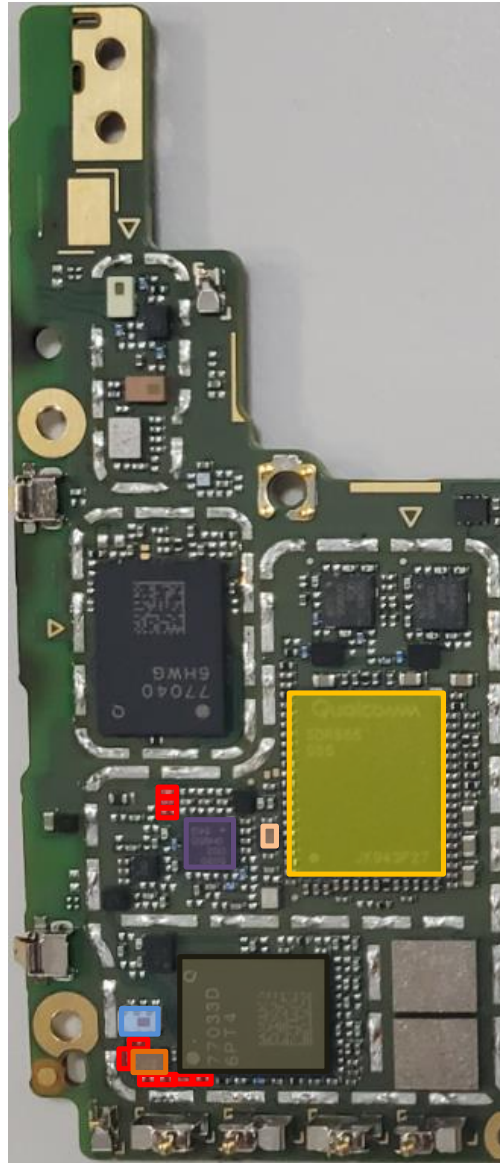
 SDR865
U4101

LB B26 PRx Path

bottom



top



LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

RF Connector LB
J4501 J4502

LB PAMID QM77033D
U4501

PRx LNA QLN5030
U5003

LB DPDT
U4502

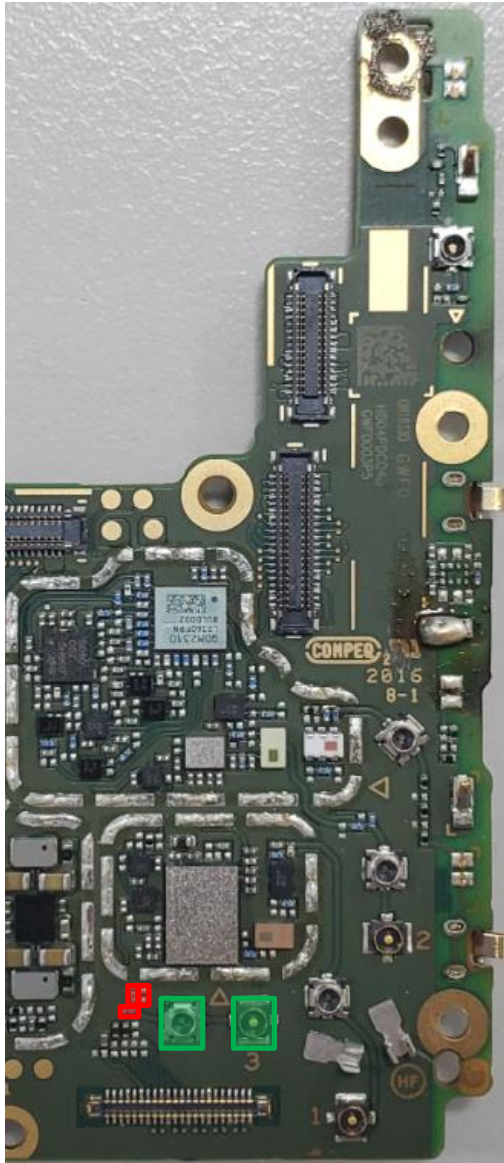
LB + B3&B7 DiPX
U4503

LB LNA OUTPUT SPDT
U5007

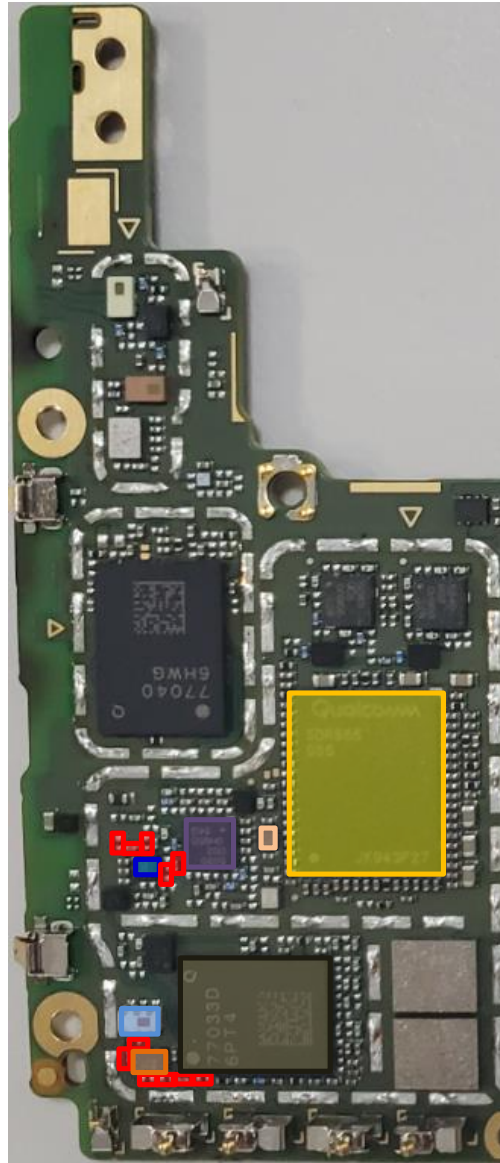
SDR865
U4101

LB B12 PRx Path

bottom



top

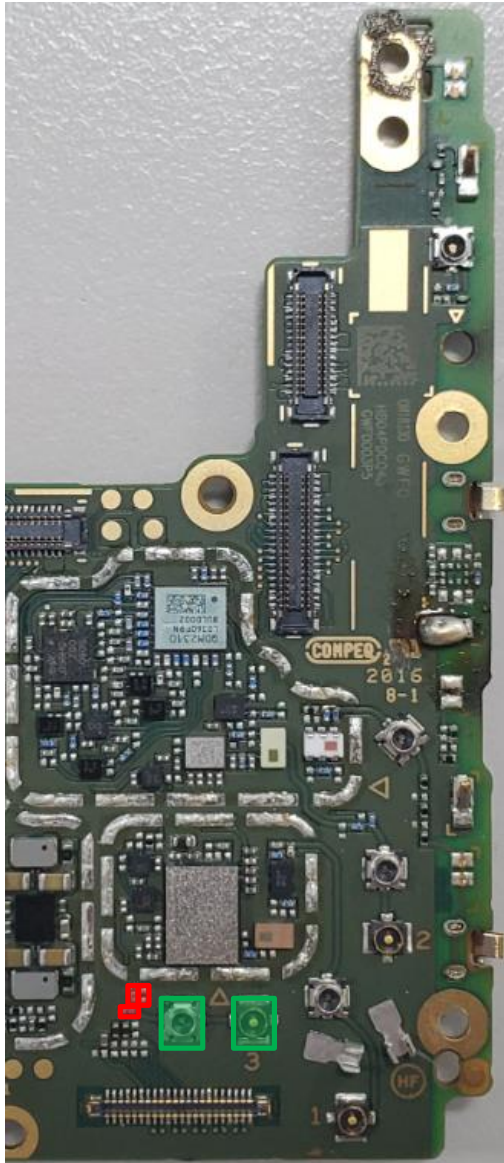


LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

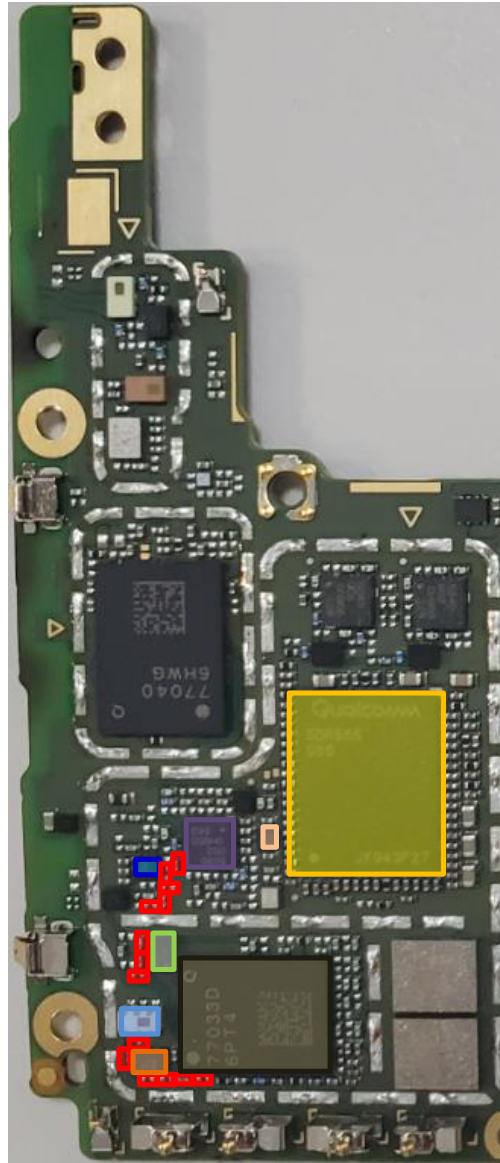
-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  PRx LNA QLN5030
U5003
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  LB SP3T
U5006
-  LB LNA OUTPUT SPDT
U5007
-  SDR865
U4101

LB B28 PRx Path

bottom



top

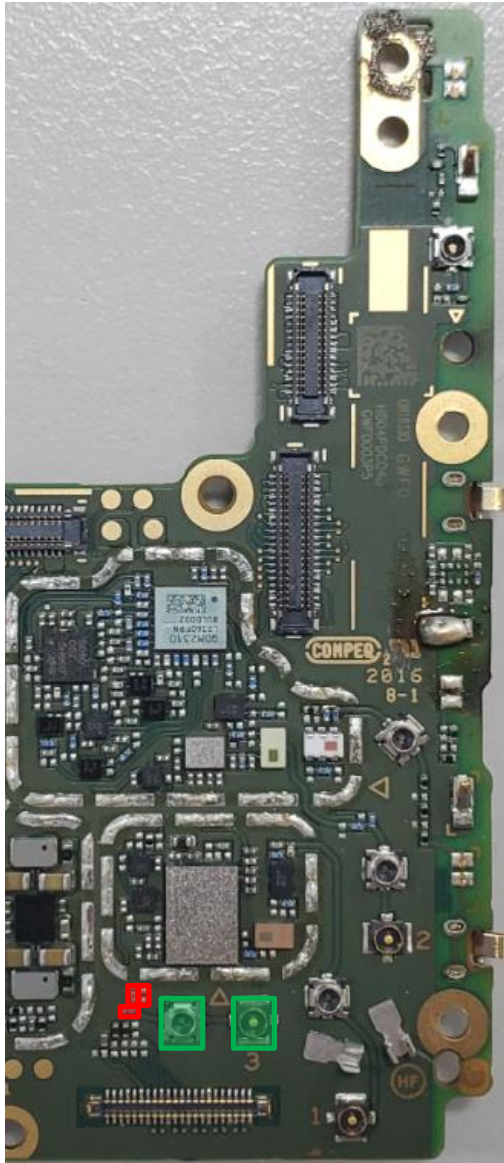


LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

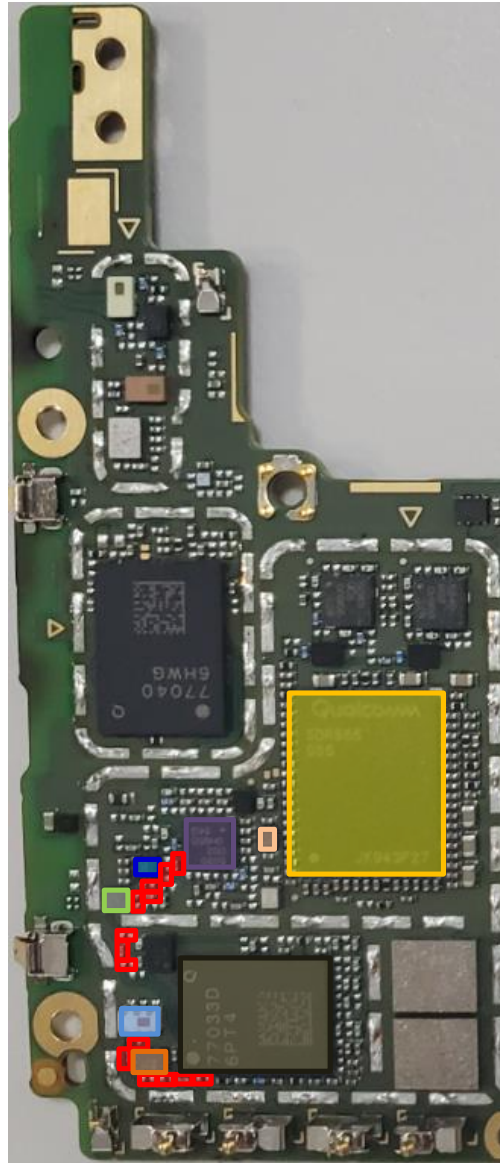
-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  PRx LNA QLN5030
U5003
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  LB SP3T
U5006
-  B28 DUX Filter
U4504
-  LB LNA OUTPUT SPDT
U5007
-  SDR865
U4101

LB B29 PRx Path

bottom



top

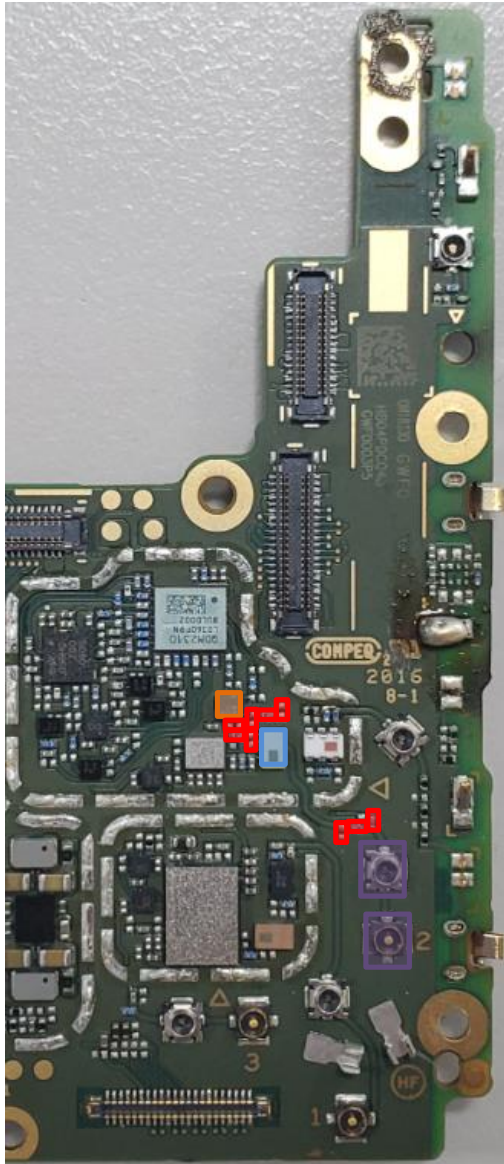


LB:B5/B8/B12/B17/B18/B19/
B20/B26/B28/B29/
N5/N8/N12/N20/N28

-  RF Connector LB
J4501 J4502
-  LB PAMID QM77033D
U4501
-  PRx LNA QLN5030
U5003
-  LB DPDT
U4502
-  LB + B3&B7 DiPX
U4503
-  LB SP3T
U5006
-  B29 Filter
U5001
-  LB LNA OUTPUT SPDT
U5007
-  SDR865
U4101

MB B1/B4 PRx Path

bottom



top

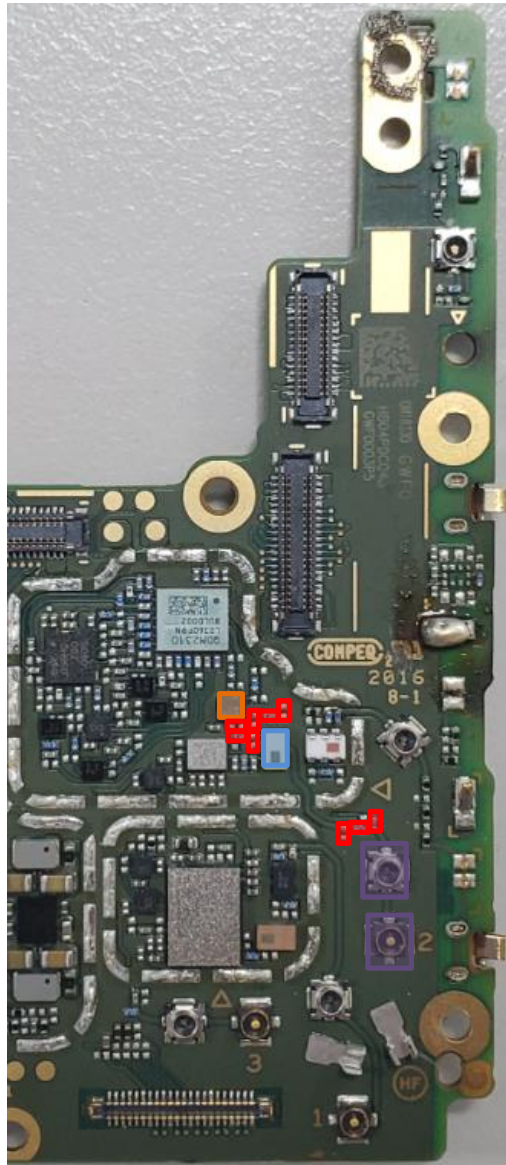


MB: B1/B2/B3/B4/B39/
N1/N2/N3

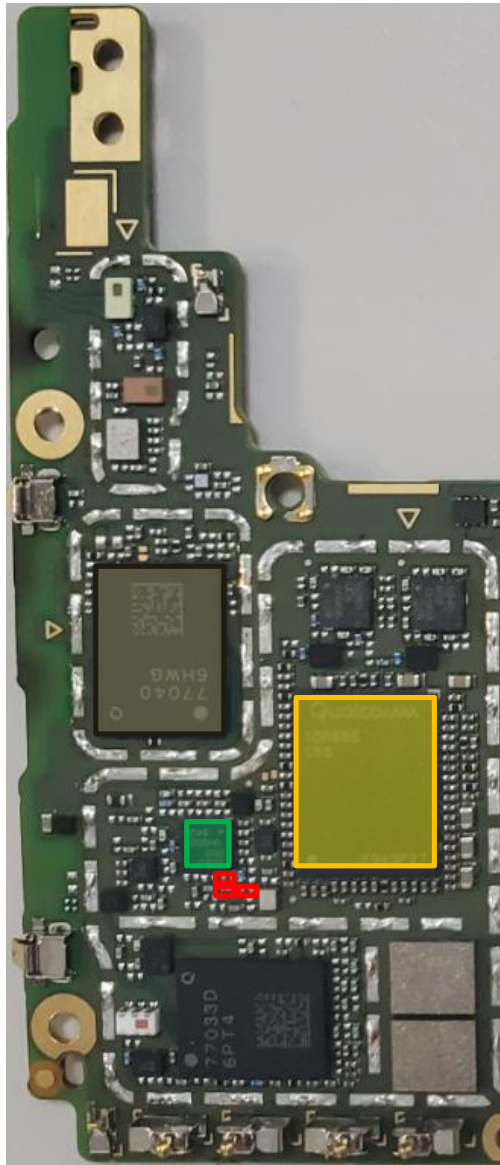
-  RF Connector MHB
CON4601 CON4602
-  PRx LNA QLN5030
U5003
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

MB B2 PRx Path

bottom



top

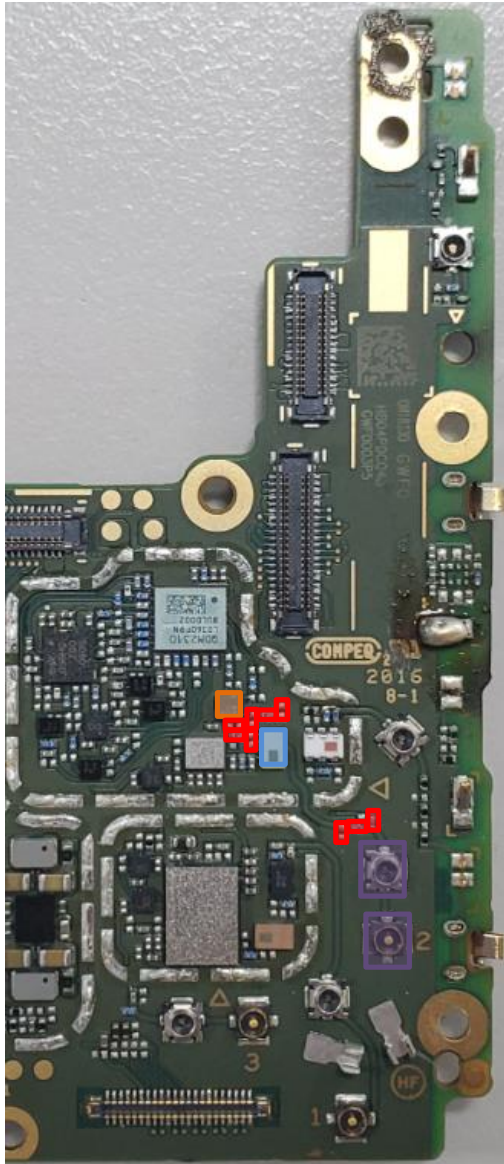


MB: B1/B2/B3/B4/B39/
N1/N2/N3

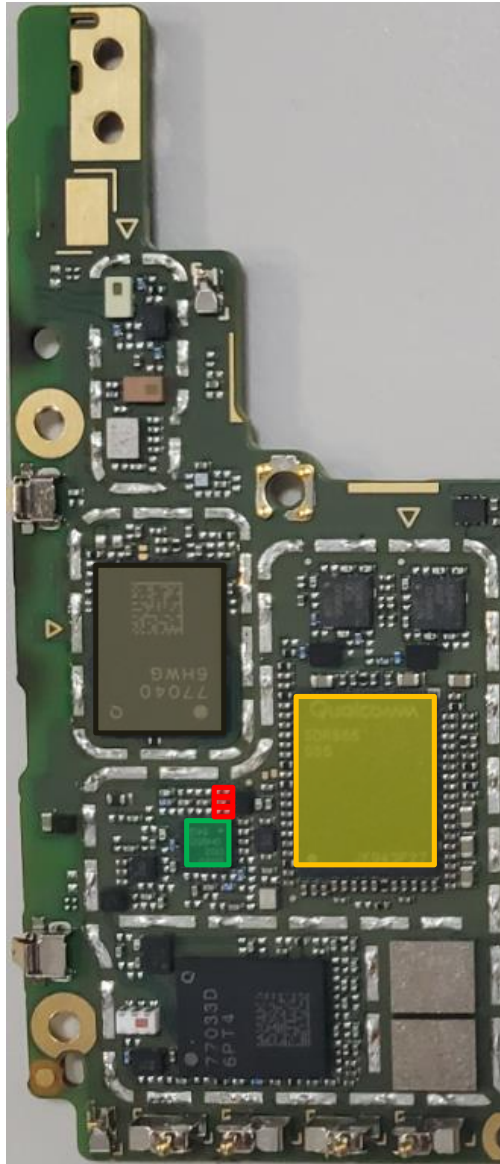
-  RF Connector MHB
CON4601 CON4602
-  PRx LNA QLN5030
U5003
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

MB B3 PRx Path

bottom



top

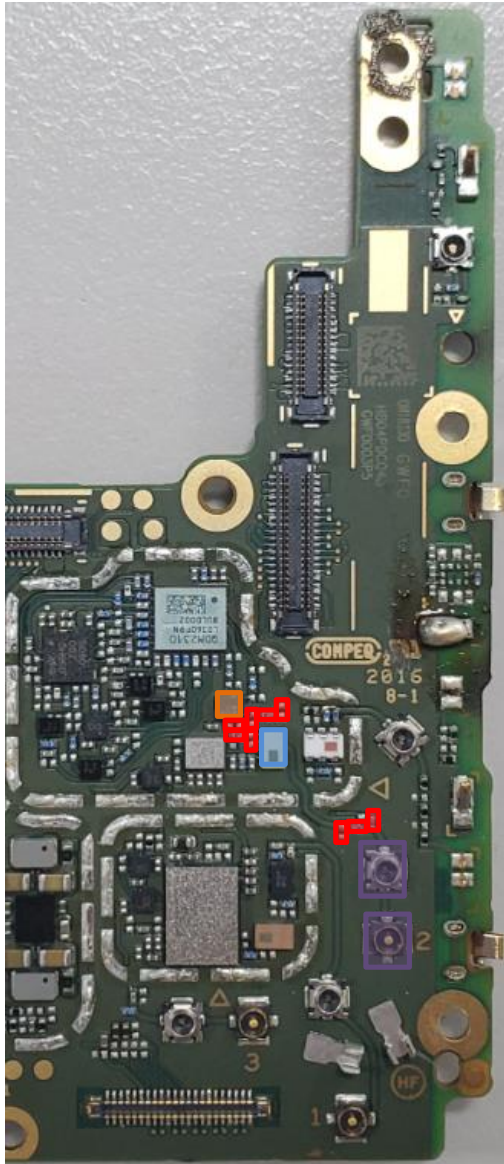


MB: B1/B2/B3/B4/B39/
N1/N2/N3

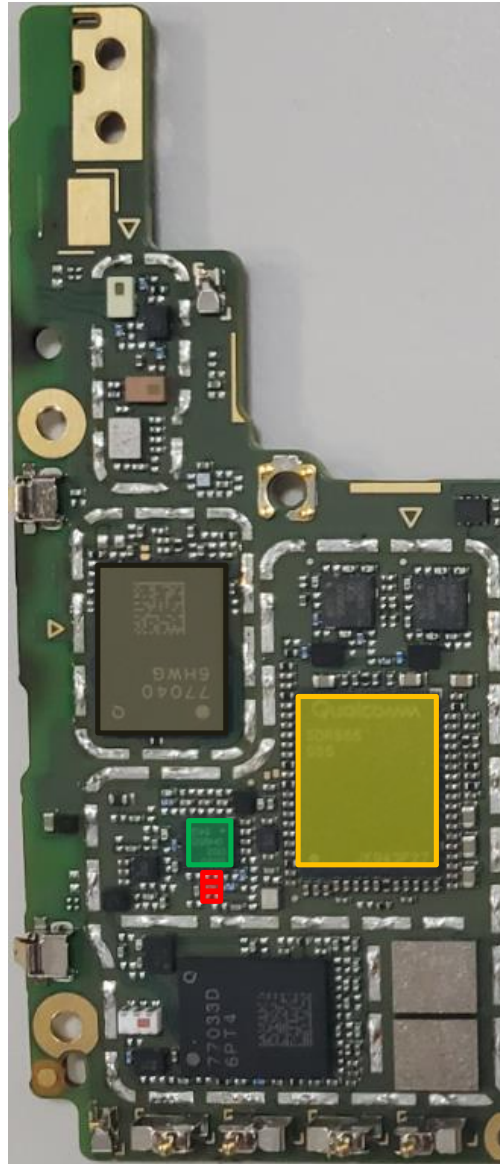
-  RF Connector MHB
CON4601 CON4602
-  PRx LNA QLN5030
U5003
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

MB B39 PRx Path

bottom



top

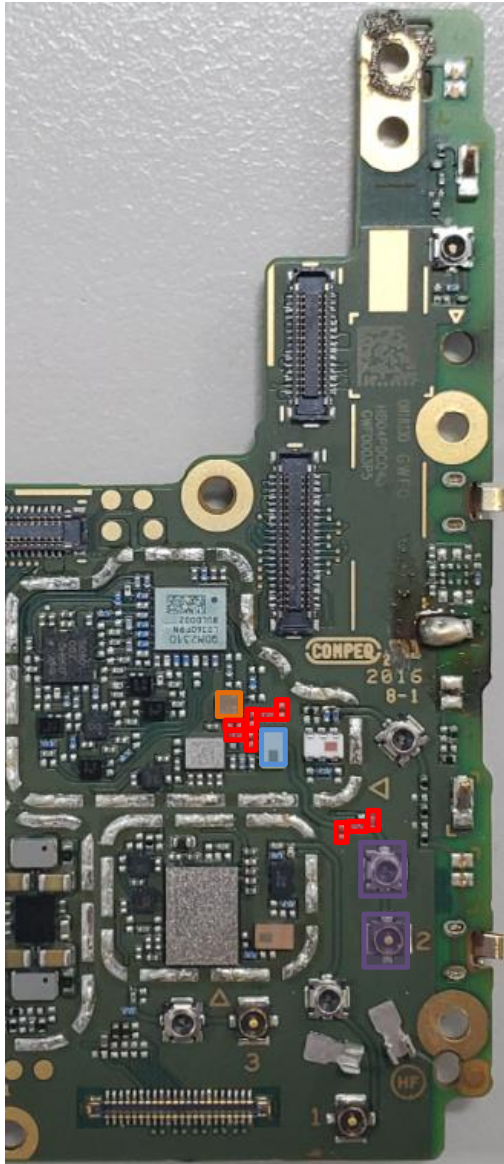


MB: B1/B2/B3/B4/B39/
N1/N2/N3

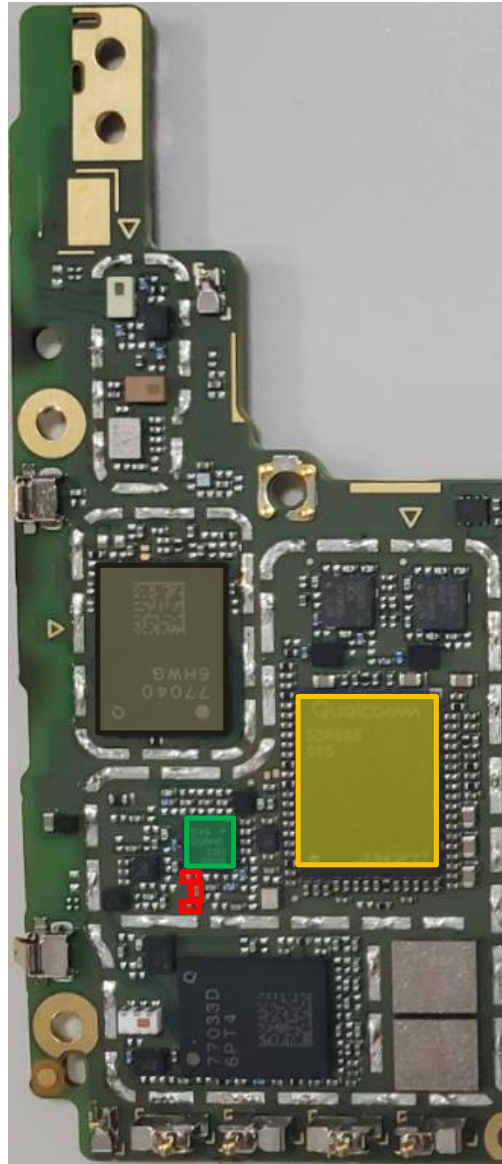
-  RF Connector MHB
CON4601 CON4602
-  PRx LNA QLN5030
U5003
-  MHB PAMID QM77040
U4601
-  MHB DPDT
U4602
-  MHB + N77 DiPX
U4610
-  SDR865
U4101

HB B7 PRx Path

bottom



top



HB : B7/B38/B40/B41/
N7/N38

 RF Connector MHB
CON4601 CON4602

 PRx LNA QLN5030
U5003

 MHB PAMID QM77040
U4601

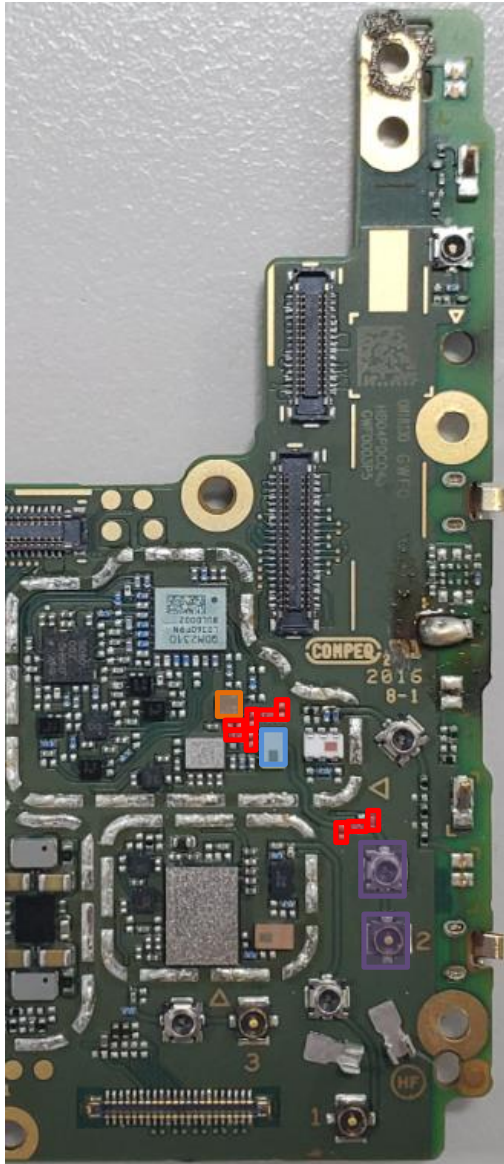
 MHB DPDT
U4602

 MHB + N77 DiPX
U4610

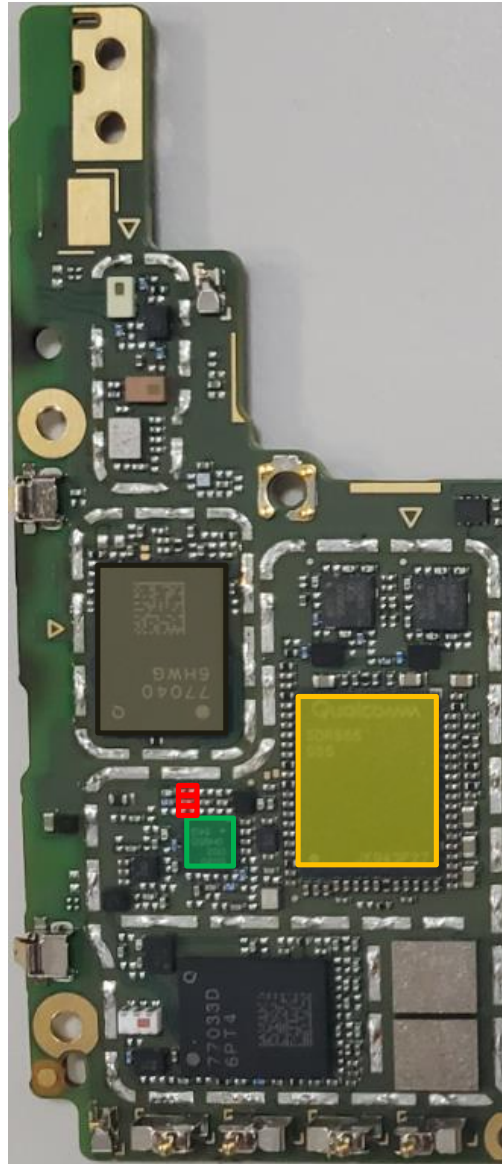
 SDR865
U4101

HB B40 PRx Path

bottom



top



HB : B7/B38/B40/B41/
N7/N38

 RF Connector MHB
CON4601 CON4602

 PRx LNA QLN5030
U5003

 MHB PAMID QM77040
U4601

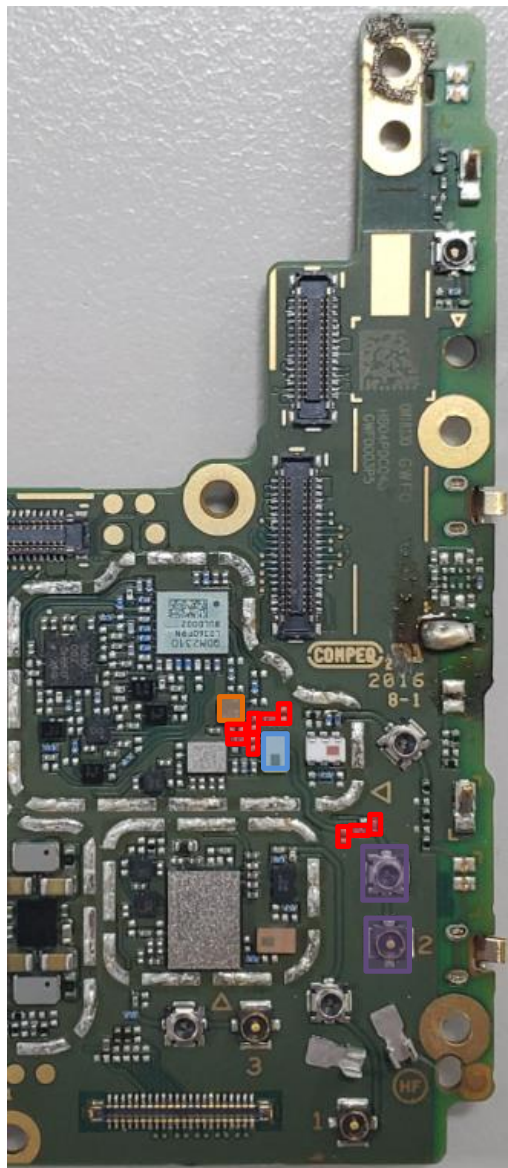
 MHB DPDT
U4602

 MHB + N77 DiPX
U4610

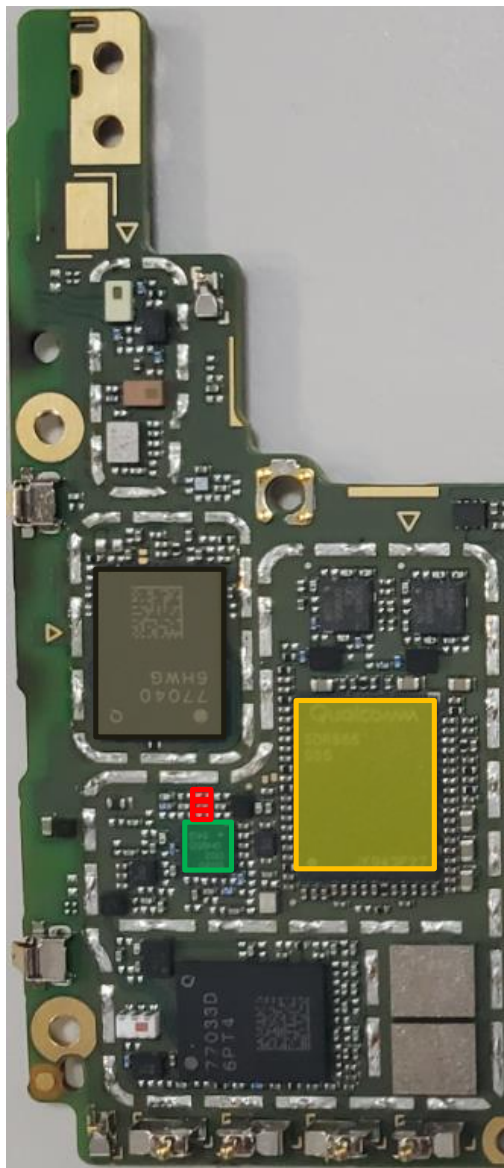
 SDR865
U4101

HB B38/B41 PRx Path

bottom



top



HB : B7/B38/B40/B41/
N7/N38

RF Connector MHB
CON4601 CON4602

PRx LNA QLN5030
U5003

MHB PAMID QM77040
U4601

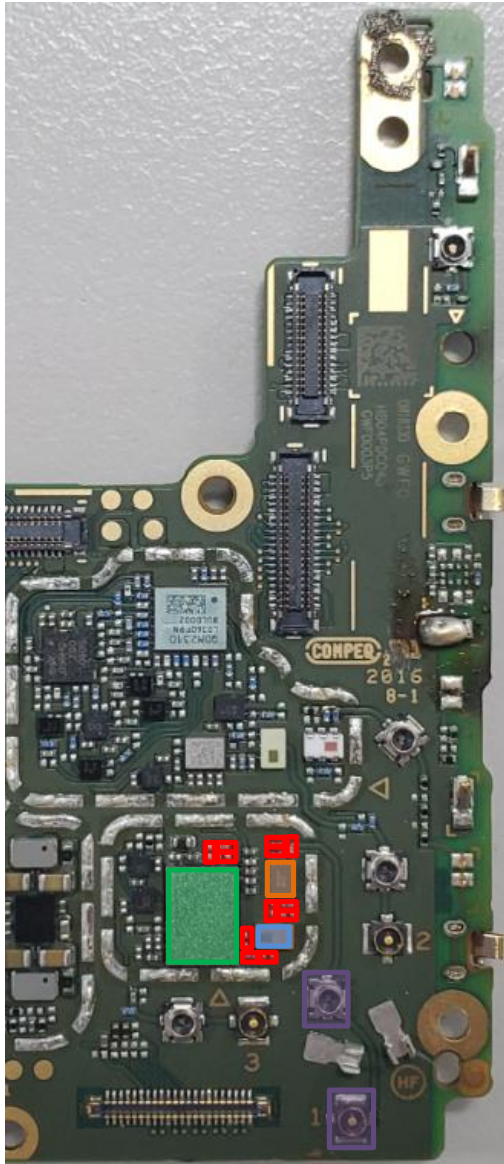
MHB DPDT
U4602

MHB + N77 DiPX
U4610

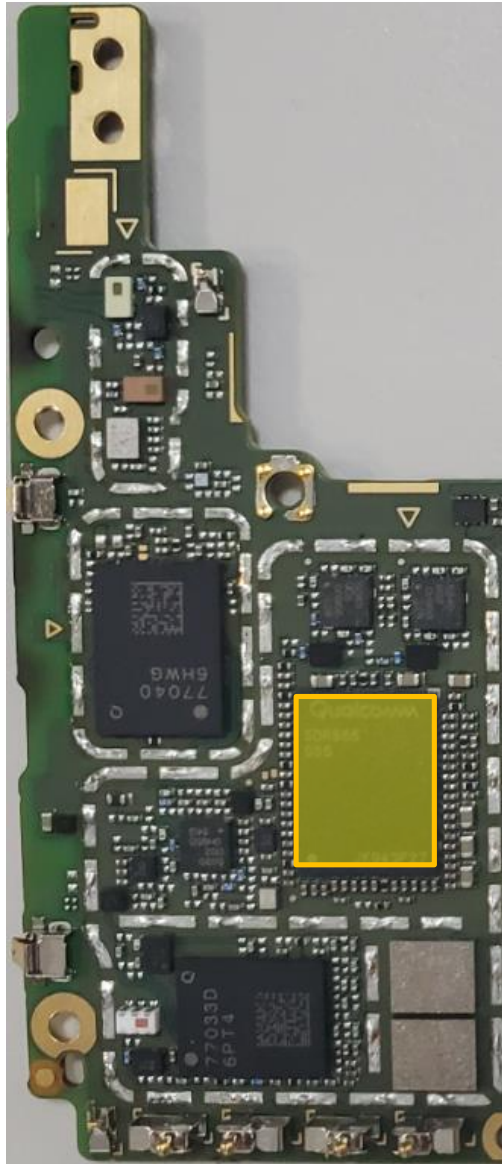
SDR865
U4101

N77 PRx Path

bottom



top



N77 : N77/N78

RF Connector N77
J4701 CON4701

N77 PA QPM5677
U4701

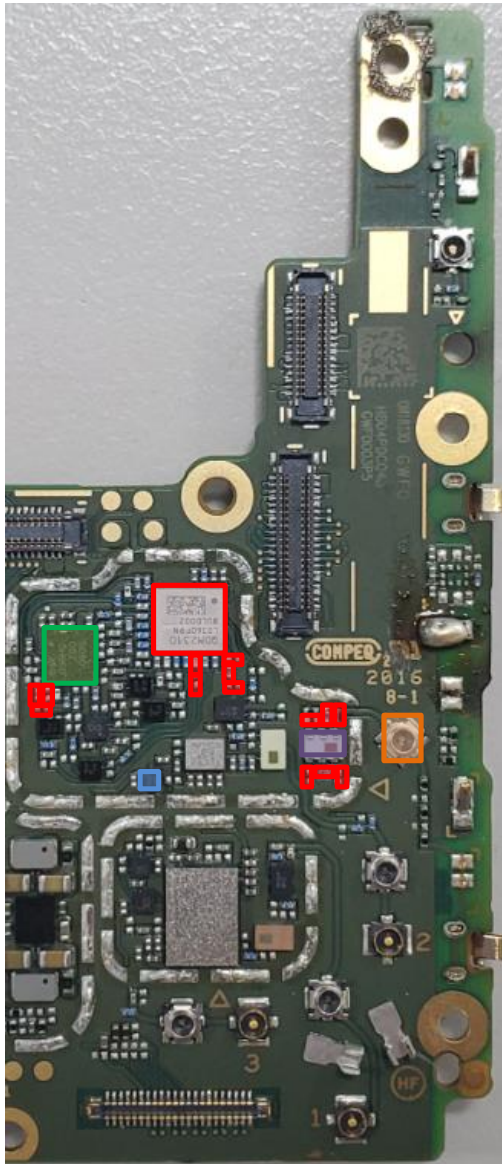
N77 DPDT
U4702

N77 BPF
U4705

SDR865
U4101

LB B8 DRx Path

bottom



top



Div. RF connector LB
CON5301



LB+B3/B7 DiPX
U5302



LB DRx DPDT
U4502



DiFEM QDM2310
U5303



DRx QLN5030
U5201



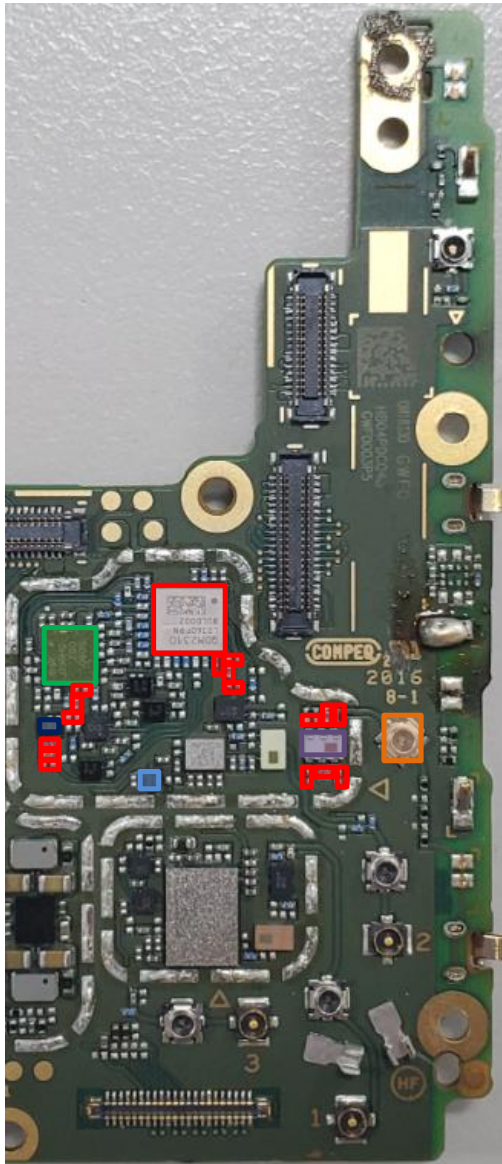
LB DRx LNA OUT DPDT
U5205



SDR865
U4101

LB B20 DRx Path

bottom



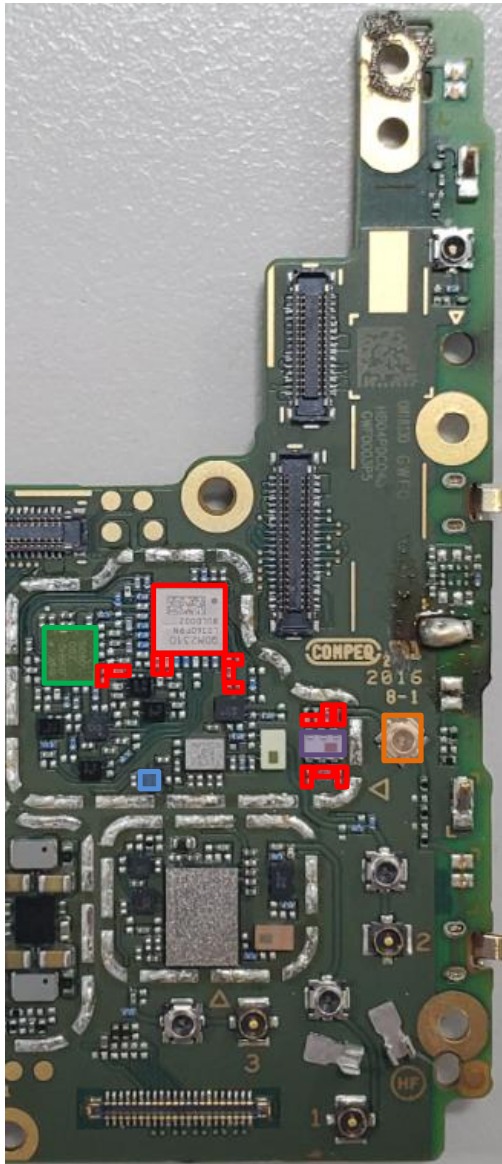
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  B20 DRx FILTER
FL5203
-  DRx QLN5030
U5201
-  LB DRx LNA OUT DPDT
U5205
-  SDR865
U4101

LB B26 DRx Path

bottom



top



Div. RF connector LB

CON5301



LB+B3/B7 DiPX

U5302



LB DRx DPDT

U4502



DiFEM QDM2310

U5303



DRx QLN5030

U5201



LB DRx LNA OUT DPDT

U5205

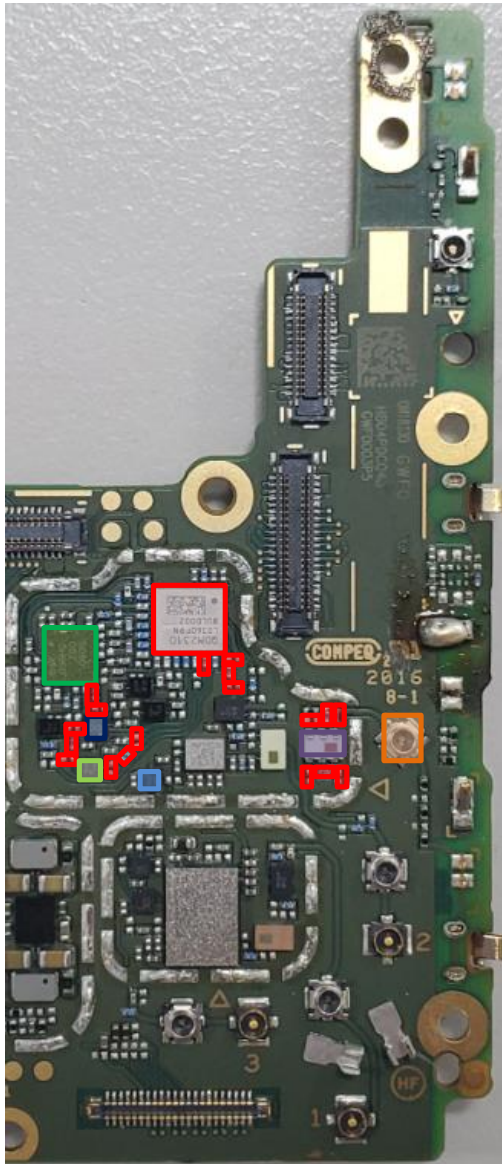


SDR865

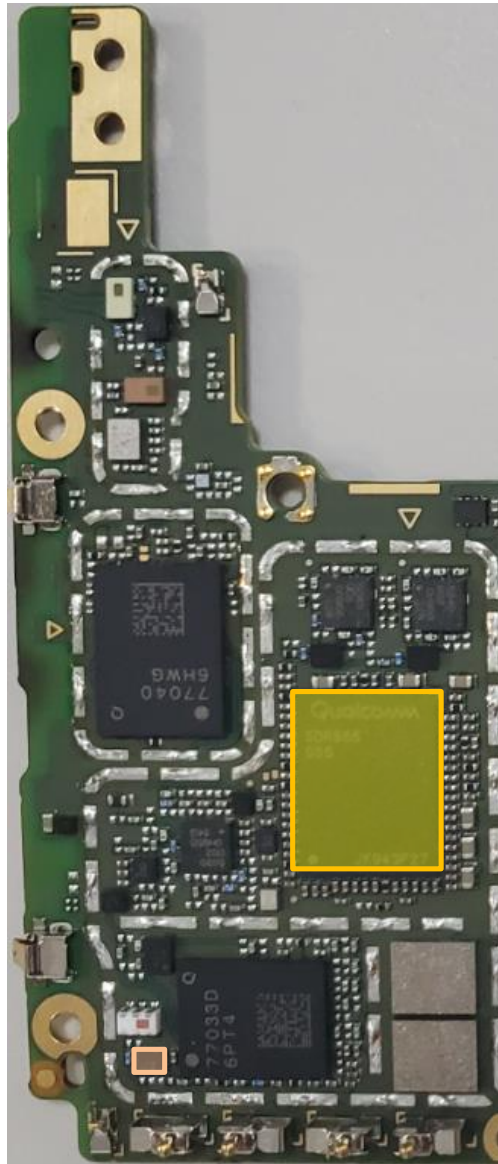
U4101

LB B12 DRx Path

bottom



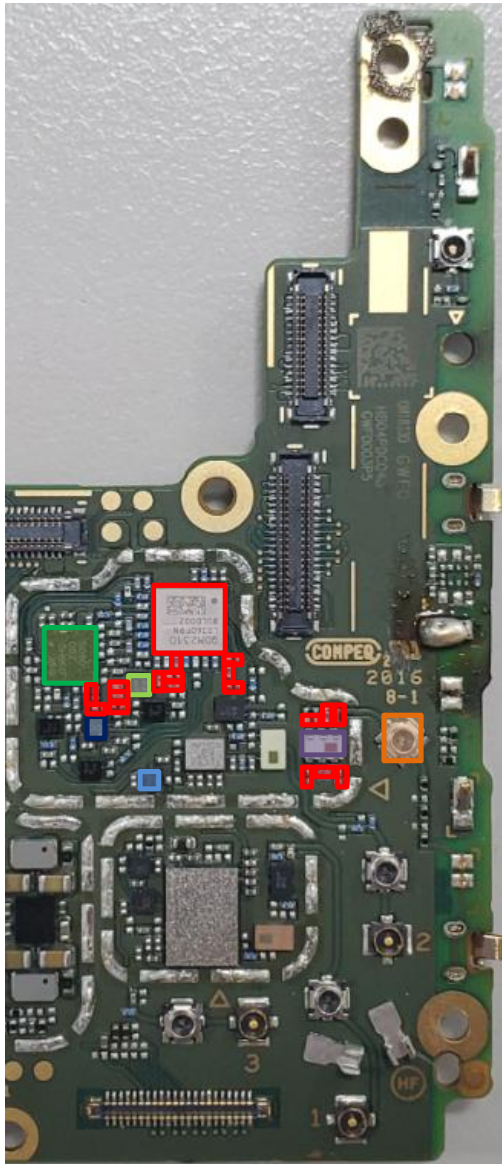
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  B12 FILTER
FL5201
-  LNA IN SP3T
U5206
-  DRx QLN5030
U5201
-  LNA OUT DPDT
U5205
-  SDR865
U4101

LB B28 DRx Path

bottom



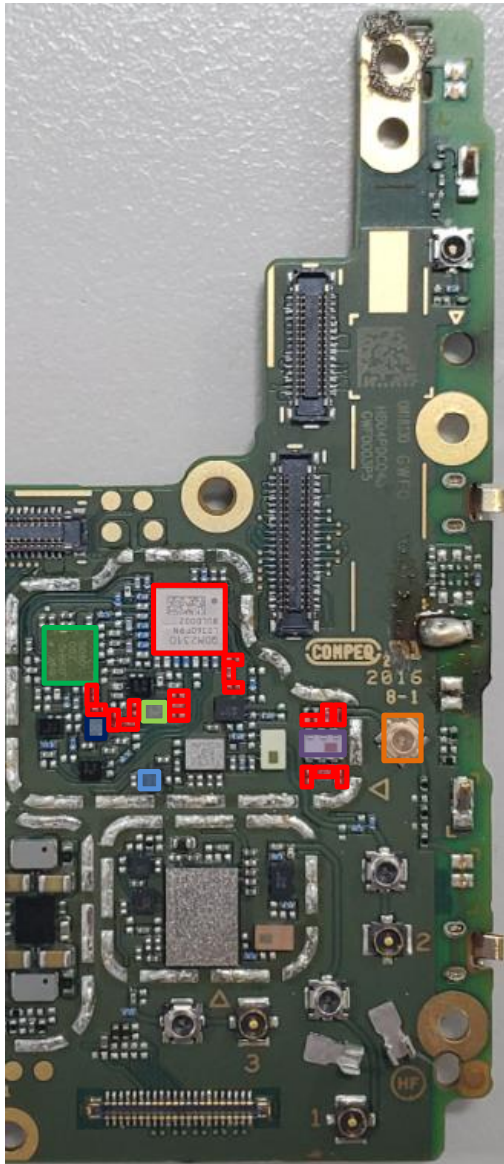
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  B28 FILTER
U5204
-  LNA IN SP3T
U5206
-  DRx QLN5030
U5201
-  LNA OUT DPDT
U5205
-  SDR865
U4101

LB B29 DRx Path

bottom



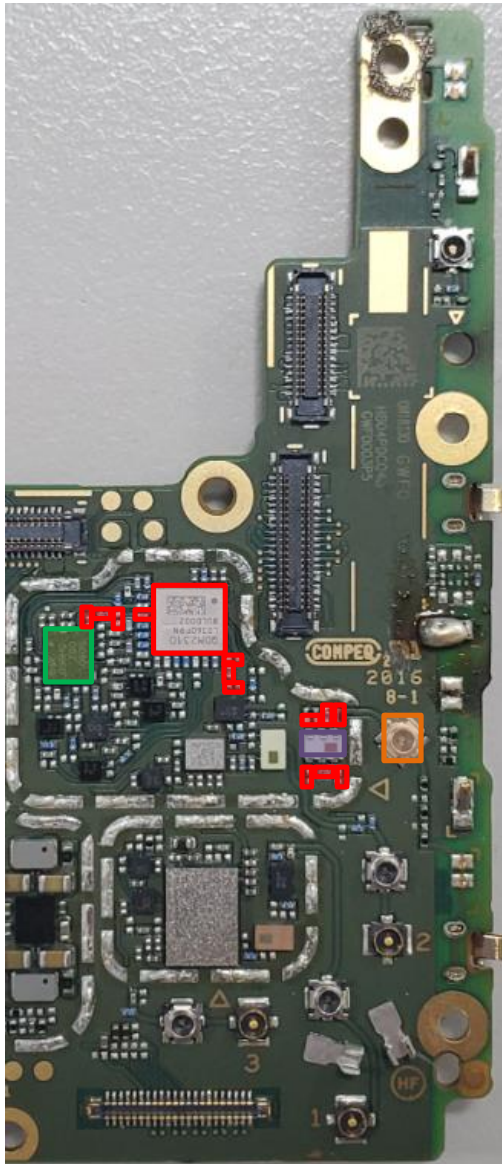
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  B29 FILTER
FL5202
-  LNA IN SP3T
U5206
-  DRx QLN5030
U5201
-  LNA OUT DPDT
U5205
-  SDR865
U4101

MB B1/B4 DRx Path

bottom



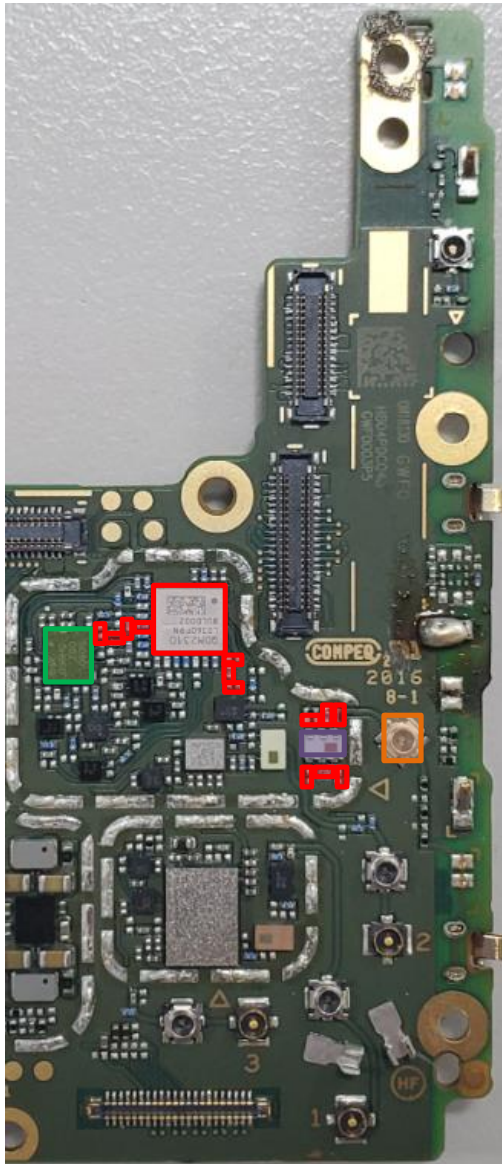
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR865
U4101

MB B2 DRx Path

bottom



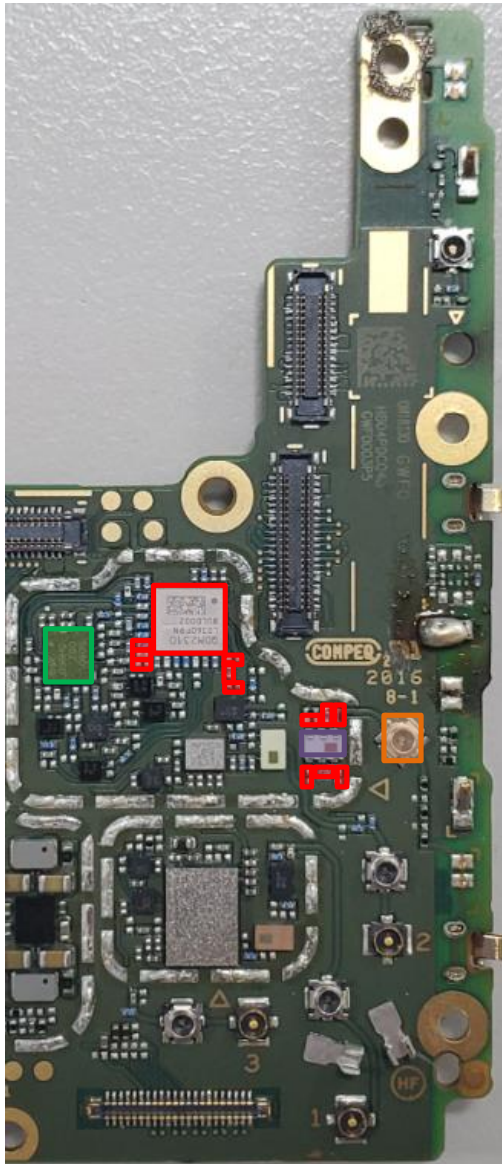
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR865
U4101

MB B3 DRx Path

bottom



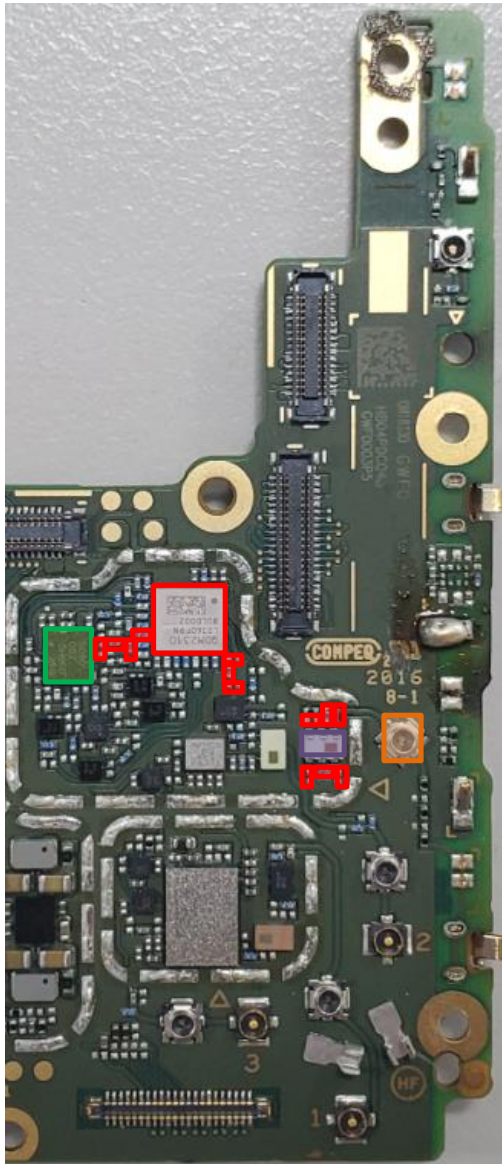
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR865
U4101

MB B39 DRx Path

bottom



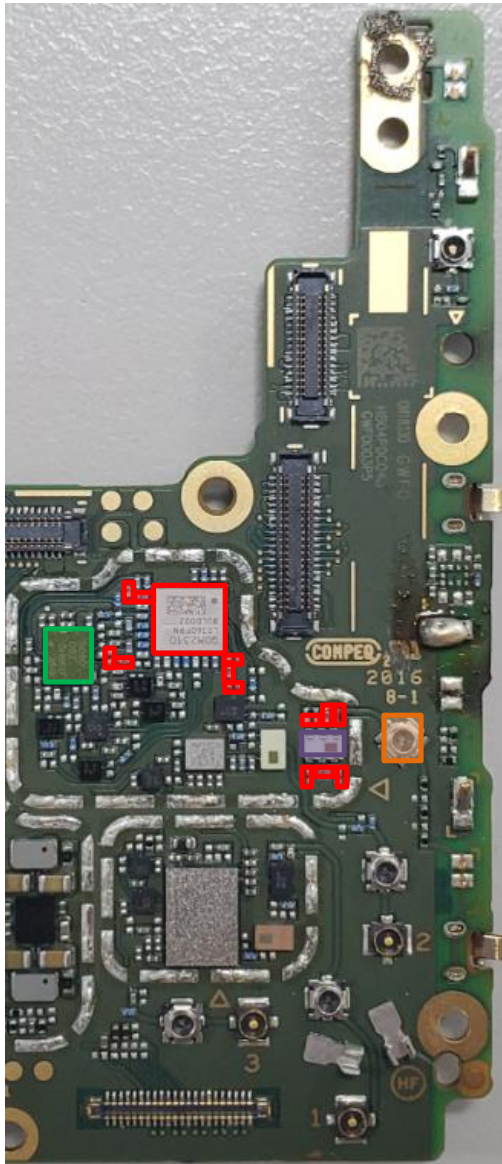
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR865
U4101

HB B7 DRx Path

bottom



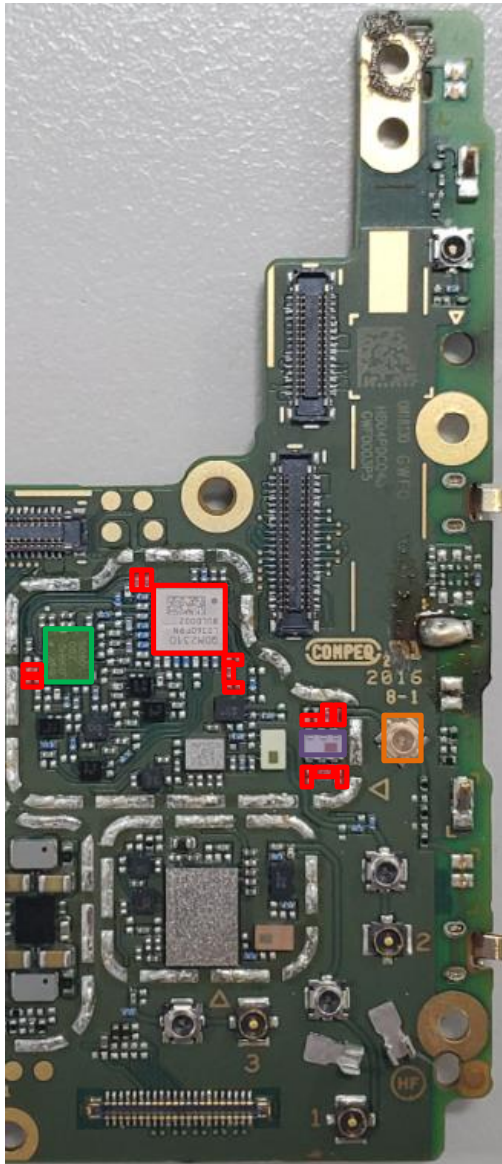
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR865
U4101

HB B40 DRx Path

bottom



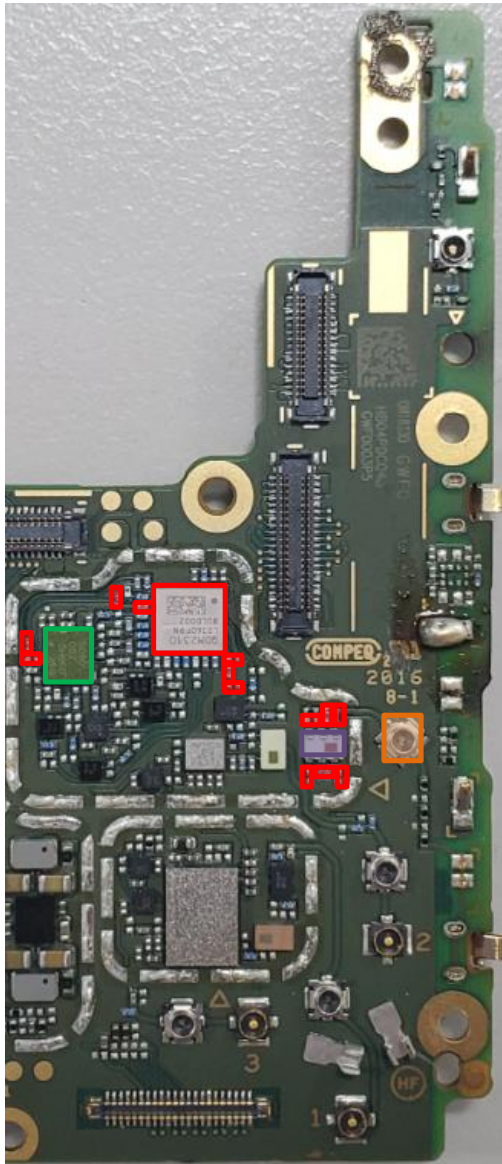
top



-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR8150
U4101

HB B38/B41 DRx Path

bottom



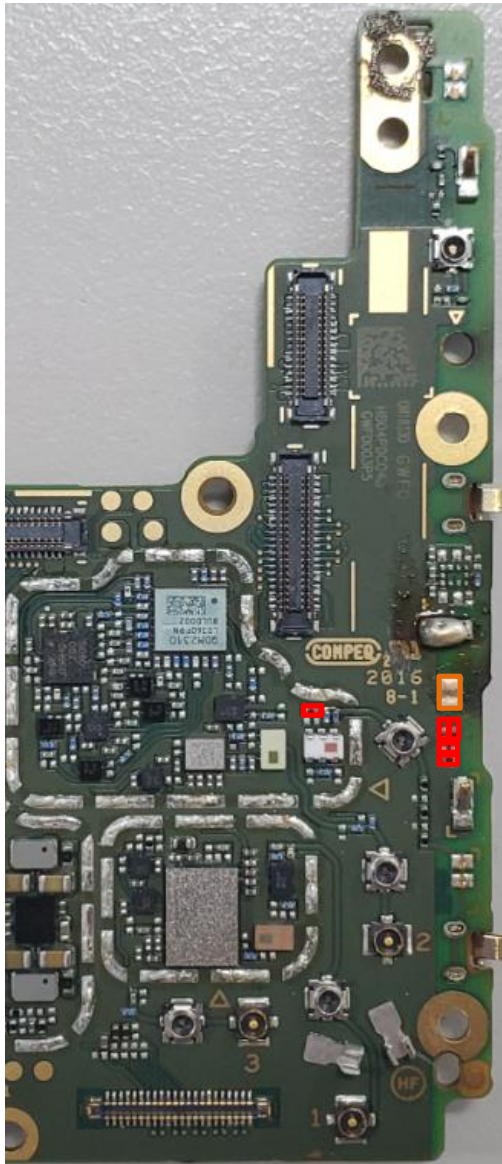
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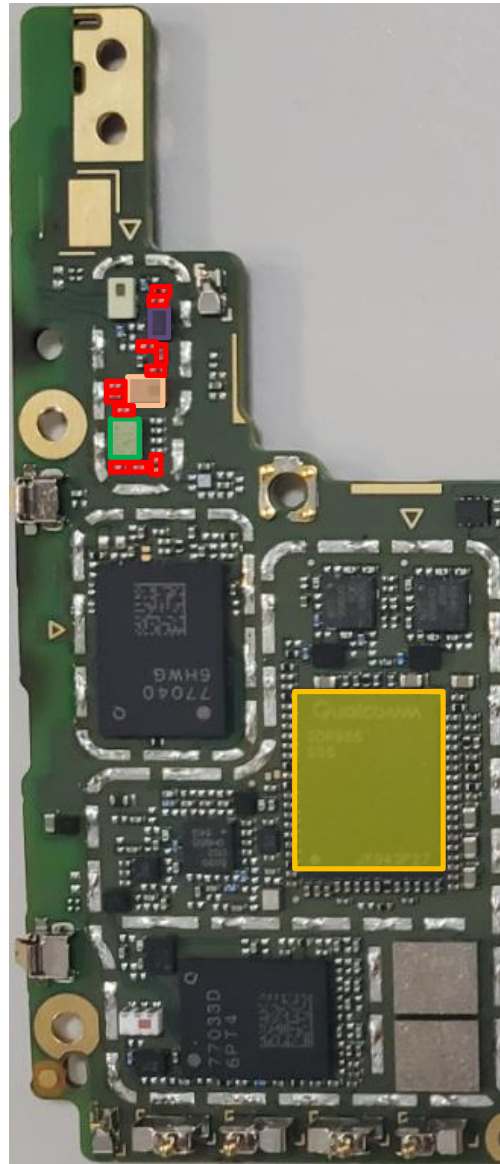
-  Div. RF connector LB
CON5301
-  LB+B3/B7 DiPX
U5302
-  LB DRx DPDT
U4502
-  DiFEM QDM2310
U5303
-  DRx QLN5030
U5201
-  SDR8150
U4101

N77 DRx Path

bottom



top



Div. RF connector LB
ANT5317

DRx N77 DPDT
U5308

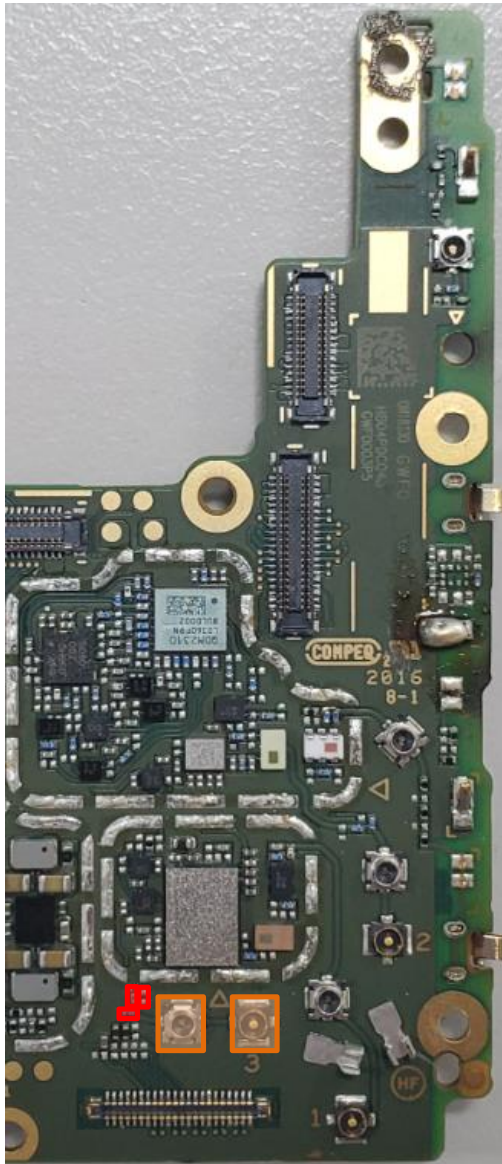
DRx N77 FILTER
U5311

DRx QDM5677
U5307

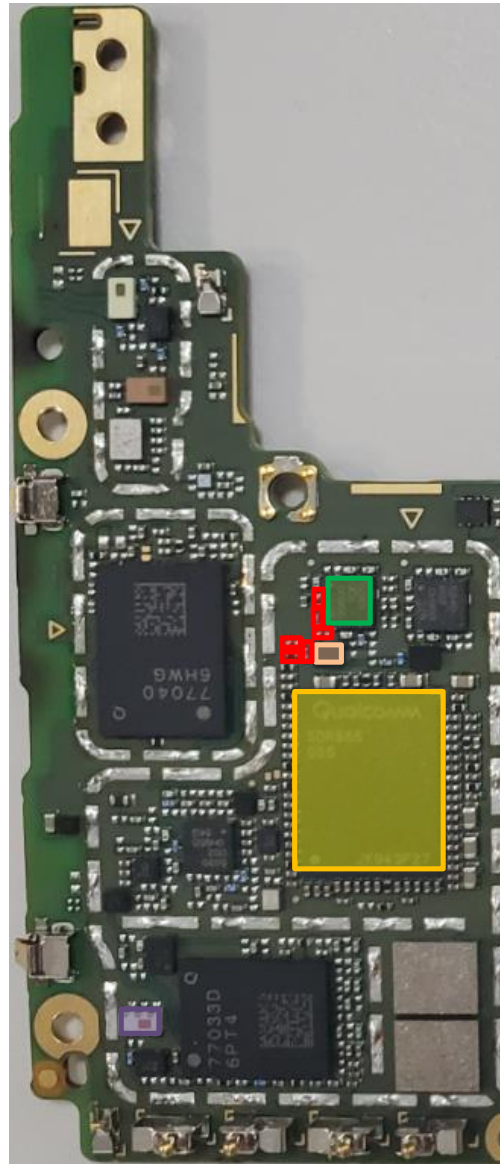
SDR8150
U4101

B3 PRx MIMO Path

bottom



top



LB + B3&B7 connector
J4501 J4502

LB + B3&B7 DiPX
U4503

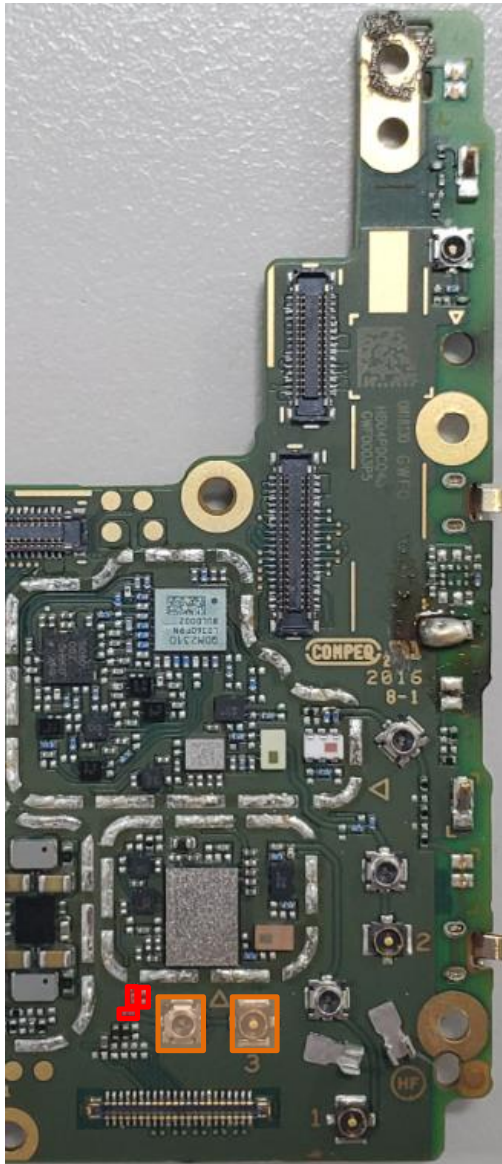
B3 + B7 DUAL SAW
U4505

PRx MIMO QLN5040
U5004

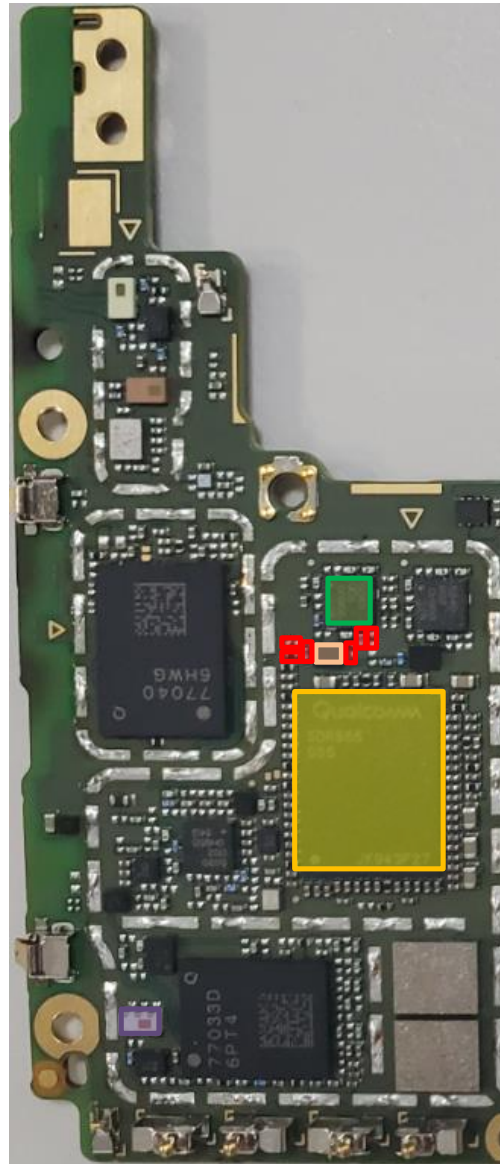
SDR8150
U4101

B7 PRx MIMO Path

bottom



top



LB + B3&B7 connector
J4501 J4502

LB + B3&B7 DiPX
U4503

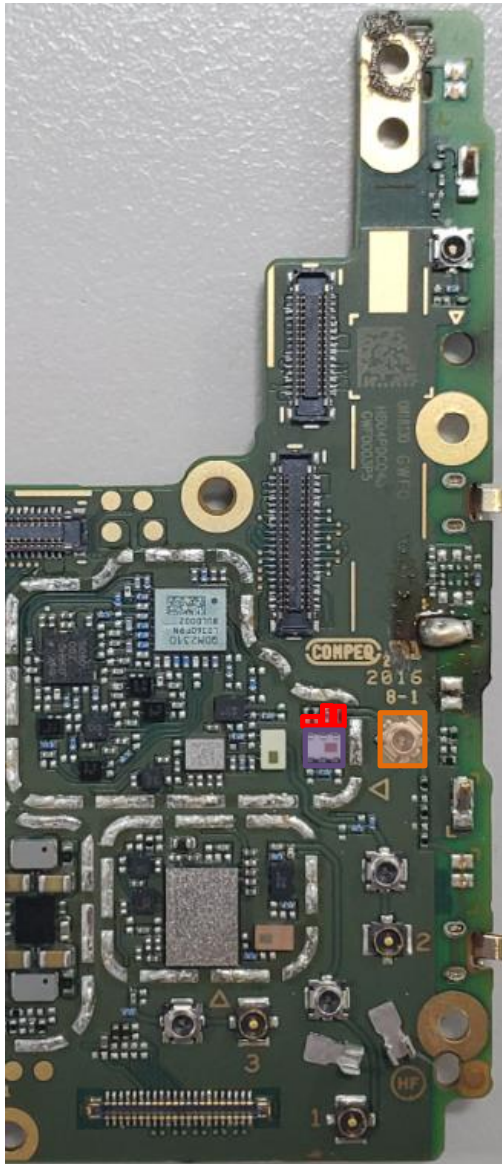
B3 + B7 DUAL SAW
U4505

PRx MIMO QLN5040
U5004

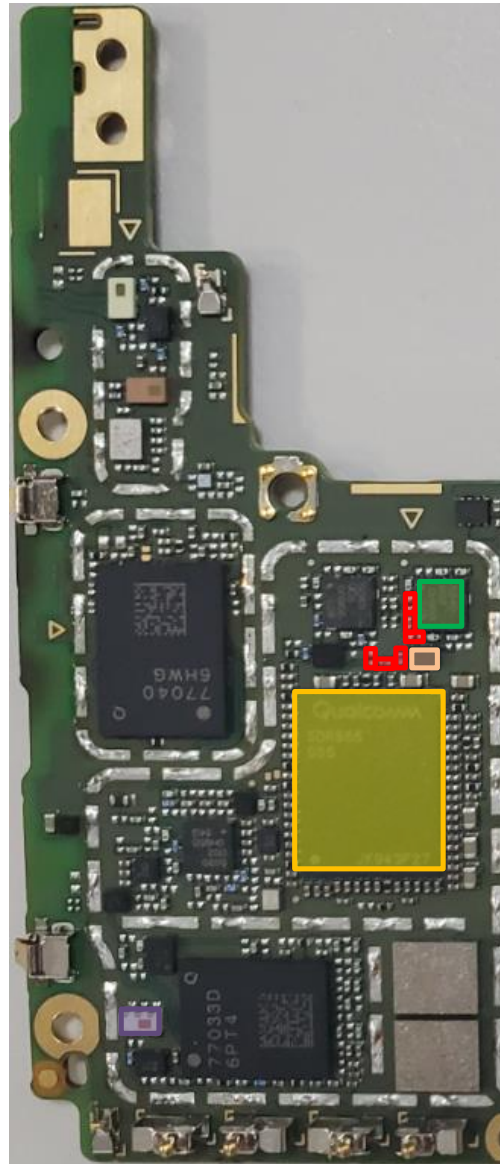
SDR8150
U4101

B3 DRx MIMO Path

bottom



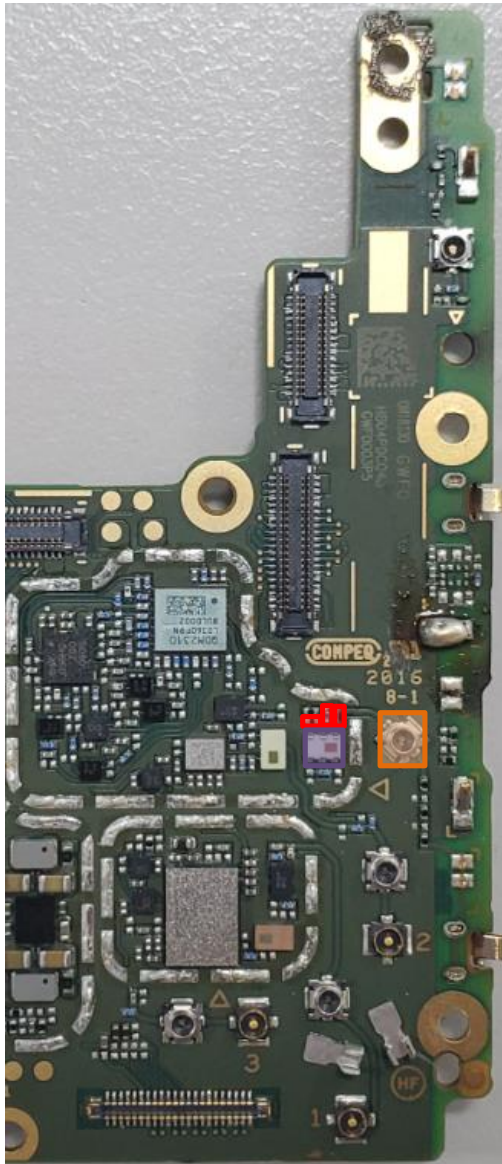
top



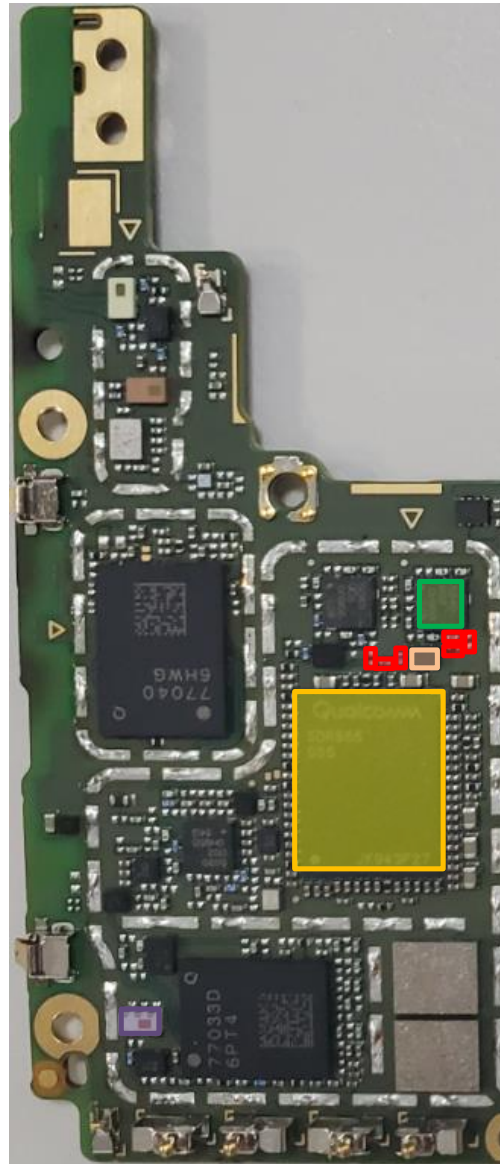
-  LB + B3&B7 connector
CON5301
-  LB + B3&B7 DiPX
U5302
-  B3 + B7 DUAL SAW
FL5301
-  DRx MIMO QLN5040
U5005
-  SDR8150
U4101

B7 DRx MIMO Path

bottom



top



LB + B3&B7 connector
CON5301

LB + B3&B7 DiPX
U5302

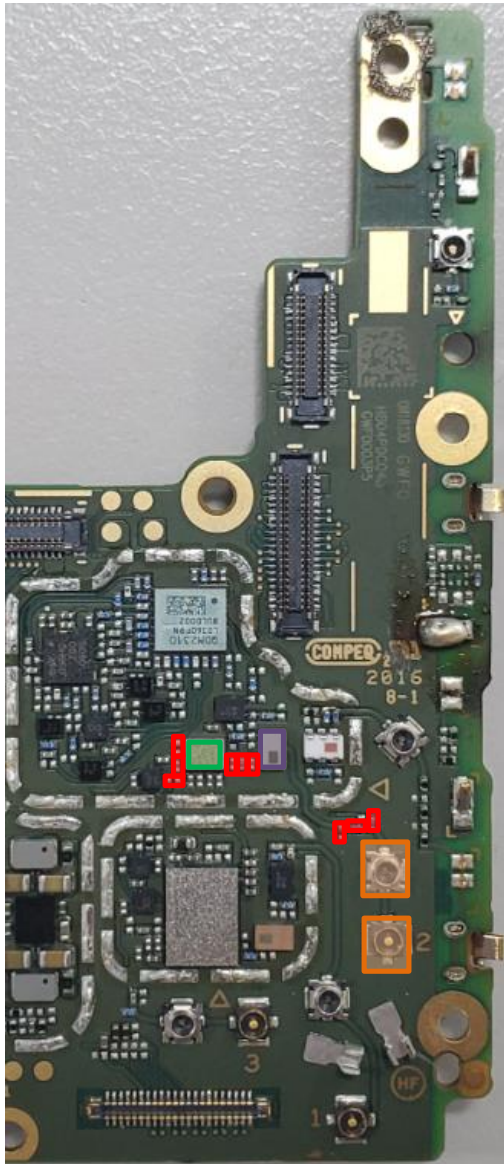
B3 + B7 DUAL SAW
FL5301

DRx MIMO QLN5040
U5005

SDR8150
U4101

N77 PRx MIMO Path

bottom



top



MHB + N77 connector
CON4602

MHB + N77 DiPX
U4610

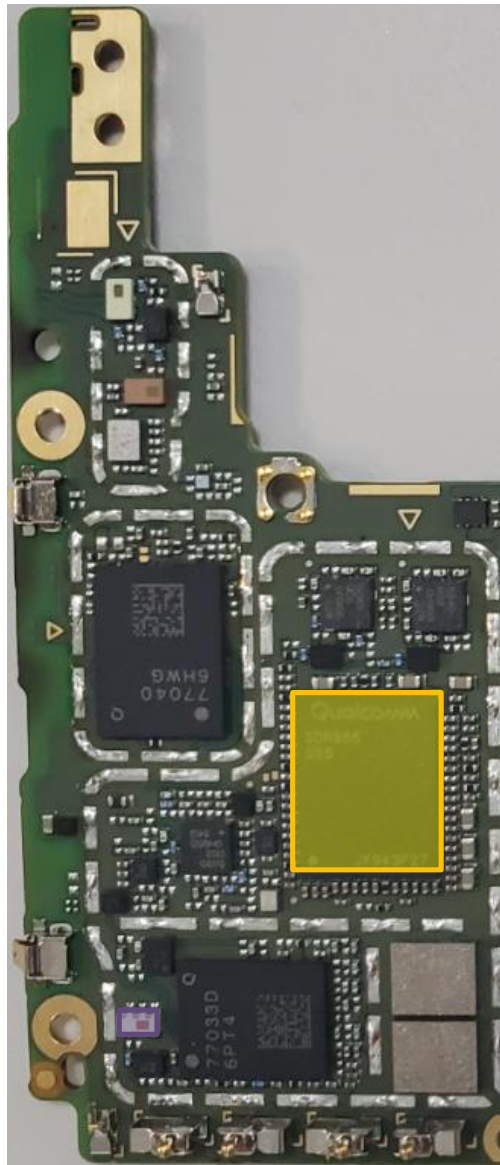
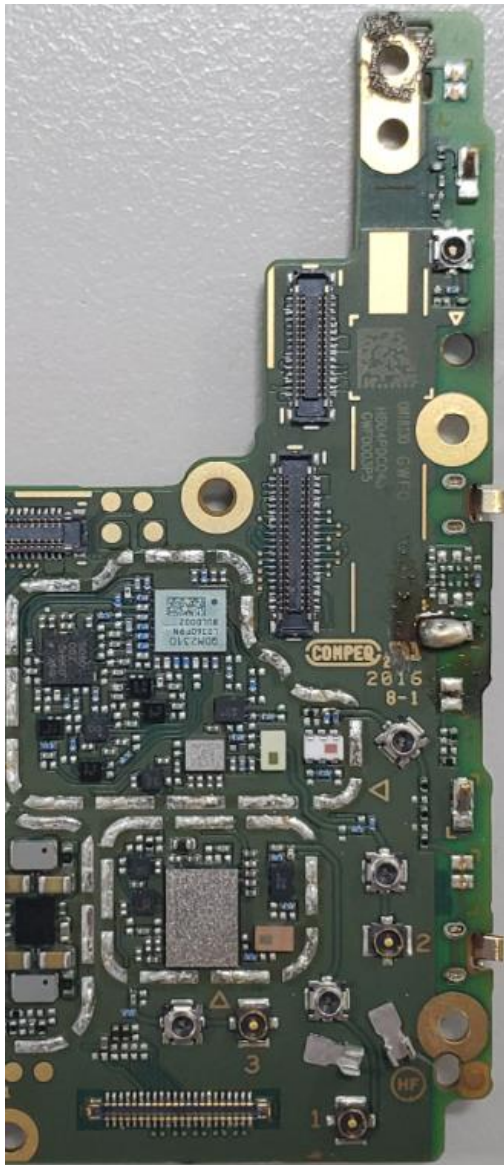
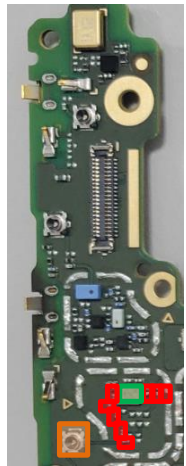
PRx MIMO QDM5677
U4604

SDR8150
U4101

N77 DRx MIMO Path

bottom

top

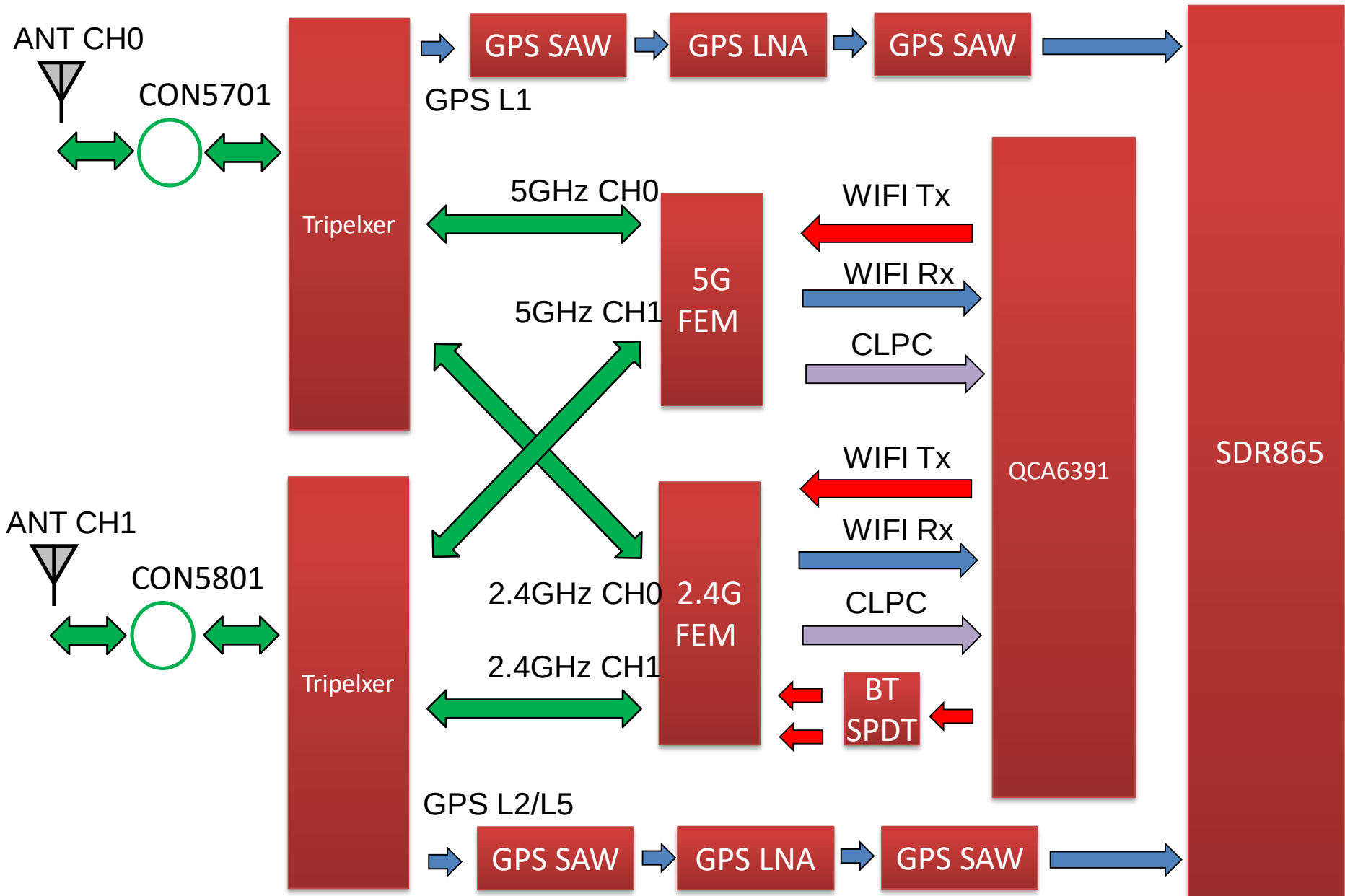


 MHB + N77 connector
CON4602

 DRx MIMO QDM5677
U4604

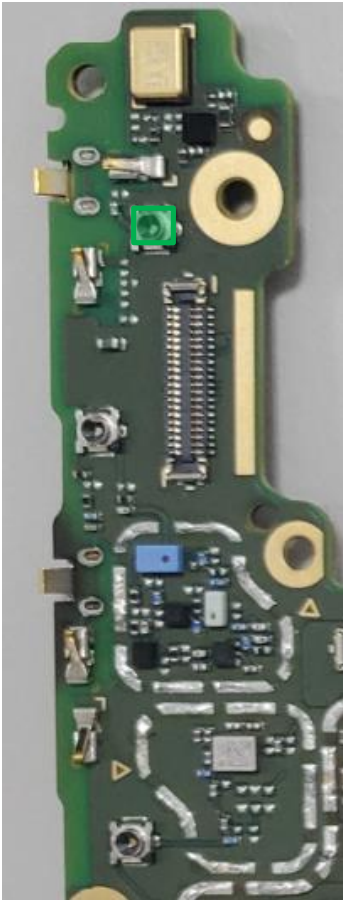
 SDR8150
U4101

WIFI/BT/GPS Block Diagram

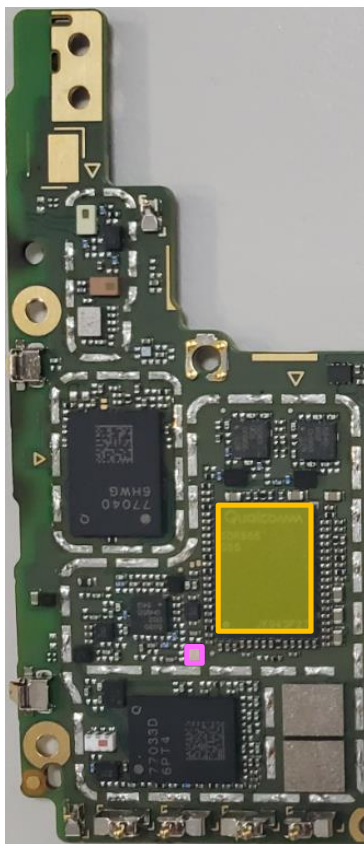


WIFI CH0 Key Components

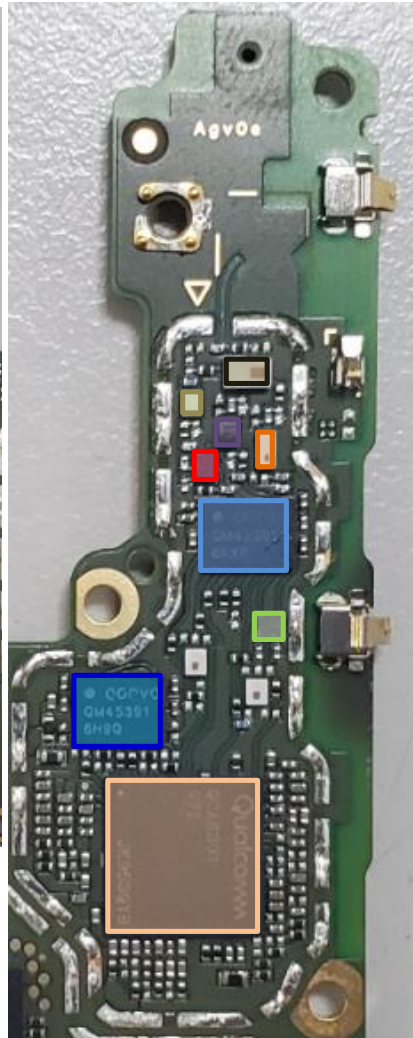
bottom



Top



Top



CH0 Connector
CON5701

CH0 TPX
U5704

CH0 2.4G FILTER
U5706

CH0 5G FILTER
U5801

2.4G FEM
U5705

5G FEM
U5802

BT SPDT
U5809

QCA6391
U5501

SDR865
U4101

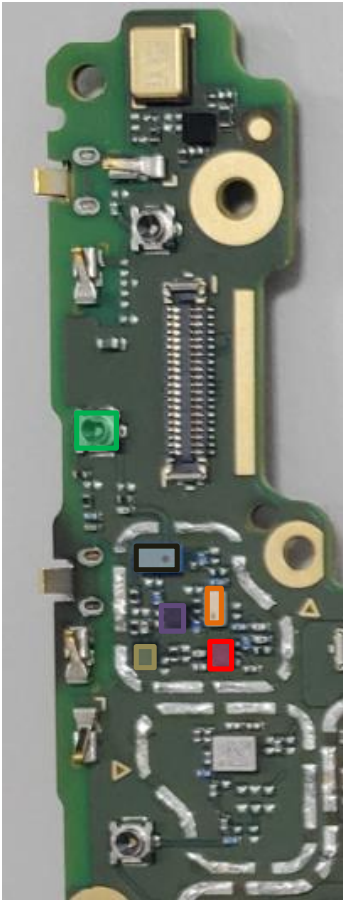
GPS L1 LNA
U5701

GPS L1 SAW
U5702

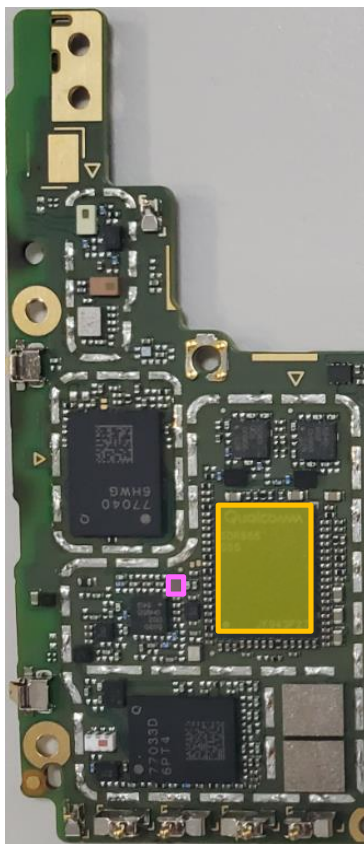
GPS L1 SAW
U5703

WIFI CH1 Key Components

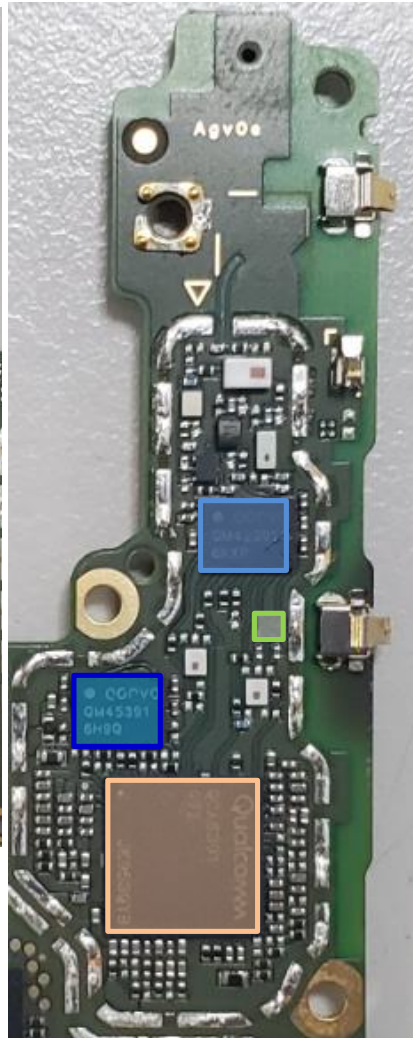
bottom



Top



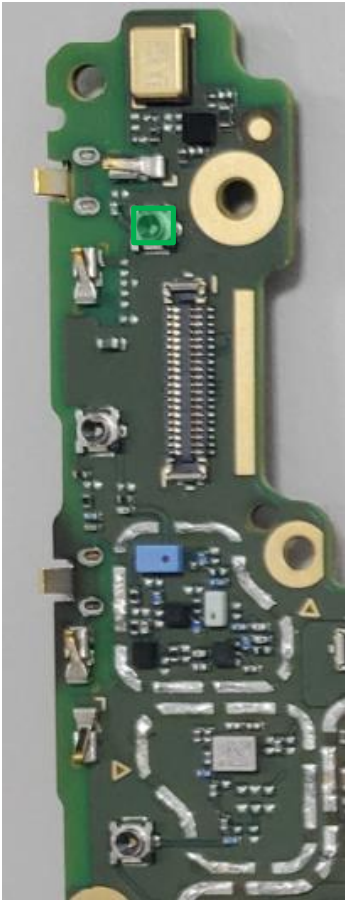
Top



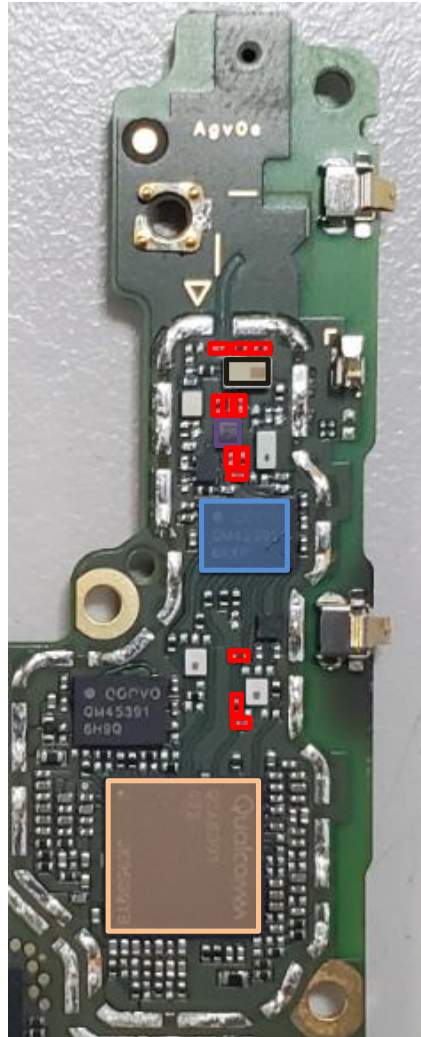
-  CH1 Connector
CON5801
-  CH1 TPX
U5804
-  CH1 2.4G FILTER
U5803
-  CH1 5G FILTER
U5805
-  2.4G FEM
U5705
-  5G FEM
U5802
-  BT SPDT
U5809
-  QCA6391
U5501
-  SDR865
U4101
-  GPS L2 LNA
U5707
-  GPS L2 SAW
U5706
-  GPS L2 SAW
U5708

WIFI CH0 2.4G Tx

bottom



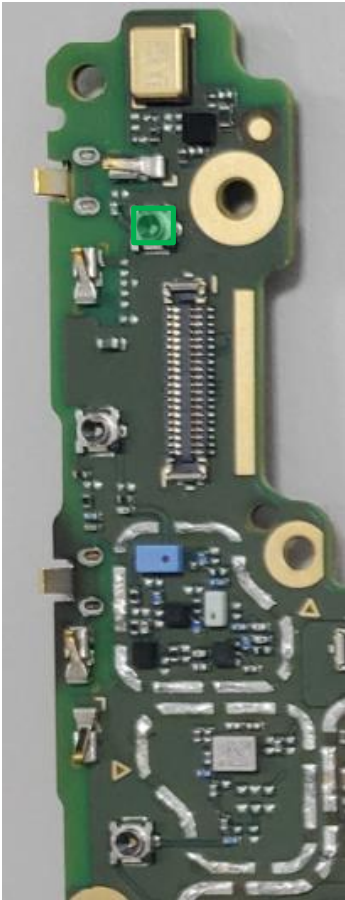
Top



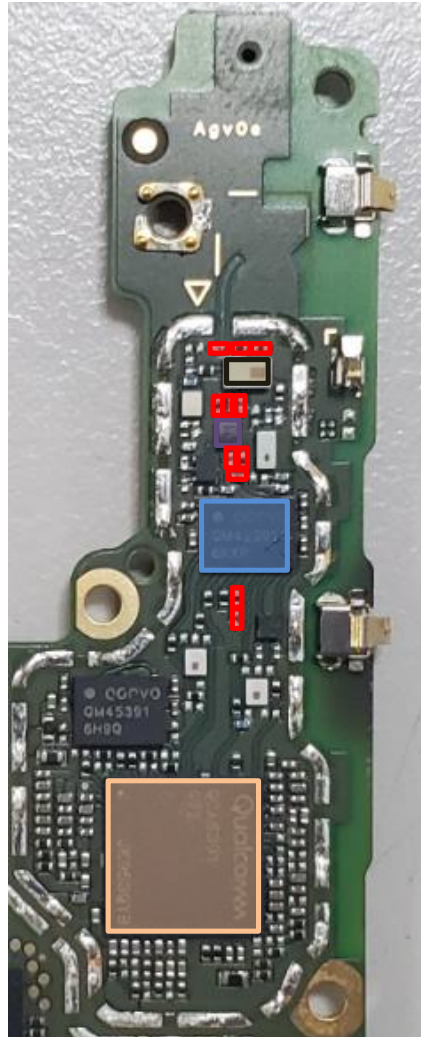
-  CH0 Connector
CON5701
-  CH0 TPX
U5704
-  CH0 2.4G FILTER
U5706
-  2.4G FEM
U5705
-  QCA6391
U5501

WIFI CH0 2.4G Rx

bottom



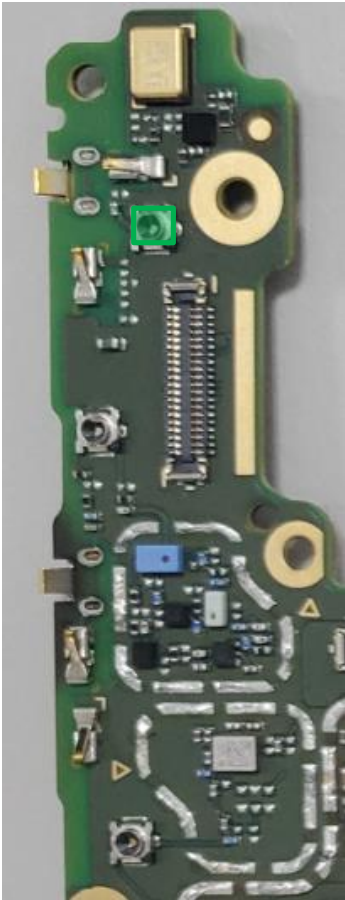
Top



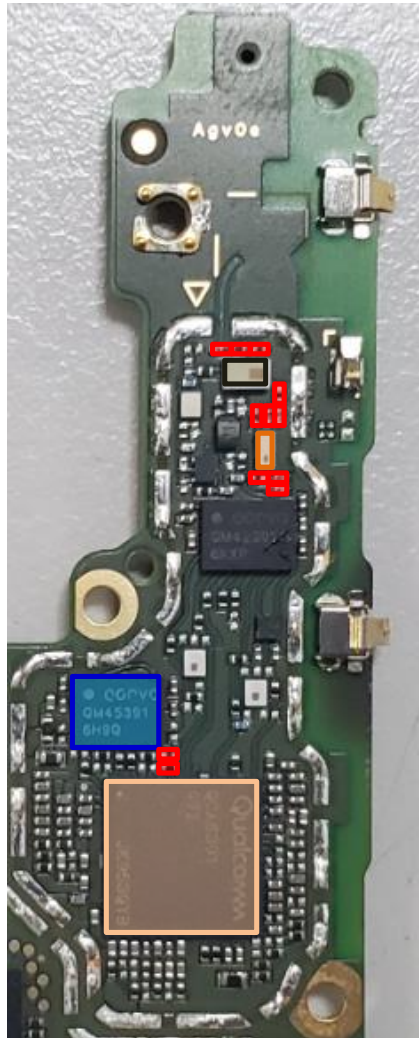
-  CH0 Connector
CON5701
-  CH0 TPX
U5704
-  CH0 2.4G FILTER
U5706
-  2.4G FEM
U5705
-  QCA6391
U5501

WIFI CH0 5G Tx

bottom



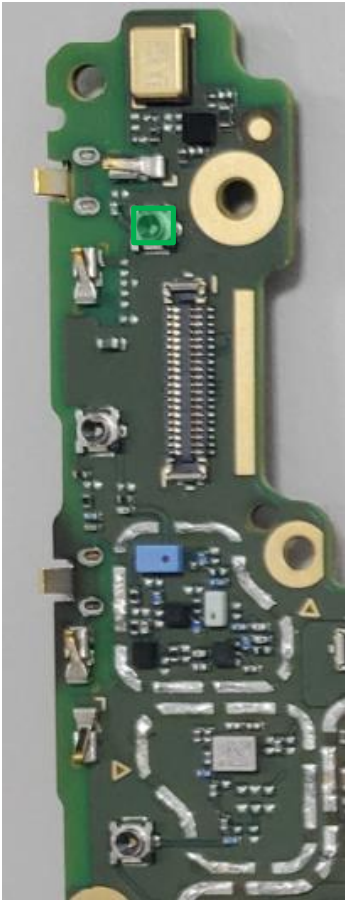
Top



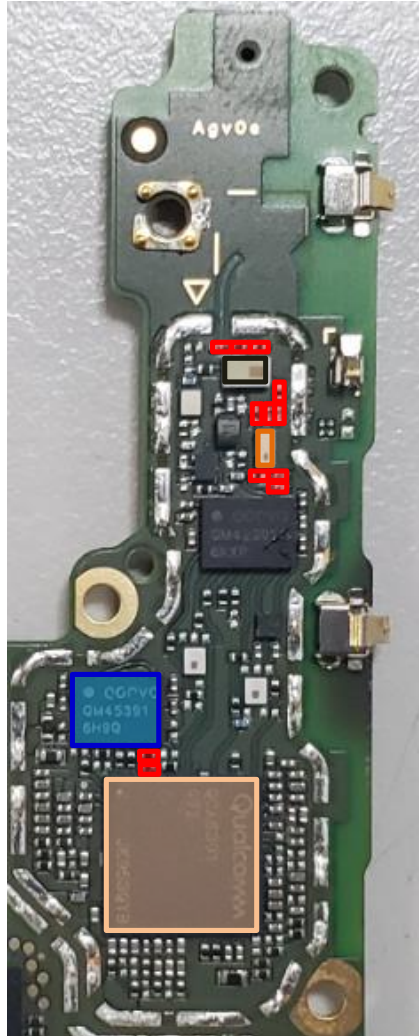
-  CH0 Connector
CON5701
-  CH0 TPX
U5704
-  CH0 5G FILTER
U5801
-  5G FEM
U5802
-  QCA6391
U5501

WIFI CH0 5G Rx

bottom



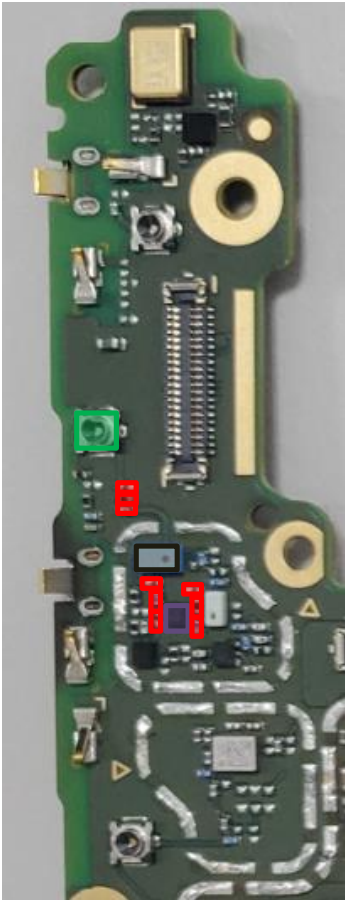
Top



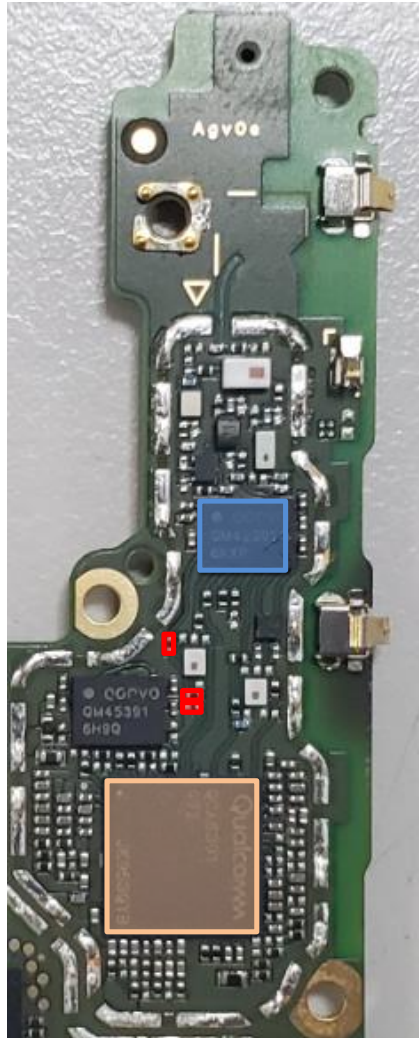
-  CH0 Connector
CON5701
-  CH0 TPX
U5704
-  CH0 5G FILTER
U5801
-  5G FEM
U5802
-  QCA6391
U5501

WIFI CH1 2.4G Tx

bottom



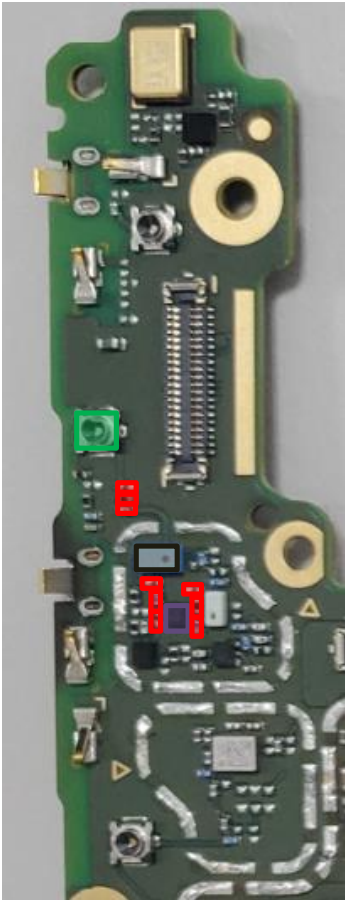
Top



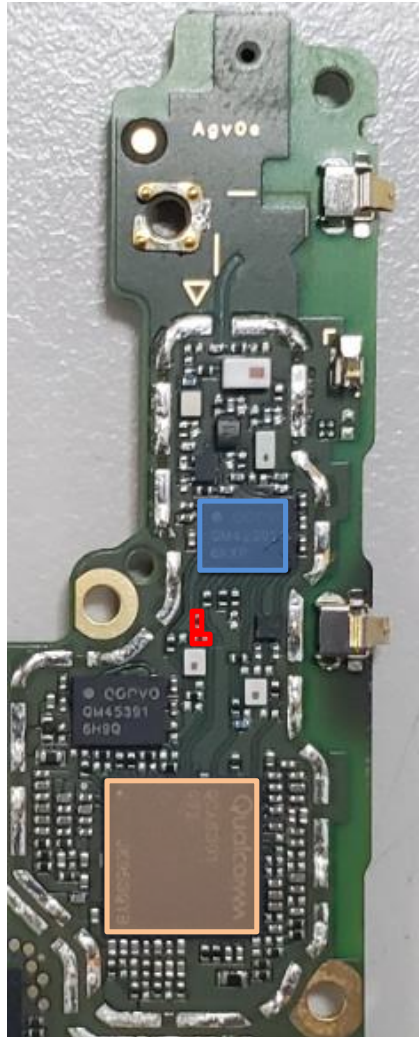
-  CH1 Connector
CON5801
-  CH1 TPX
U5804
-  CH1 2.4G FILTER
U5803
-  2.4G FEM
U5705
-  QCA6391
U5501

WIFI CH1 2.4G Rx

bottom



Top

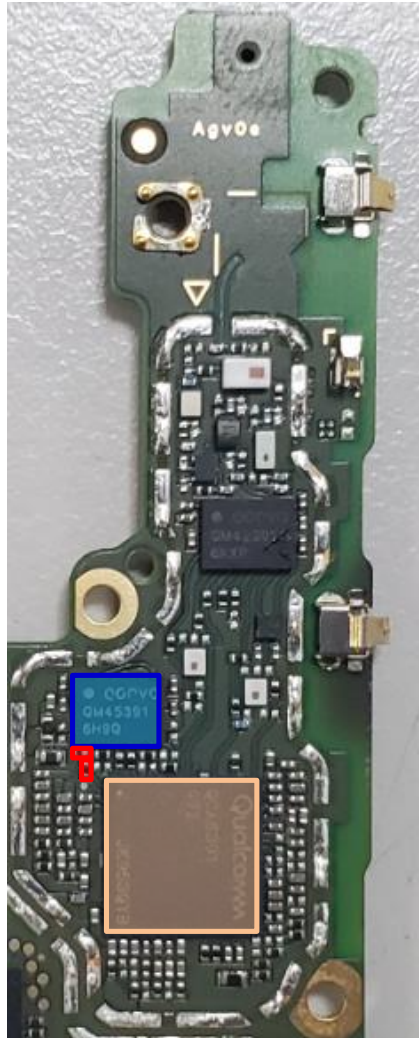
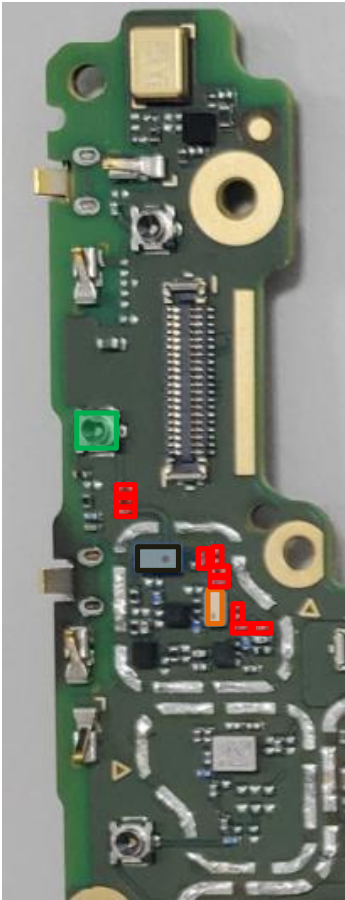


-  CH1 Connector
CON5801
-  CH1 TPX
U5804
-  CH1 2.4G FILTER
U5803
-  2.4G FEM
U5705
-  QCA6391
U5501

WIFI CH1 5G Tx

bottom

Top



CH1 Connector
CON5801

CH1 TPX
U5804

CH1 5G FILTER
U5805

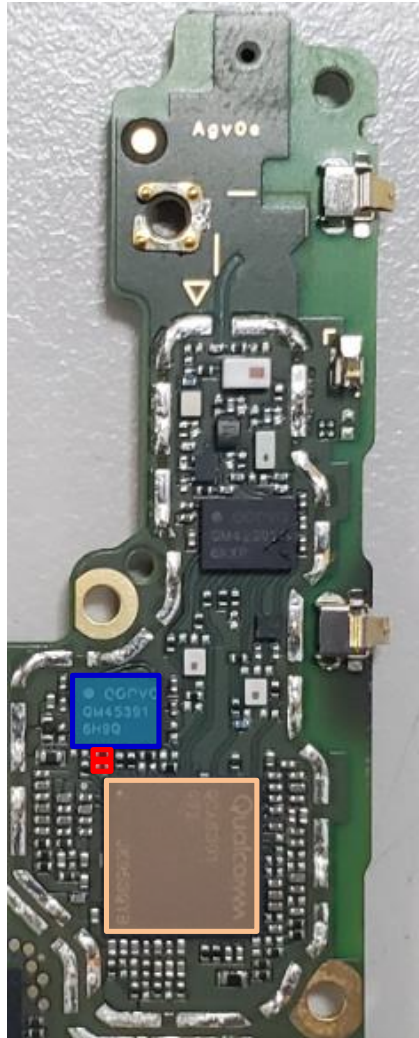
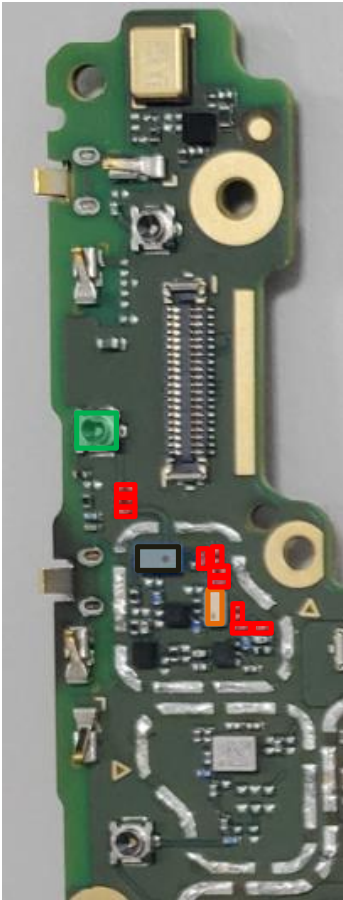
5G FEM
U5802

QCA6391
U5501

WIFI CH1 5G Rx

bottom

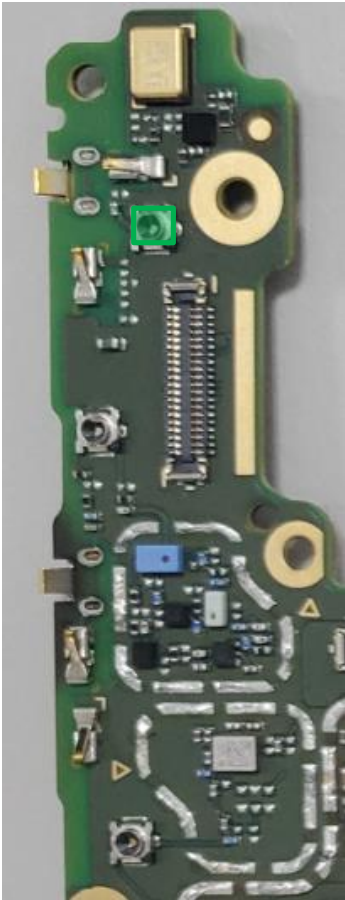
Top



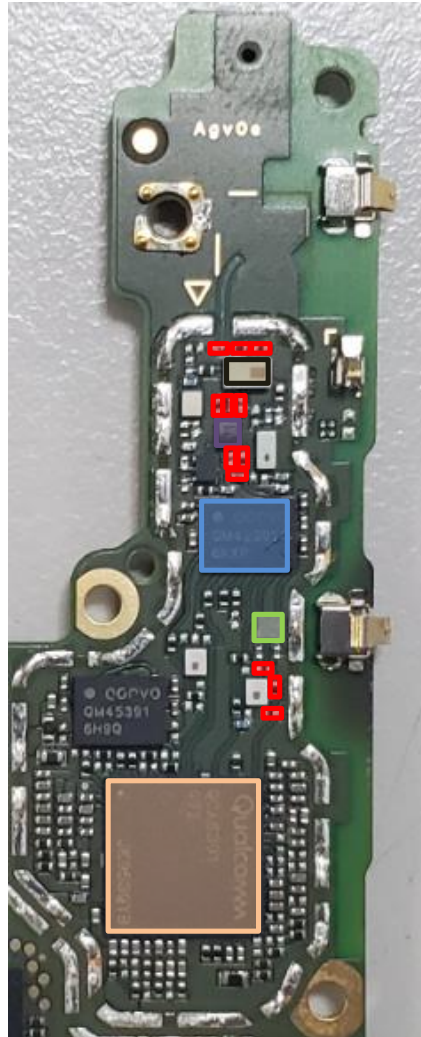
-  CH1 Connector
CON5801
-  CH1 TPX
U5804
-  CH1 5G FILTER
U5805
-  5G FEM
U5802
-  QCA6391
U5501

BT CH0 2.4G Tx

bottom



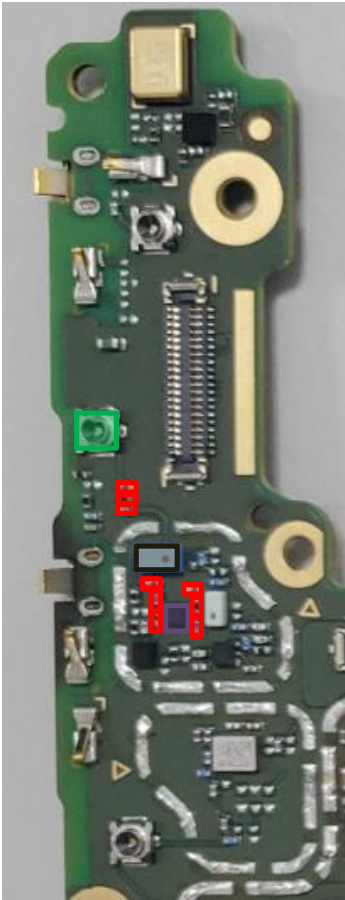
Top



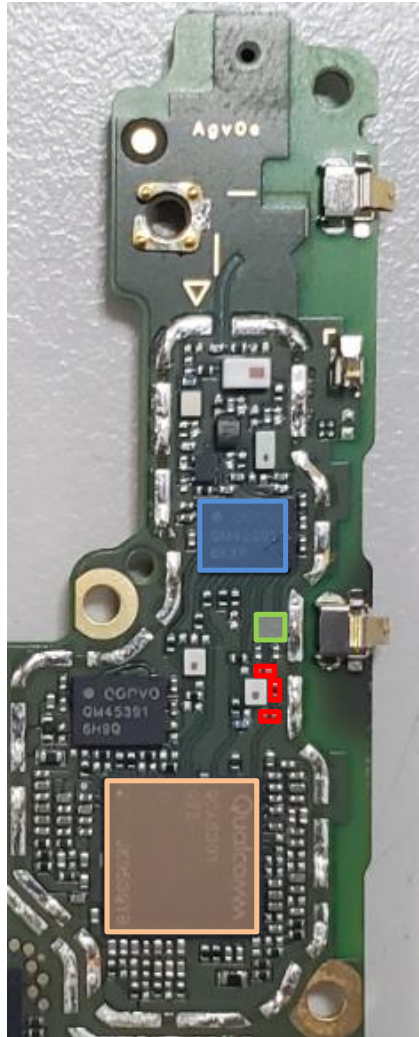
-  CH0 Connector
CON5701
-  CH0 TPX
U5704
-  CH0 2.4G FILTER
U5706
-  2.4G FEM
U5705
-  BT SPDT
U5809
-  QCA6391
U5501

BT CH1 2.4G Tx

bottom



Top



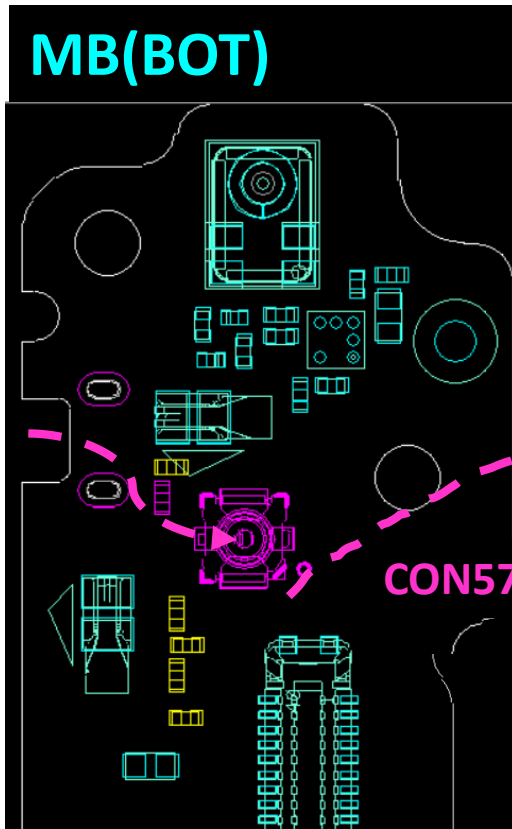
-  CH1 Connector
CON5801
-  CH1 TPX
U5804
-  CH1 2.4G FILTER
U5803
-  2.4G FEM
U5705
-  BT SPDT
U5809
-  QCA6391
U5501

ZS670KL Trouble Shooting Guide

GPS/NFC

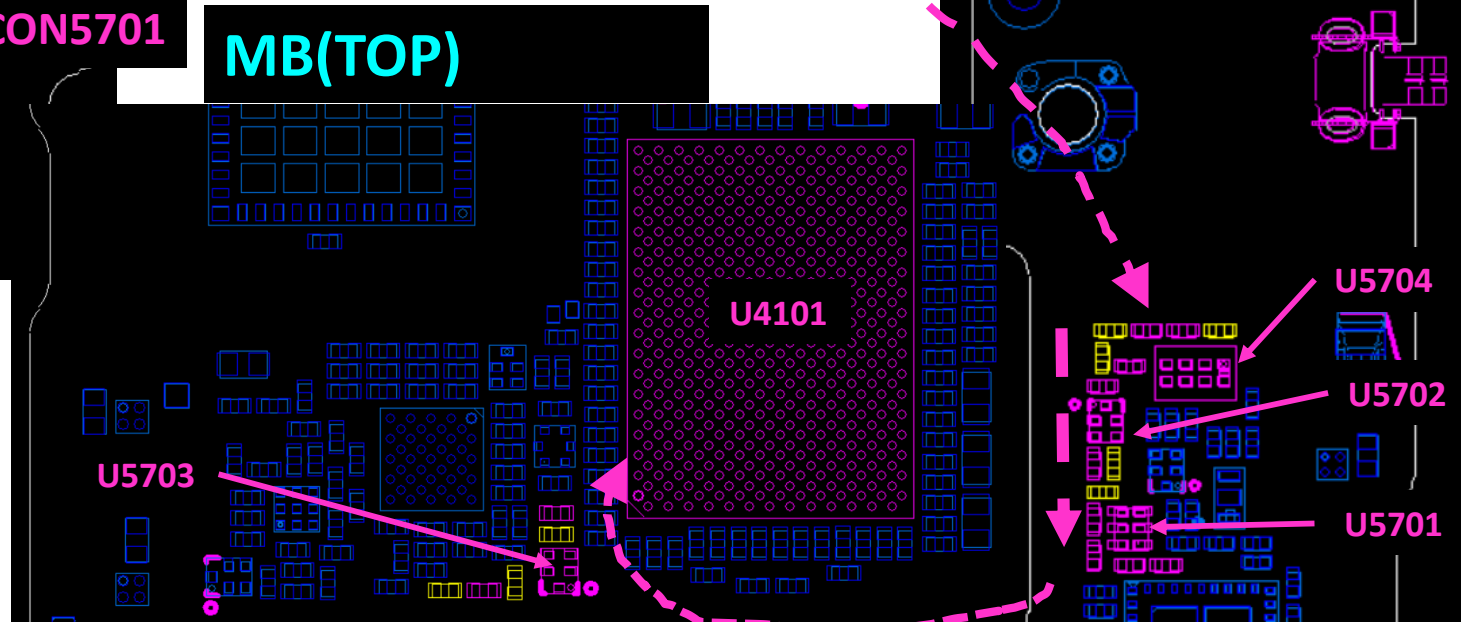
GPS_L1

MB(BOT)



CON5701

MB(TOP)



CON5701: RF SWITCH CONNECTOR

U5704: TRIPLEXER/TP2012-A1255MST

U5702: Pre SAW/SAFFB1G56KB0F0AR1X

U5701: LNA/WS7918DC-6/TR

U5703: Post SAW/SAFFB1G56KB0F0AR1X

U4101: Transceiver/SDR865

MB(TOP)

U5704

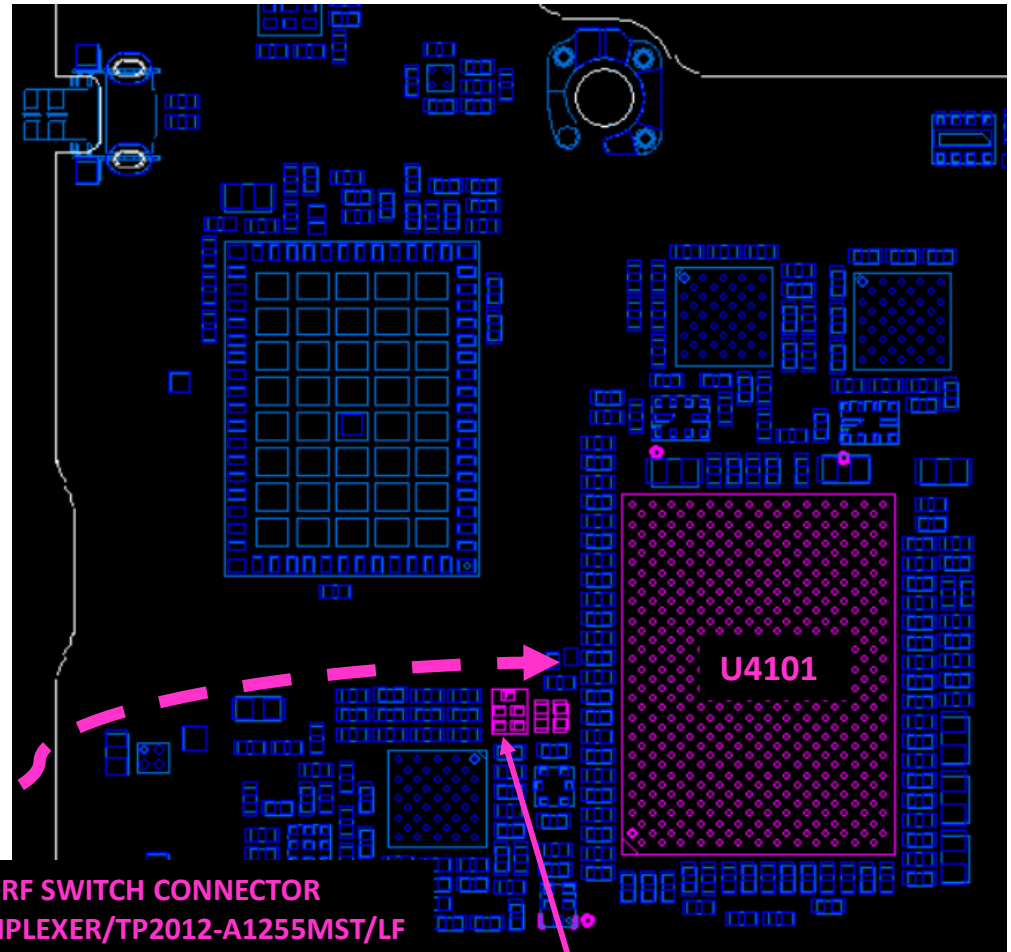
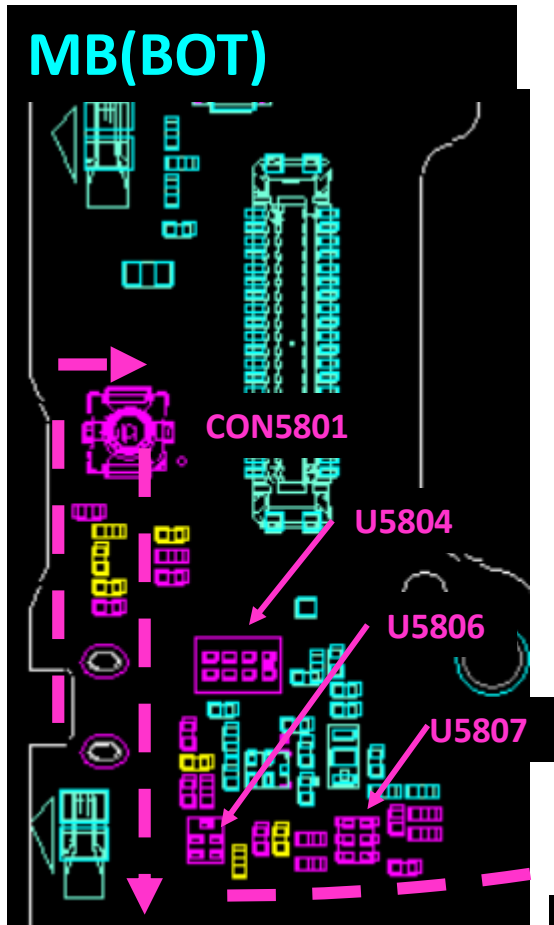
U5702

U5701

U5703

U4101

GPS_L2/L5



CON5801: RF SWITCH CONNECTOR
U5804: TRIPLEXER/TP2012-A1255MST/LF
U5806: Pre SAW/B39122B8884P810
U5807: LNA/WS7918DC-6/TR
U5808: Post SAW/B39122B8884P810
U4101: Transceiver/SDR865

NFC

U301: NFC CHIP-NXP/NQ310A1EV/C101Y

U302: DC/DC CONVERTER/FAN48615UC11X

X301: XTAL 27.12MHZ/X2C027120BA1H-U

